ADVANCED GRAPH THEORY

Course No: **MMTHDGT325**	Total Credits: **04**
Examination:	Total Marks: 100
(a). Assessment	Max. Marks: 28
(b). Theory	Max. Marks: 72
Time Duration: 2.5 hrs	Min. Pass Marks: 40

Objectives: To expose the student to the various concepts of graph theory to model many types of relations and processes in physical, biological, social and information systems.

UNIT -I

Colorings

Vertex coloring, chromatic number $\chi(G)$, bounds for $\chi(G)$, Brook's theorem, edge coloring, Vizing's theorem, map coloring, six color theorem, five color theorem, every graph is four colourable iff every cubic bridgeless plane map is 4-colorable, every planar graph is 4-colorable iff $\chi'(G) = 3$, Heawood map-coloring theorem, uniquely colorable graphs

UNIT -II

Matchings

Matchings and 1-factors, Berge's theorem, Hall's theorem, 1-factor theorem of Tutte, antifactor sets, f-factor theorem, f-factor theorem implies 1-factor theorem, Erdos-Gallai theorem follows from f-factor theorem, degree factors, k-factor theorem, factorization of K_n.

UNIT -III

Edge graphs and eccentricity sequences

Edge graphs, Whitney's theorem on edge graphs, Beineke's theorem, edge graphs of trees, edge graphs and traversability, total graphs, eccentricity sequences and sets, Lesniak's theorem for trees, construction of trees, neighbourhoods, Lesniak theorem graphs.

UNIT -IV

Groups in graphs and graph spectra

Automorphism groups of graphs, graphs with a given group, Frucht's theorem, Cayley digraph, spectrum of a graph, spectrum of some graphs-regular graph, complement of a graph, edge graph, complete graph, complete bipartite, cycle and path, Laplacian spectrum, energy of a graph, Laplacian energy.

Recommended Books:

1. R. Balakrishnan, K. Ranganathan, A Text Book of Graph Theory, Springer-Verlag, New York
2. B. Bollobas, Extremal Graph Theory, Academic Press.
3. F. Harary, Graph Theory, Addison-Wesley.
4. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science, Prentice Hall
5. S. Pirzada, An Introduction to Graph Theory, Universities Press, Orient Blackswan, 2012
6. W. T. Tutte, Graph Theory, Cambridge University Press.
7. D. B. West, Introduction to Graph Theory, Prentice Hall

ADVANCED GRAPH THEORY (MMTHDGT325)

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Chapter 1

Coloring of Graphs

INTRODUCTION

The rich theory of graph coloring traces its origins to 1852, when Francis Guthrie, while coloring a map of the counties of England, conjectured that four colors suffice to distinguish adjacent regions on any planar map. This deceptively simple query, popularized by Augustus De Morgan and Arthur Cayley, became known as the **Four Color Problem** and served as the catalyst for the development of chromatic graph theory. Early attempts to resolve it led to significant milestones: while Alfred Kempe's 1879 proof was later found to be flawed, his techniques were salvaged by Percy Heawood in 1890 to prove the **Five Color Theorem** and to generalize the problem to surfaces of higher genus, resulting in the **Heawood Map Coloring Theorem**.

As the field matured, the focus expanded from planar maps to the coloring of vertices and edges in general graphs. In 1941, R.L. Brooks provided the fundamental connection between the maximum degree $\Delta(G)$ and the **chromatic number** $\chi(G)$, characterizing the rare instances where the greedy bound is sharp. Parallel to this, the study of **edge coloring** was propelled by P.G. Tait, who in 1880 demonstrated that the Four Color Theorem is equivalent to the statement that every cubic bridgeless plane map has an edge chromatic index of 3 (i.e., $\chi'(G) = 3$). This line of inquiry culminated in **Vizing's Theorem** (1964), which famously classified all graphs into just two classes based on their edge chromatic number. The centuries-old quest finally concluded in 1976 when Appel and Haken provided a computer-assisted proof of the Four Color Theorem, yet the theoretical machinery developed along the way—from uniqueness of colorings to spectral bounds—remains a cornerstone of modern combinatorics.

1.1 Vertex Coloring

Definition 1.1.1. Let $G = (V, E)$ be an undirected graph. A **vertex coloring** of G is an assignment of colors to the vertices of G such that no two adjacent vertices have the same color. More formally, it is a function $c : V \to \{1, 2, \ldots, k\}$ for some positive integer k, such that if $(u, v) \in E$, then $c(u) \neq c(v)$.

The colors can be any set of distinct labels, but for convenience, we usually represent them by positive integers. A coloring using k colors is called a k**-coloring**.

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The colors can be any set of distinct labels, but for convenience, we usually represent them by positive integers. A coloring using k colors is called a k**-coloring**.

Definition 1.1.2. The **chromatic number** of a graph G, denoted by $\chi(G)$, is the minimum number of colors needed for a proper vertex coloring of G. In other words, $\chi(G)$ is the smallest integer k for which a k-coloring of G exists.

- If G contains no edges (an empty graph), then $\chi(G) = 1$.
- If G contains at least one edge, then $\chi(G) \geq 2$.
- For a complete graph K_n, $\chi(K_n) = n$ since every pair of vertices is adjacent and thus must have distinct colors.
- For a cycle graph C_n with $n \geq 3$:
  - If n is even, $\chi(C_n) = 2$.
  - If n is odd, $\chi(C_n) = 3$.

Proof. Label the vertices of C_n by $v_1, v_2, \ldots, v_n$ and color the vertices $(v_1, v_2, \ldots, v_{n-1}$ using color c_1 if i is odd and by color c_2 if n is even until all verices $(v_1, v_2, \ldots, v_n$ have been colored. Now, if $n-1$ is odd then v_{n-1} has color c_1 so we can assign color c_2 to vertex v_n. But if $n-1$ is even, then we are forced to use a different color, say c_3, for vertex v_n.

Theorem 1.1.3. *Every tree with $n \geq 2$ vertices is 2-chromatic.*

Proof. Let T be a tree with $n \geq 2$ vertices. Since $n \geq 2$, T must contain at least one edge, so $\chi(T) \geq 2$. We need to show that T can be 2-colored.

Choose an arbitrary vertex v in T and assign it color 1. For any other vertex $u \neq v$, assign u color 1 if its distance from v is even, and assign it color 2 if its distance from v is odd. Formally, let $d(v, u)$ be the distance (number of edges in the unique path) between v and u. Define a coloring $c : V(T) \to \{1, 2\}$ as follows: $c(u) = 1$ if $d(v, u)$ is even. $c(u) = 2$ if $d(v, u)$ is odd.

We must show that this is a proper 2-coloring. Consider any edge $(x, y) \in E(T)$. The vertices x and y must be adjacent. In any path from v to x and from v to y, the path from v to x and the path from v to y will differ by exactly one edge if x and y are adjacent. Specifically, if P_{vx} is the unique path from v to x, and P_{vy} is the unique path from v to y. If x and y are adjacent, then either P_{vx} includes y and the edge (y, x) or P_{vy} includes x and the edge (x, y). This implies that $d(v, x)$ and $d(v, y)$ must differ by exactly 1. For example, if $d(v, y) = d(v, x) + 1$, then $d(v, x)$ is even, $d(v, y)$ is odd, or vice versa. Therefore, one vertex will have an even distance from v and the other will have an odd distance from v. This means $c(x) \neq c(y)$ for any adjacent vertices x, y. Thus, T is 2-colorable. Since $\chi(T) \geq 2$, we conclude that $\chi(T) = 2$. ■

Theorem 1.1.4. *A graph G is bicolourable (2-chromatic) if and only if it has no odd cycles.*

Proof. Let G be a connected graph with cycles of only even length and let T be a spanning tree in G. Then, by Theorem 1.1.3, T can be coloured with two colours. Now add the chords to T one by one. As G contains cycles of even length only, the end vertices of every chord get different colours of T . Thus G is coloured with two colours and hence is 2-chromatic. Conversely, let G be bicolourable, that is, 2-chromatic. We prove G has even cycles only. Assume to the contrary that G has an odd cycle. Then G is 3-chromatic, a contradiction. Hence G has no odd cycles. ■

Theorem 1.1.5. *A nonempty graph G is bicolourable if and only if G is bipartite.*

Proof. Let G be a bipartite graph and let $V = V_1 \cup V_2$ be the bipartition of $V(G)$. Now assigning colour 1 to all vertices in V_1 and colour 2 to all vertices in V_2 gives a 2-colouring of G. Since G is nonempty, $\chi(G) = 2$.

Conversely, let G be bicolourable, that is, G has a 2-colouring. Denote by V_1 the set of all those vertices coloured 1 and by V_2 the set of all those vertices coloured 2. Then no two vertices in V_1 are adjacent and no two vertices in V_2 are adjacent. Thus any edge in G joins a vertex in V_1 and a vertex in V_2. Hence G is bipartite with bipartition $V = V_1 \cup V_2$. ■

Theorem 1.1.6 (Konig). *A graph is bicolourable if and only if it has no odd cycles.*

Proof. Let G be a connected graph with cycles of only even length and let T be a spanning tree in G. Then T can be coloured with two colours. Now add the chords to T one by one. As G contains cycles of even length only, the end vertices of every chord get different colours of T . Thus G is coloured with two colours and hence is $2-$chromatic. Conversely, let G be bicolourable, that is, $2-$chromatic. We prove G has even cycles only. Assume to the contrary that G has an odd cycle. Then G is at least $3-$chromatic, a contradiction. Hence, G has no odd cycles. ∎

Theorem 1.1.7. *For any graph G, $\chi(G) \leq \Delta(G) + 1$.*

Proof. Let G be any graph with n vertices. To prove the result, we induct on n. For $n = 1$, $G = K_1$ and $\chi(G) = 1$ and $\Delta(G) = 0$. Therefore the result is true for $n = 1$.

Assume that the result is true for all graphs with $n - 1$ vertices and therefore by induction hypothesis,

$$\chi(G - v) \leq \Delta(G - v) + 1 \leq \Delta(G) + 1$$

because $\Delta(G - v) \leq \Delta(G)$. This shows that $G - v$ can be coloured by using $\Delta(G) + 1$ colours. Since $\Delta(G)$ is the maximum degree of a vertex in G, vertex v has at most $\Delta(G)$ neighbours in G. Thus these neighbours use up at most $\Delta(G)$ colours in the colouring of $G - v$. Therefore, there is at least one colour not used by v's neighbours and that can be used to colour v giving a $\Delta(G) + 1$ colouring for G.

∎

Definition 1.1.8 (Color Critical Graphs)**.** If G is a $k-$chromatic graph and $\chi(G - v) = k - 1$ for every vertex $v \in V(G)$, then G is said to be $k-$critical.

Theorem 1.1.9. *If G is a k-critical graph, then*

a. *G is connected,*

b. *$\delta(G) \geq k - 1$,*

c. *G has no pair of subgraphs G_1 and G_2 for which $G = G_1 \cup G_2$, and $G_1 \cap G_2$ is a complete graph,*

d. *$G - v$ is connected for every vertex v of G, provided $k > 1$.*

Proof.

a. Assume G is not connected. Since $\chi(G) = k$, there is a component G_1 of G such that $\chi(G_1) = k$. If v is any vertex of G which is not in G_1, then G_1 is a component of the subgraph $G - v$. Therefore, $\chi(G - v) = \chi(G_1) = k$. This contradicts the fact that G is k-critical.

b. Assume to the contrary that v be a vertex of G so that $d(v) < k - 1$. Since G is k-critical, the subgraph $G - v$ has a $(k - 1)$-colouring. As v has at most $k - 2$ neighbours, these neighbours use at most $k - 2$ colours in this $(k - 1)$-colouring of $G - v$. Now, colour v with the unused colour and this gives a $(k - 1)$-colouring of G. This contradicts the given assumption that $\chi(G) = k$. Hence every vertex v has degree at least $k - 1$.

c. Let $G = G_1 \cup G_2$, where G_1 and G_2 are subgraphs with $G_1 \cap G_2 = K_r$. Since G is k-critical, therefore G_1 and G_2 both have chromatic number at most $k - 1$. Consider a $(k - 1)$-colouring of G_1 and a $(k - 1)$-colouring of G_2. As $G_1 \cap G_2$ is complete, in the overlap, every vertex in $G_1 \cap G_2$ has a different colour (in each of the $(k - 1)$-colourings). This implies that colours in the $(k - 1)$-colouring of G_2 can be rearranged such that it assigns the same colour to each vertex in $G_1 \cap G_2$, as is given by the colouring of G_1. Combining the two colourings then produces a $(k - 1)$-colouring of all of G. This is impossible, since $\chi(G) = k$. Thus no subgraphs of the type G_1 and G_2 exist.

d. Assume $G - v$ is disconnected, for some vertex v of G. Then $G - v$ has a subgraph H_1 and H_2 with $H_1 \cup H_2 = G - v$ and $H_1 \cap H_2 = \emptyset$. Let G_1 and G_2 be the subgraphs of G, where G_1 is induced by H_1 and v, while G_2 is induced by H_2 and v. Then $G = G_1 \cup G_2$ and $G_1 \cap G_2 = K_1$ (with K_1 as a single vertex). This contradicts (c) and thus $G - v$ is connected.

∎

Corollary 1.1.10. *A $k-$chromatic graph has at least k vertices of degree at least $k-1$ each.*

Proof. Consider G with chromatic number k. Remove all vertices from G to obtain a subgraph H such that H is $k-$critical. Then, $\chi(H) = k$ and every vertex in H has degree at least $k-1$. This completes the proof. ■

Theorem 1.1.11. *Let G be a graph and $k = \max\{\delta(G') : G'$ is a subgraph of $G\}$. Then $\chi(G) = k - 1$.*

Proof. Let H be a k-minimal subgraph of G. Then H is a subgraph of G and therefore $\delta(H) \leq k$. Using Theorem 1.1.9, we have, $\delta(H) \geq \chi(H) - 1 = \chi(G) - 1$. Thus, $\chi(G) \leq \delta(H) + 1 = k + 1$. ■

Theorem 1.1.12. *Let G b a graph with degree sequence $[d_i]_1^n$ such that $d_1 \geq d_2 \geq \cdots \geq d_n$. Then, $\chi(G) \leq \max\{\min\{i, d_i + 1\}\}$.*

Proof. Let G be k-chromatic. Then, by Theorem 1.1.11, G has at least k vertices of degree at least $k-1$. Therefore, $d_k \geq k-1$ and $\max\{\min\{i, d_i+1\}\} \geq \min\{k, d_k+1\} = k = \chi(G)$. ■

Algorithm 1.1.13 (The Greedy Coloring Algorithm)**.** The **greedy coloring algorithm** is a simple method for graph vertex coloring that processes vertices in a specific order and assigns each vertex the smallest possible color not used by its already-colored neighbors.

1. Order the vertices $v_1, v_2, \ldots, v_n$
2. For $i = 1$ to n:

   Assign v_i the smallest color $c \in \mathbb{N}$ such that no adjacent vertex already has color c.

The algorithm produces a proper coloring but may not use the minimum number of colors (chromatic number $\chi(G)$). The number of colors used depends heavily on the vertex ordering. (See Figures 1.1, 1.1).

Algorithm 1.1.14. A **Kempe chain** is a maximal connected component of a graph induced by vertices colored with two given colors.

The **Kempe chain argument** is a recoloring technique used in graph coloring where colors are swapped along an alternating path between two colors. By swapping these colors along the entire component, one obtains another proper coloring.

This method was famously used in:

- The proof of the **Five Color Theorem** for planar graphs
- Alfred Kempe's flawed but influential attempt at the **Four Color Theorem**
- Various graph coloring algorithms and proofs

How to use Kempe Chain Argument: Let G be a graph that uses at least two colors for its proper coloring. Choose two colors, say i & j and construct a subgraph H of G induced by the colors i & j. Let K be a connected component in H. Interchange the colors i and j in K. This produces a new coloring of G. In this process, the subgraph K of G is known as Kempe chain and this technique to obtain a new coloring of G is called the Kempe chain argument. (See Figures 1.1, 1.1).

Theorem 1.1.15 (Brooks' Theorem)**.** *Let G be a connected graph with maximum degree Δ. If G is neither a complete graph nor an odd cycle, then $\chi(G) \leq \Delta$.*

Proof. We proceed by induction on the number of vertices n. The base cases for $n \leq 3$ are easily verified. Now assume the theorem holds for all graphs with fewer than n vertices.

Consider two cases:

Case 1: G is not regular.

Then there exists a vertex v with $\deg(v) < \Delta$. Remove v to form $G' = G - v$. Since G is connected, each component of G' has maximum degree $\leq \Delta$ and is neither a complete graph nor an odd cycle (as these are regular). By the induction hypothesis, each component can be properly colored with

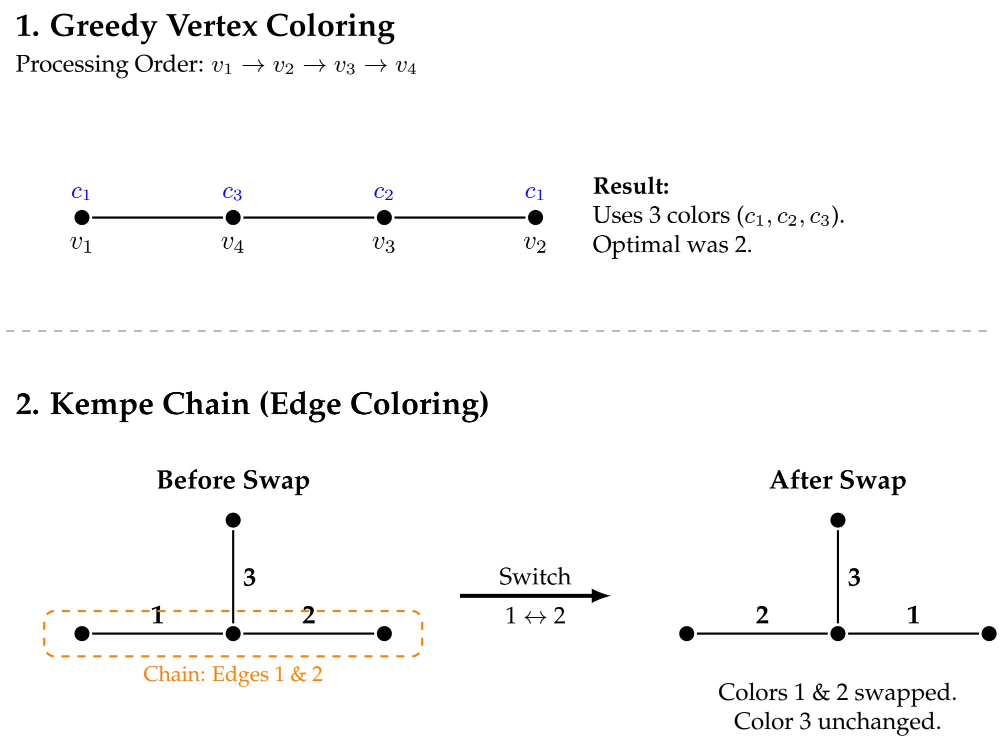


Figure 1.1: Top: Greedy Vertex Coloring. Bottom: Kempe Chain Argument on Edge Coloring.

at most Δ colors. Since v has at most $\Delta - 1$ neighbors, we can color v with a color not used by its neighbors. Thus, $\chi(G) \leq \Delta$.
Case 2: G is Δ-regular.
Since G is not complete, there exists a vertex v with neighbors u and w such that $u \not\sim w$. Consider $G' = G - v$. By induction hypothesis, G' can be properly colored with Δ colors.

If the neighbors of v use at most $\Delta - 1$ colors, we can color v directly. Otherwise, the neighbors use all Δ colors. Let u and w be colored 1 and 2 respectively. Consider the subgraph H induced by vertices colored 1 or 2. Let C_u be the component of H containing u. If $w \notin C_u$, swapping colors 1 and 2 in C_u frees color 1 for v. If $w \in C_u$, then there exists a path from u to w with alternating colors 1 and 2. This path, together with edges v–u and v–w, forms an odd cycle. Since G is not an odd cycle, there must exist another vertex x adjacent to v with color 3. Now consider colors 1 and 3: if the components containing u and x are distinct, we can swap to free color 1. If not, then similarly, the path from u to x in the 1-3 component forms an odd cycle with v. But then the 1-2 and 1-3 components overlap, forcing a structure that implies G is complete—a contradiction. Thus, recoloring is always possible, and $\chi(G) \leq \Delta$. ∎

Theorem 1.1.16. *For any graph G, $\frac{n}{\chi(G)} \leq \alpha_0(G) \leq n - \chi(\overline{G}) + 1$.*

Proof. If $\chi(G) = k$, then V can be partitioned into k colour classes $V_1, V_2, \ldots, V_k$, each of which is an independent set of vertices. If $|V_i| = n_i$, then every $n_i \leq \alpha_0$, so that $n = \sum n_i \leq k\alpha_0$. This proves the middle inequality: $\frac{n}{k} \leq \alpha_0$, or $\frac{n}{\chi(G)} \leq \alpha_0(G)$. Now, let S be a maximal independent set containing α_0 vertices. Clearly, $\chi(G - S) \geq \chi(G) - 1$. As $G - S$ has $n - \alpha_0$ vertices, $\chi(G - S) \leq n - \alpha_0$. Thus, $\chi(G) - 1 \leq \chi(G - S) \leq n - \alpha_0$. Rearranging, $\chi(G) \leq n - \alpha_0 + 1$. This proves the last inequality: $\alpha_0(G) \leq n - \chi(G) + 1$. As G has a complete subgraph of order $\chi(\overline{G})$, $\chi(G) \geq \alpha_0(\overline{G})$, or $\frac{n}{\chi(G)} \leq \frac{n}{\alpha_0(\overline{G})}$, proving the first inequality. ∎

Theorem 1.1.17. *For any graph of order n,*

$$2\sqrt{n} \leq \chi + \overline{\chi} \leq n + 1, \text{ and } n \leq \chi\overline{\chi} \leq \left(\frac{n+1}{2}\right)^2, \tag{1.1}$$

where $\chi = \chi(G)$ and $\overline{\chi} = \chi(\overline{G})$.

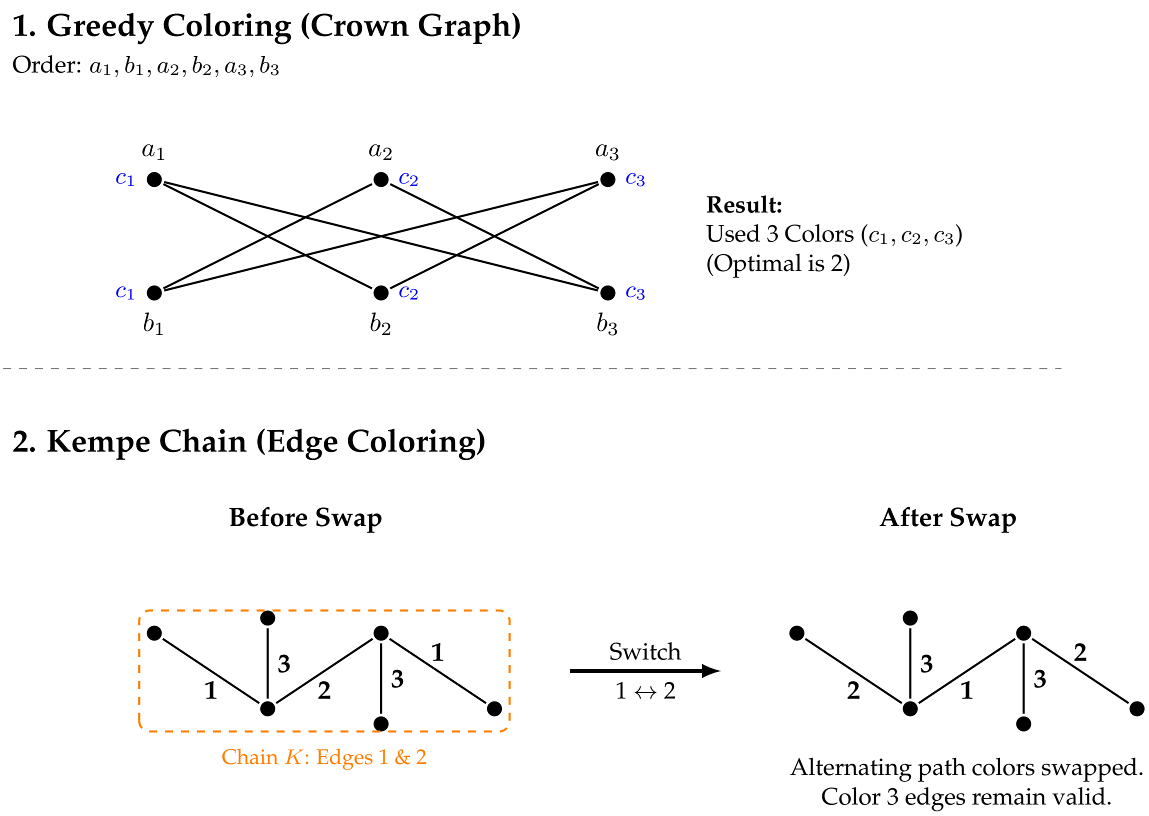


Figure 1.2: Top: Greedy coloring on a Crown Graph (6 vertices) forcing a sub-optimal 3rd color. Bottom: Kempe Chain argument on a larger path, swapping edge colors 1 and 2.

Proof. Let G be k-chromatic and let $v_1, v_2, \ldots, v_k$ be the colour classes of G, where $|V_i| = n_i$. Then $\sum n_i = n$ and $\max n_i \geq n/k$. Since each V_i induces a complete subgraph of $\overline{G}$, $\overline{\chi} \geq \max n_i/n_i$ so that $\overline{\chi} \geq n/n_i$. Thus for each i, $n_i \geq n/\overline{\chi}$. Since the geometric mean of two positive numbers is always less or equal to their arithmetic mean, it follows that $\chi + \overline{\chi} \geq 2\sqrt{n}$.

To prove $\chi + \overline{\chi} \leq n + 1$, we induct on n. Clearly, the equality holds for $n = 1$. We assume that $\chi(G) + \overline{\chi}(G) \leq n$ for all graphs G having fewer than n vertices. Let H and $\overline{H}$ be complementary graphs with n vertices and let v be a vertex of H. Then $G = H - v$ and $\overline{G} = \overline{H} - v$ are complementary graphs with $n - 1$ vertices. Let the degree of v in H be d, so that degree of v in $\overline{H}$ is $n - d - 1$. Clearly, $\chi(H) \leq \chi(G) + 1$ and $\overline{\chi}(H) \leq \overline{\chi}(G) + 1$. If either $\chi(H) < \chi(G) + 1$ or $\overline{\chi}(H) < \overline{\chi}(G) + 1$, then $\chi(H) + \overline{\chi}(H) \leq n + 1$. If $\chi(H) = \chi(G) + 1$ and $\overline{\chi}(H) = \overline{\chi}(G) + 1$, then the removal of v from H, producing G, decreases the chromatic number, so that $d \geq \chi(G)$. Similarly, $n - d - 1 \geq \overline{\chi}(G)$. Thus, $\chi(G) + \overline{\chi}(G) \leq n - 1$. Therefore we always have $\chi(H) + \overline{\chi}(H) \leq n + 1$. Applying now the inequality $4\chi\overline{\chi} \leq (\chi + \overline{\chi})^2$, we get $\chi\overline{\chi} \leq \left(\frac{n+1}{2}\right)^2$. ■

2nd Proof. Let G be a graph of order n. Let $\chi = \chi(G)$ and $\overline{\chi} = \chi(\overline{G})$.

(i) $n \leq \chi\overline{\chi}$**:** Consider a χ-coloring of G with color classes $V_1, \ldots, V_\chi$. Each V_i is an independent set in G. Consider a $\overline{\chi}$-coloring of $\overline{G}$ with color classes $\overline{V}_1, \ldots, \overline{V}_{\overline{\chi}}$. Each $\overline{V}_j$ is an independent set in $\overline{G}$, which means it is a clique in G. Every vertex $v \in V(G)$ belongs to some V_i and some $\overline{V}_j$. If u, v are two distinct vertices such that they are in the same color class V_i in G AND in the same color class $\overline{V}_j$ in $\overline{G}$, then u and v are not adjacent in G (because V_i is an independent set) and u and v are not adjacent in $\overline{G}$ (because $\overline{V}_j$ is an independent set). This is impossible, as any two distinct vertices must be adjacent in either G or $\overline{G}$. Therefore, each vertex must be assigned a unique ordered pair of colors (i, j) where $1 \leq i \leq \chi$ and $1 \leq j \leq \overline{\chi}$. The total number of such unique pairs is $\chi\overline{\chi}$. Since there are n vertices, $n \leq \chi\overline{\chi}$.

(ii) $2\sqrt{n} \leq \chi + \overline{\chi}$**:** Using the AM-GM inequality: $\frac{\chi+\overline{\chi}}{2} \geq \sqrt{\chi\overline{\chi}}$. From the previous part, $n \leq \chi\overline{\chi}$, so $\sqrt{n} \leq \sqrt{\chi\overline{\chi}}$. Combining these, $\frac{\chi+\overline{\chi}}{2} \geq \sqrt{n}$, which implies $2\sqrt{n} \leq \chi + \overline{\chi}$.

(iii) $\chi + \overline{\chi} \leq n + 1$**:** This is a standard result, often attributed to Nordhaus and Stewart (1963). The maximum value of $\chi(G) + \chi(\overline{G})$ is $n + 1$, achieved by complete graphs (K_n) and empty graphs ($\overline{K_n}$). For K_n, $\chi(K_n) = n$ and $\overline{\chi}(K_n) = 1$, so $\chi + \overline{\chi} = n + 1$. For $\overline{K_n}$, $\chi(\overline{K_n}) = 1$ and $\overline{\chi}(\overline{K_n}) = n$, so $\chi + \overline{\chi} = n + 1$.

(iv) $\chi\overline{\chi} \leq \left(\frac{n+1}{2}\right)^2$**:** From $\chi + \overline{\chi} \leq n + 1$, and using the property that for any real numbers a, b, $4ab \leq (a + b)^2$: $4\chi\overline{\chi} \leq (\chi + \overline{\chi})^2 \leq (n + 1)^2$. Dividing by 4, we get $\chi\overline{\chi} \leq \frac{(n+1)^2}{4} = \left(\frac{n+1}{2}\right)^2$. ■

1.2 Edge Colouring

An edge colouring of a non-empty graph G is a labelling $f : E(G) \to \{1, 2, \ldots\}$ such that adjacent edges are assigned different colours. The labels are called *colours*. A k-edge colouring of G is an edge colouring of G that uses exactly k different colours, and in this case, G is said to be k-edge colourable.

This definition implies that a k-edge colouring of a graph $G(V, E)$ partitions the edge set E into k independent sets $E_1, E_2, \ldots, E_k$ such that $E = E_1 \cup E_2 \cup \ldots \cup E_k$. These independent sets E_i, for $1 \leq i \leq k$, are called the *colour classes*. The function $f : E(G) \to \{1, 2, \ldots, k\}$ such that $f(e) = i$ for each $e \in E_i$ (where $1 \leq i \leq k$) is called the *colour function*. The minimum number k for which a k-colouring of G exists is called the *edge chromatic number* (or *edge chromatic index*) and is denoted by $\chi'(G)$.

We have the following observations:

1. If H is a subgraph of a graph G, then $\chi'(H) \leq \chi'(G)$.

2. For any graph G, $\chi'(G) \geq \Delta(G)$, where $\Delta(G)$ is the maximum degree of G.

If v is any vertex of G with degree $d(v) = \Delta(G)$, then the $\Delta(G)$ edges incident with v must all have different colours in any edge colouring of G.

3. For a cycle graph C_n of order n:

$$\chi'(C_n) = \begin{cases} 2, & \text{if } n \text{ is even,} \\ 3, & \text{if } n \text{ is odd.} \end{cases}$$

Definition: Let i be a colour used in an edge colouring of a graph G. If there is an edge coloured i incident to a vertex v of G, we say i is *present* at v. If there is no edge coloured i incident to v, we say i is *absent* from v.

Theorem 1.2.1 (König, 1916)**.** *For a non-empty bipartite graph G, $\chi'(G) = \Delta(G)$.*

Proof. The proof is by induction on the number of edges of G. If G has only one edge, the result is trivial since $\chi'(G) = 1$ and $\Delta(G) = 1$.

Assume that G has more than one edge, and that the theorem holds for all non-empty bipartite graphs having fewer edges than G. Since we know $\Delta(G) \leq \chi'(G)$ (Observation 2), it suffices to prove that G has a $\Delta(G)$-edge colouring. Let $\Delta(G) = k$.

Consider an arbitrary edge $e = uv$ in G. The graph $G - e$ is bipartite and has fewer edges than G. By the induction hypothesis, $G - e$ has a $\Delta(G - e)$-edge colouring. Since $\Delta(G - e) \leq \Delta(G) = k$, $G - e$ has a k-edge colouring. We will show how to extend this k-edge colouring of $G - e$ to a k-edge colouring of G by assigning a colour to e.

Since $d_G(u) \leq k$ and $d_G(v) \leq k$, in the k-edge colouring of $G - e$, there is at least one colour absent at vertex u and at least one colour absent at vertex v.

1. **Case 1: There is a colour c that is absent at both u and v.** In this case, we can assign colour c to edge $e = uv$. This results in a k-edge colouring of G.

2. **Case 2: There is no colour absent at both u and v.** This implies that for every colour c, if c is absent at u, it must be present at v, and vice versa. Let i be a colour absent at v (and thus present at u) and j be a colour absent at u (and thus present at v).

   Consider the subgraph $H(i, j)$ formed by all edges in $G - e$ coloured either i or j. Let K be the connected component of $H(i, j)$ containing u. We claim that v does not belong to K.

   Proof of claim: Assume, for contradiction, that v belongs to K. Then there exists a path P in K between u and v. Since u and v are adjacent in G (via edge e), they belong to different partitions in the bipartite graph G. Therefore, any path between u and v in G must have odd length. The path P is composed of alternately coloured edges i and j. Since colour i is present at u, the first edge of P incident to u must be coloured i. As P has an odd number of edges and alternates colours, its last edge incident to v must also be coloured i. This implies that colour i is present at v, which contradicts our initial assumption that i is absent from v. Thus, v does not belong to the Kempe chain K.

   Now, we apply the Kempe chain argument to the component K. We interchange the colours i and j on all edges in K.

   - At vertex u: Since $u \in K$ and the colour i was present at u, after the recolouring, colour j becomes present at u and colour i becomes absent at u.
   - At vertex v: Since $v \notin K$, the colours of edges incident to v are unaffected by this interchange. Therefore, i remains absent from v.

   After this recolouring, colour i is now absent at both u and v. We are now back to Case 1, where we can assign colour i to edge $e = uv$. This yields a k-edge colouring of G.

Therefore, in all cases, G has a $\Delta(G)$-edge colouring. ∎

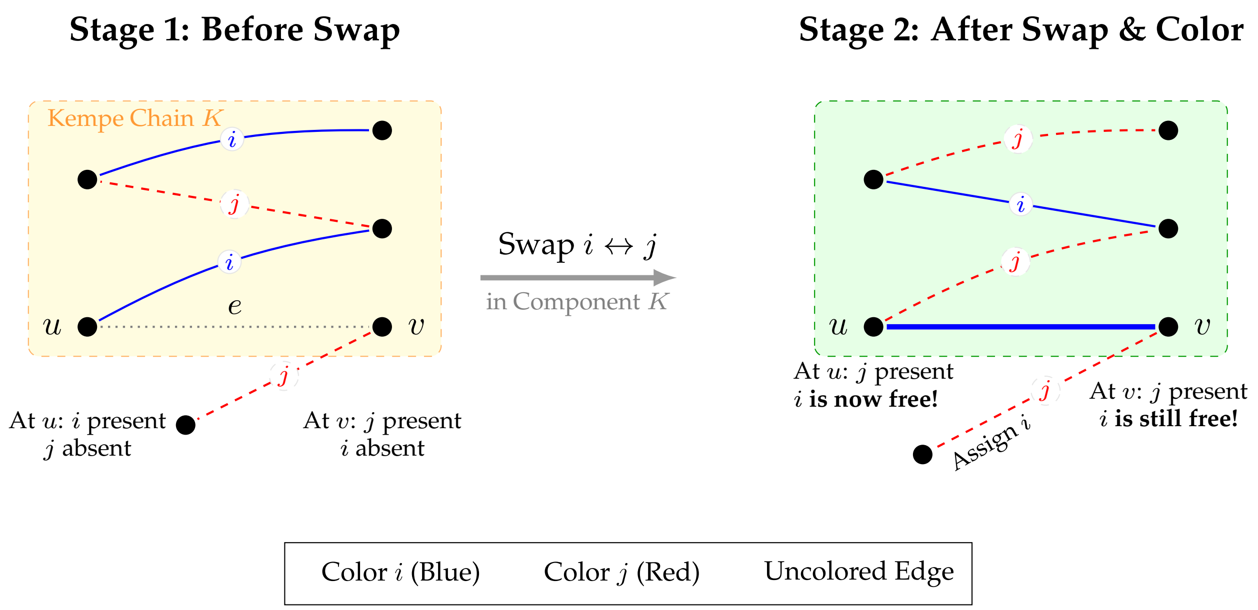


Figure 1.3: Visualization of the Kempe Chain swap. By drawing the highlight background first and using standard arrow syntax, this code is robust and compatible with standard LaTeX installations.

Theorem 1.2.2. *If $G = K_n$ is a complete graph with n vertices, $n \geq 2$, then*

$$\chi'(G) = \begin{cases} n-1, & \text{if } n \text{ is even,} \\ n, & \text{if } n \text{ is odd.} \end{cases}$$

Proof. Let $G = K_n$ be a complete graph with n vertices.

Case 1: n **is odd.** We know that $\chi'(G) \geq \Delta(G) = n-1$. We need to show that $\chi'(K_n) = n$. Assume for contradiction that K_n has an $(n-1)$-edge colouring. In an edge colouring, edges of a single colour form a matching. The maximum size of a matching in K_n when n is odd is $(n-1)/2$. If K_n has an $(n-1)$-edge colouring, its $n(n-1)/2$ edges must be partitioned into $n-1$ colour classes. If each colour class contains at most $(n-1)/2$ edges (the maximum possible for a matching in K_n when n is odd), then the total number of edges would be at most $(n-1) \times \frac{n-1}{2} = \frac{(n-1)^2}{2}$. However, K_n has $n(n-1)/2$ edges. Comparing the total number of edges: $\frac{(n-1)^2}{2} < \frac{n(n-1)}{2}$ for $n \geq 2$. This is a contradiction. Therefore, K_n cannot be $(n-1)$-edge colourable, meaning $\chi'(K_n) \geq n$. Since $\chi'(K_n) \geq \Delta(K_n) = n-1$, and we have shown it must be greater than $n-1$, it must be at least n. A simple geometric construction can show that n colours are sufficient. Arrange the n vertices of K_n in a regular polygon. Assign a colour to each edge of the polygon. For each edge on the boundary, colour all edges parallel to it with the same colour. This uses $(n-1)/2$ colours for edges perpendicular to a line from a vertex to the midpoint of the opposite side. This type of construction, properly executed, shows n colours are sufficient. More simply, consider the edges incident to one vertex v_0. These $n-1$ edges require $n-1$ distinct colours. The remaining edges can then be coloured in a way that respects the $n-1$ initial colours plus one additional colour. This type of construction can be more formally described to show that n colours are sufficient. Thus, $\chi'(K_n) = n$ for odd n.

Case 2: n **is even.** We know that $\chi'(G) \geq \Delta(G) = n-1$. We will show that K_n can be $(n-1)$-edge coloured. Let v be an arbitrary vertex of K_n. Consider the graph $K_n - v$, which is a complete graph on $n-1$ vertices. Since n is even, $n-1$ is odd. From Case 1, we know that $\chi'(K_{n-1}) = n-1$. So, $K_n - v$ has an $(n-1)$-edge colouring.

Now, we extend this colouring to K_n by colouring the $n-1$ edges incident to v. In any $(n-1)$-edge colouring of K_{n-1}, at each vertex $w \in V(K_{n-1})$, there is exactly one colour absent from the edges incident to w. This is because $d_{K_{n-1}}(w) = n-2$, and there are $n-1$ total colours. Furthermore, for distinct vertices $w_1, w_2 \in V(K_{n-1})$, the colours absent at w_1 and w_2 must be distinct. If they were the same, say colour c, then c would be absent from both w_1 and w_2. The edge w_1w_2 could then be coloured c, but this is not guaranteed to be consistent with other assignments. A more common argument is that one can obtain an $(n-1)$-edge colouring by taking an $(n-1)$-edge colouring of $K_n - v$, and for each vertex $w \in V(K_{n-1})$, colour the edge vw with the unique colour that is *not* used on any edge incident to w in the colouring of K_{n-1}. Since there are $n-1$ vertices in K_{n-1}, and each has degree $n-2$, there is exactly one colour missing at each vertex. It can be shown that these missing colours are distinct for distinct vertices. Therefore, each edge vw can be assigned a unique colour without conflict at v or w. This construction provides an $(n-1)$-edge colouring for K_n. Thus, $\chi'(K_n) = n-1$ for even n. ■

Theorem 1.2.3 (Vizing). *For any graph G, $\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$.*

Proof. Let G be a simple graph, we always have $\Delta(G) = \chi'(G)$. To prove $\chi'(G) \leq \Delta(G) + 1$, we use induction on the number of edges of G. Let $\Delta(G) = k$. If G has only one edge, then $k = 1 = \chi'(G)$. Therefore assume that G has more than one edge and that the result is true for all graphs having fewer edges than G. Let $e = v_1v_2$ be an edge of G. Then by induction hypothesis the subgraph $G - e$ has $(k+1)$-edge colouring and let the colours used be $1, 2, \ldots, k+1$. Since $d(v_1) \leq k$ and $d(v_2) \leq k$, out of these $k+1$ colours at least one colour is absent from v_1 and at least one colour is absent from v_2. If there is a common colour absent from both v_1 and v_2, then we use this to colour e and get a $(k+1)$-colouring of G. Therefore in this case, $\chi'(G) \leq k+1$.

Now assume that whenever j is absent from v_1 but present at v_2 and there is a colour, say 2, absent from v_2 but present at v_1. We start from v_1 and we construct a sequence of distinct vertices $v_1, v_2, \ldots, v_j$, where each v_i for $i \geq 2$ is adjacent to v_1. Let v_1v_3 be coloured 2. This v_3 exists, because 2 is present at v_1. We observe that not all the $k+1$ colours are present at v_3 and assume that the colour 3 is absent from v_3. But the colour 3 is present at v_1 and choose the vertex v_4 so that v_1v_4 is coloured 3. Continuing in this way, we choose a new colour i absent from v_i but present at v_1, so that v_1v_{i+1} is the edge coloured i. In this way, we get a sequence of vertices $v_1, v_2, v_3, \ldots, v_{j-1}, v_j$ such that

a. v_i is adjacent to v_1 for each $i > 1$,

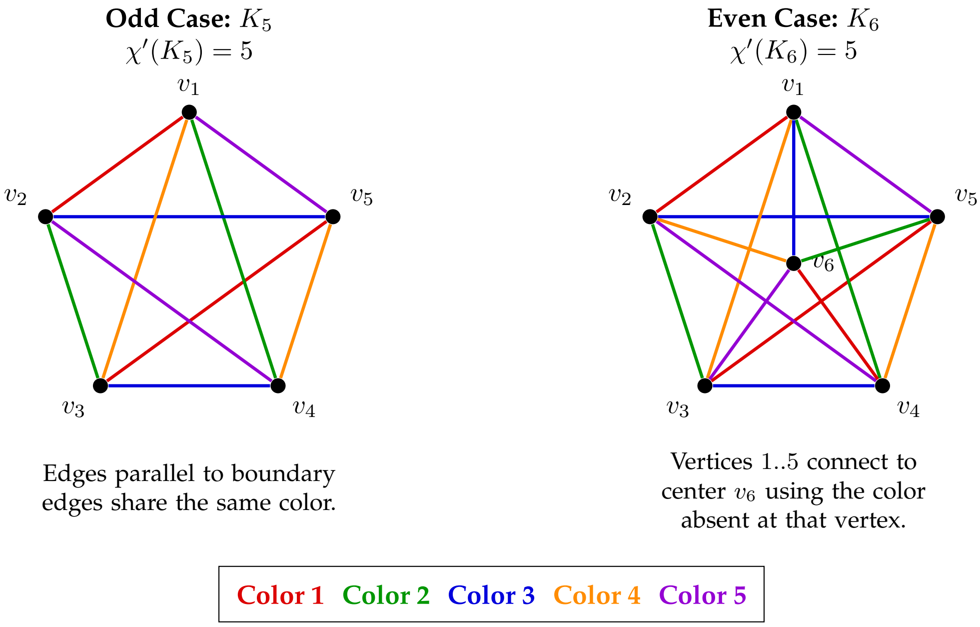


Figure 1.4: Coloring of K_5 and K_6

b. the colour i is absent from each v_i for $i = 1, 2, \ldots, j-1$ and

c. the edge v_1v_{i+1} is coloured i for each $i = 1, 2, \ldots, j-1$.

As $d(v_1) \leq k$, the sequence $v_1, v_2, v_3, \ldots, v_{j-1}, v_j$ has at most $k+1$ terms, that is, $j \leq k+1$. Now assume that $v_1, v_2, \ldots, v_j$ is a longest such sequence, that is, the sequence for which it is not possible to find a new colour j, absent from v_j, together with a new edge v_1v_{j+1} is coloured j. We first assume that for some colour j absent from v_j, there is no edge of that colour present at v_1. We remove the colour $j-1$ from v_1v_j and recolour the edges v_1v_i with i, $i = 2, 3, \ldots, j-1$. Since i was absent from v_i for each $i = 2, \ldots, j-1$, this gives a $(k+1)$-colouring of the subgraph $G - v_1v_j$. Now as the colour j is absent from both v_1 and v_j, recolour v_1v_j by the colour j. This gives a $(k+1)$-colouring of G.

Now assume that whenever j is absent from v_j, j is present at v_1. If v_1v_{j+1} is a new neighbour of v_1 so that v_1v_{j+1} is coloured by j, then we have extended our sequence to $v_1, v_2, \ldots, v_{j+1}$ which is a contradiction to the assumption that $v_1, v_2, \ldots, v_j$ is the longest sequence. Thus one of the edges $v_1v_3, \ldots, v_1v_j - 1$ is to be coloured by j, say v_1v_ℓ with $3 \leq \ell \leq j-1$. Now colour $e = v_1v_2$ by 2, and for $i = 3, \ldots, \ell-1$, recolour each of the edges v_1v_i by i while unaltering the colours of the edges v_1v_i for $i = \ell+1, \ldots, j$. Removing the colour j from v_1v_ℓ, we have a $(k+1)$-colouring of the edge deleted subgraph $G - v_1v_\ell$. Let $H(1, j)$ represent the subgraph of G induced by the edges coloured 1 or j in this partial colouring of G. Since the degree of every vertex in $H(1, j)$ is either 1 or 2, each component of $H(1, j)$ is either a path or a cycle. As 1 is absent from v_1 and j is absent from both v_1 and v_j, it follows that all these three vertices do not belong to the same connected component of $H(1, j)$. Therefore, if K and L represent the corresponding Kempe chains containing v_j and v_ℓ respectively, then either $v_1 \notin K$ or $v_1 \notin L$. Let $v_1 \notin L$. Then interchanging the colours of L, the Kempe chain argument gives a $(k+1)$-colouring of $G - v_1v_\ell$ in which 1 is missing from both v_1 and v_j. Colouring v_1v_j by 1 gives a $(k+1)$-colouring of G. Now, let $v_1 \notin K$. Colour the edge v_1v_ℓ by ℓ, recolour the edges v_1v_i by i, for $i = \ell, \ldots, j-1$, and remove the colour $j-1$ from v_1v_j. Then, from the definition of the sequence $v_1, v_2, \ldots, v_j$, we get a $(k+1)$-colouring of $G - v_1v_j$ without affecting two coloured subgraph $H(1, j)$. Using Kempe chain argument to interchange the colours of K, we obtain a $(k+1)$-colouring of $G - v_1v_j$ in which 1 is absent from both v_1 and v_j. Therefore, again colouring v_1v_j by 1 gives $(k+1)$-colouring of G. ■

2nd Proof. The lower bound $\Delta(G) \leq \chi'(G)$ is straightforward. At any vertex v with degree $\Delta(G)$, there are $\Delta(G)$ edges incident to it. Since all these edges must receive distinct colors in a proper edge coloring, at least $\Delta(G)$ colors are necessary. Thus, $\chi'(G) \geq \Delta(G)$.

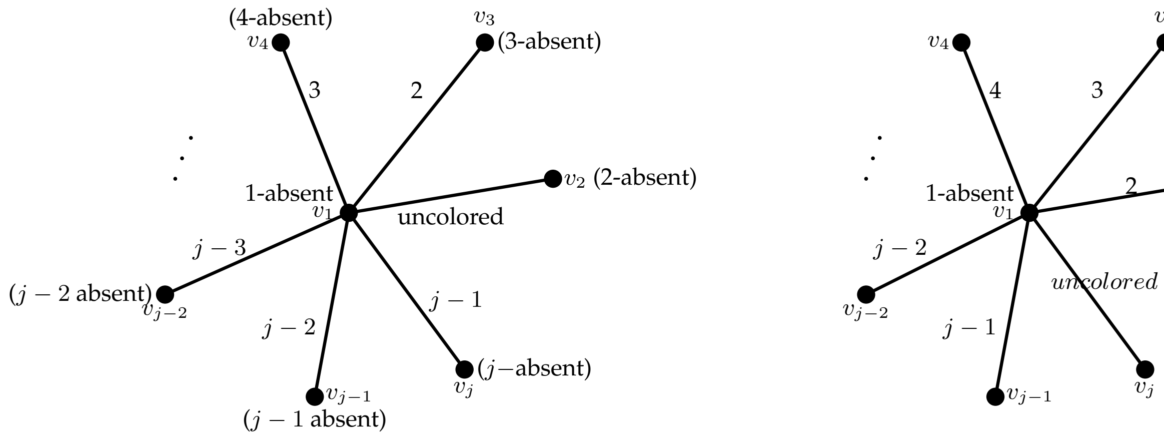


Figure 1.5: Vizing's Theorem

The more challenging part is to prove the upper bound $\chi'(G) \leq \Delta(G) + 1$. This part of the proof typically involves a constructive algorithm or an inductive argument. Here, we provide a high-level constructive proof sketch.

Let $\Delta = \Delta(G)$. We want to show that G can be $(\Delta + 1)$-edge-colored. We proceed by induction on the number of edges m in G. The base case is trivial for $m = 0$ or $m = 1$.

Assume that any graph with fewer than m edges can be $(\Delta+1)$-edge-colored. Consider a graph G with m edges. Let $e = uv$ be an uncolored edge in G. We try to color e with a color $c \in \{1, \ldots, \Delta+1\}$. If we can color all edges one by one, we will have a $(\Delta + 1)$-edge-coloring.

Consider a partial edge coloring c' of G, where some edges might be uncolored. For any vertex $x \in V(G)$:

- Let $C(x)$ be the set of colors used on edges incident to x.
- Let $M(x)$ be the set of missing colors at x, i.e., $M(x) = \{1, \ldots, \Delta + 1\} \setminus C(x)$.

Since the degree of any vertex is at most Δ, and we have $\Delta + 1$ available colors, each vertex x must have at least one missing color, i.e., $M(x) \neq \emptyset$.

Now, let $e_0 = uv$ be an uncolored edge. We want to assign a color to e_0. If there is a color $c \in M(u) \cap M(v)$, we can assign $c(e_0) = c$. If no such color exists, then $M(u)$ and $M(v)$ are disjoint. Since $|M(u)| \geq 1$ and $|M(v)| \geq 1$, we must have: $|C(u)| \leq \Delta$ and $|C(v)| \leq \Delta$. Also, $|M(u)| \geq (\Delta+1)-\Delta = 1$ and $|M(v)| \geq (\Delta + 1) - \Delta = 1$.

If $M(u) \cap M(v) = \emptyset$, we can try to "rearrange" colors to make a color available for uv. Let $a \in M(u)$ and $b \in M(v)$. Since $a \notin M(v)$, there must be an edge vw colored a. Since $b \notin M(u)$, there must be an edge ux colored b. We consider an a, b-alternating path or cycle starting from v. An a, b-alternating path (or "fan") at a vertex x is a path where edges alternate between colors a and b.

The proof involves a technique called "color shifting" or "fan coloring". Let $a \in M(u)$ and $b \in M(v)$. Since $a \notin M(v)$, there exists a vertex w such that the edge vw is colored a. Consider the path P starting at w that alternates between colors a and b. If this path P does not reach u, we can swap the colors a and b along P. This frees up color a at v (if w is the endpoint that has an edge colored a) and color b at the other end. This modification allows us to make a available at v, so we can then color uv with a. The critical part is showing that such a path P will always exist and that swapping colors along it does not create new conflicts and ultimately allows uv to be colored. The argument considers various cases for the a, b-alternating paths/cycles stemming from v. The key idea is that since the number of colors available at each vertex is $\Delta + 1$ (one more than its maximum possible degree), there's always "just enough" flexibility to resolve conflicts locally by judiciously choosing an alternating path and flipping its colors. This ensures that no matter how complex the graph, an edge uv can always find a color without exceeding $\Delta + 1$ colors.

By systematically applying this procedure (e.g., iteratively trying to color uncolored edges, and if a conflict arises, using color shifting), we can always find a proper $(\Delta + 1)$-edge-coloring for G. ■

Theorem 1.2.4. *Let G be a graph of order n and let $\overline{G}$ be the complement of G. Then*

1. $n-1 \leq \chi'(G)+\chi'(\overline{G}) \leq 2(n-1)$,
   $0 \leq \chi'(G)\chi'(\overline{G}) \leq (n-1)^2$, *for even* n,
2. $n \leq \chi'(G)+\chi'(\overline{G}) \leq 2n-3$,
   $0 \leq \chi'(G)\chi'(\overline{G}) \leq (n-1)(n-2)$, *for odd* n.

Further, the bounds are the best possible for every positive integer $n(\neq 2)$.

Proof. Let G be a graph of order n and let $\overline{G}$ be the complement of G. Clearly,

$$\Delta(G) \geq n-1-\Delta(\overline{G}), \quad \text{so that} \quad \Delta(G)+\Delta(\overline{G}) \geq n-1.$$

Therefore, combining with $\chi'(G) \geq \Delta(G)$, we get

$$\chi'(G)+\chi'(\overline{G}) \geq n-1.$$

This proves the lower bound for $\chi'(G)+\chi'(\overline{G})$ for both even and odd n.

Part 1. (Even n) The upper bound $\chi'(G)+\chi'(\overline{G}) \leq 2(n-1)$ comes from Vizing's theorem, which states $\chi'(G) \leq \Delta(G)+1$. Thus, $\chi'(G) \leq n-1+1 = n$ (since $\Delta(G) \leq n-1$). Similarly, $\chi'(\overline{G}) \leq n$. So, $\chi'(G)+\chi'(\overline{G}) \leq 2n$. The tighter upper bound $2(n-1)$ is generally true if neither G nor $\overline{G}$ is a complete graph (or empty graph). For n even, $\chi'(K_n) = n-1$ and $\chi'(\overline{K_n}) = \chi'(E_n) = 0$ (for E_n being the empty graph). In this case $\chi'(K_n)+\chi'(\overline{K_n}) = n-1$, which satisfies $n-1 \leq n-1 \leq 2(n-1)$. The upper bound for $\chi'(G)\chi'(\overline{G})$ is $(n-1)^2$. This happens, for example, if $\chi'(G) = n-1$ and $\chi'(\overline{G}) = n-1$.

Part 2. (Odd n) We claim $n \leq \chi'(G)+\chi'(\overline{G})$. If n is odd, we have $\chi'(G)+\chi'(\overline{G}) \geq n$. Consider the case where $G = K_n$ (for odd n). Then $\chi'(K_n) = n$. Its complement $\overline{K_n}$ is the empty graph E_n, so $\chi'(E_n) = 0$. Thus $\chi'(K_n)+\chi'(\overline{K_n}) = n+0 = n$. This shows the lower bound is attainable. Similarly, if $G = E_n$, then $\chi'(E_n) = 0$ and $\chi'(\overline{E_n}) = \chi'(K_n) = n$. So the sum is again n. If $\chi'(G)+\chi'(\overline{G}) < n$ for odd n, it would imply that both $\chi'(G)$ and $\chi'(\overline{G})$ are less than n. But for odd n, $\chi'(K_n) = n$. If G (or $\overline{G}$) is K_n, then its edge chromatic number is n, so the sum would be at least n. Hence, $\chi'(G)+\chi'(\overline{G}) \geq n$.

Obviously, $\chi'(G)\chi'(\overline{G}) \geq 0$. It can be seen that the lower bounds are attained in complete graphs K_n.

We now prove that $\chi'(G)+\chi'(\overline{G}) \leq 2n-3$ and $\chi'(G)\chi'(\overline{G}) \leq (n-1)(n-2)$.

Clearly, for $n = 1, 3$, the inequalities $\chi'(G)+\chi'(\overline{G}) \leq 2n-3$ are true. So, let $n \geq 5$. If $\Delta(G)+\Delta(\overline{G}) \leq 2n-5$, then by Vizing's theorem, we get $\chi'(G)+\chi'(\overline{G}) \leq 2n-3$.

Otherwise, we have the following cases:

i. $\Delta(G) = n-1$ and $\Delta(\overline{G}) = n-2$. So G has a pendant vertex v. Then $\Delta(G-v) \geq n-2$ and $\chi'(G-v) \geq n-2$. But $\chi'(\overline{G-v}) \leq \chi'(K_{n-1}) = n-2$. Therefore, $\chi'(G-v) = n-2$. As $\overline{G}$ is the disjoint union of an isolated vertex and a subgraph of K_{n-1}, $\chi'(\overline{G}) = n-2$. Hence, $\chi'(G)+\chi'(\overline{G}) = 2n-3$.

ii. $\Delta(G) = n-2$ and $\Delta(\overline{G}) = n-2$. Again, G has a pendant vertex v and as before, $\chi'(G-v) \leq n-2$ and $\chi(\overline{G}) \leq n-2$. Similarly, $\chi'(\overline{G}) = n-2$. Thus, $\chi'(G)+\chi'(\overline{G}) = 2n-4 < 2n-3$.

iii. $\Delta(G) = n-1$ and $\Delta(\overline{G}) = n-3$. In this case, G has a vertex v of degree two and so $\chi'(G-v) = n-2$.

   Let vu and vw be the edges incident with v in G, with $d(u) = n-1$. In $(n-2)$-edge colouring of $G-v$, change the colour i of an edge $uu'(\neq vw)$ to a new colour $n-1$, and now colour vu by i and vw by $n-1$. This gives an $(n-1)$-colouring of G. Now, by Vizing's theorem, $\chi'(\overline{G}) \leq n-2$.

   Together, we get $\chi'(G)+\chi'(\overline{G}) \leq 2n-3$. Since $\chi'(G)+\chi'(\overline{G}) \leq 2n-3$, clearly we have $\chi'(G)\chi'(\overline{G}) \leq (n-1)(n-2)$. We observe that in graph $K_{1,n-1}$, the upper bounds are attained.

   a. Let n be even. Then $\chi'(G)+\chi'(\overline{G}) \geq n-1$. Also, $\chi'(G)\chi'(\overline{G}) \geq 0$. The lower bounds are attained for complete graphs. To get the upper bounds, since G and $\overline{G}$ are subgraphs of K_n and $\chi'(K_n) = n-1$ for all even n, $\chi'(G)+\chi'(\overline{G}) \leq 2(n-1)$ and $\chi'(G)\chi'(\overline{G}) \leq (n-1)^2$. These upper bounds are attained in the complete bipartite graphs $K_{1,n-1}, n \neq 2$.

■

A more complete argument for the upper bound for odd n involves considering critical graphs and the maximum possible degrees. Vizing's theorem states that $\chi'(G)$ is either $\Delta(G)$ or $\Delta(G)+1$. Thus, $\chi'(G) \leq \Delta(G)+1$ and $\chi'(\overline{G}) \leq \Delta(\overline{G})+1$. So $\chi'(G)+\chi'(\overline{G}) \leq \Delta(G)+\Delta(\overline{G})+2$. We know that $\Delta(G)+\Delta(\overline{G}) \leq n-1+n-1 = 2n-2$. The specific bound $2n-3$ for odd n is a tighter result that relies on the properties of degrees in complementary graphs. For odd n, it can be shown that if G is a graph of order $n \geq 3$ such that $\Delta(G) < n-1$ and $\Delta(\overline{G}) < n-1$, then $\chi'(G)+\chi'(\overline{G}) \leq 2n-4$. If either G or $\overline{G}$ is K_n, then the sum is n. The maximum of $\Delta(G)+\Delta(\overline{G})$ is $2n-2$ if $n-1$ is achieved by both (e.g., K_n and E_n do not both have $\Delta = n-1$). For any graph G, $\Delta(G)+\Delta(\overline{G}) \geq n-1$. It is also known that $\Delta(G)+\Delta(\overline{G}) \leq 2n-2$. The $2n-3$ bound for odd n relies on the fact that if G is not K_n or E_n, then $\chi'(G) \leq n-1$ and $\chi'(\overline{G}) \leq n-1$, which gives the sum $\leq 2n-2$. For n odd, it is possible for both $\chi'(G)$ and $\chi'(\overline{G})$ to be equal to $\Delta(G)+1$ and $\Delta(\overline{G})+1$ respectively. A detailed proof showing $2n-3$ is the sharp upper bound is quite involved and often requires casework on graph structure. The maximum value for $\chi'(G)\chi'(\overline{G})$ for odd n is given as $(n-1)(n-2)$. This bound is achieved when $\chi'(G) = n-1$ and $\chi'(\overline{G}) = n-2$ (or vice versa), which implies specific graph structures.

The problem of determining the maximum value of $\chi'(G)+\chi'(\overline{G})$ and $\chi'(G)\chi'(\overline{G})$ is part of a larger area of research in graph theory concerning graph invariants of complementary graphs. The bounds stated are known results from this area.

1.3 Region Colouring (Map Colouring)

A region colouring of a planar graph is a labelling of its regions $f : R(G) \to \{1,2,\ldots\}$; the labels called *colours*, such that no two adjacent regions get the same colour. A k-region colouring of a planar graph G consists of k different colours and G is then called k-region colourable. From the definition, it follows that the k-region colouring of a planar graph G partitions the region set R into k independent sets $R_1, R_2, \ldots, R_k$, so that $R = R_1 \cup R_2 \cup \ldots \cup R_k$. The independent sets are called the *colour classes*, and the function $f : R(G) \to \{1,2,\ldots,k\}$ such that $f(r) = i$, for each $r \in R_i, 1 \leq i \leq k$, is called the *colour function*. The minimum number k for which there is a k-region colouring of the planar graph G is called the *region-chromatic number* of G, and is denoted by $\chi''(G)$. The colouring of regions is also called *map colouring*, because of the fact that in an atlas different countries are coloured such that countries with common boundaries are shown in different colours.

Theorem 1.3.1. *1. A planar graph G is k-region colourable if and only if its dual G^* is k-vertex colourable.*

2. If G is a plane connected graph without loops, then G has a k-vertex colouring if and only if its dual G^* has a k-region colouring.

Proof. a. Let the regions and edges of G be respectively denoted by $r_1, \ldots, r_t$ and $e_1, \ldots, e_m$. Let the vertices of G^* be $r_1^*, \ldots, r_t^*$ and edges be $e_1^*, \ldots, e_m^*$. Then the vertices and edges of G^* are in one-to-one correspondence with the regions and edges of G, and two vertices r^* and s^* in G^* are joined by an edge e^* if and only if the corresponding regions r and s in G have the corresponding edge e as a common edge on their boundary.

Let G be k-region colourable. We colour the vertices in G^* such that each vertex in G^* gets the same colour as assigned to the region r in G. Since the vertices r^* and s^* are only adjacent in G^* if the corresponding regions r and s are adjacent in G, G^* is k-vertex colourable.

Conversely, let G^* be k-vertex colourable. Now colour the regions of G such that the region r in G gets the same colour as the vertex r^* in G^*. This gives a k-region colouring of G, since the regions r and s are adjacent in G only if the corresponding vertices r^* and s^* are adjacent in G^*.

b. Since G has no loops its dual G^* has no bridges, and therefore G^* is planar. Thus by (a), G^* is k-region colourable if and only if the double dual G^{**} is k-vertex colourable. Since G is connected, G is isomorphic to G^{**} and hence the result follows.

■

Theorem 1.3.2. *Every planar graph is 6-colourable.*

Proof. Follows easily from the fact that every planar graph has a vertex of degree ≤ 5. ■

Theorem 1.3.3 (Heawood)**.** *Every planar graph is 5-colourable.*

Proof. Let G be a planar graph with n vertices. We use induction on n, the order of G. The result is obvious for $n \leq 5$. So, let $n \geq 6$. Assume the result to be true for all planar graphs with fewer than n vertices.

Let G' be the graph obtained from G by deleting the vertex v and removing all the edges incident with v. The graph G' with order $n-1$ is clearly planar and by induction hypothesis is 5-colourable. Let the colours used to colour G' be c_1, c_2, c_3, c_4, c_5. We know for a planar graph with $n \geq 6$ vertices, there exists a vertex, say v, such that $d(v) \leq 5$. Thus v has utmost five neighbours in G and all of these neighbours are the already coloured vertices in G'. If in G' less than five colours are used to colour these neighbours, then the 5-colouring of G is obtained by using the colouring for G' on all vertices, and by colouring v with the colour not used to colour the neighbours of v.

Let all the five colours be used in G', to colour the neighbours of v. This implies that there are exactly five neighbours of v, say u_1, u_2, u_3, u_4 and u_5. Assume without loss of generality that u_i is coloured with c_i, for each i, $1 \leq i \leq 5$.

Consider all the vertices of G' that are coloured with colour c_1 and c_3. If u_1 and u_3 are in different components of the Kempe subgraph $H(c_1, c_3)$ induced by those vertices coloured c_1 and c_3, then using the Kempe chain argument to interchange the colours c_1 and c_3 in the component containing u_1 leaves c_1 unused on the set $\{u_1, u_2, u_3, u_4, u_5\}$. We colour v by c_1 to get a 5-colouring of G. Finally, if u_1 and u_3 are in the same component of $H(c_1, c_3)$, and u_2 and u_4 are in the same component of $H(c_2, c_4)$, then the $c_1 - c_3$ Kempe chain from u_1 to u_3 crosses the $c_2 - c_4$ Kempe chain from u_2 to u_4. This is impossible as the graph is a plane graph. ■

Theorem 1.3.4. *Every planar graph is four colourable if and only if every cubic bridgeless planar graph is 4-colourable.*

Proof. We observe that every planar graph is four colourable if and only if every bridgeless planar graph (without cut edges) is four colourable. This is because if G has a bridge e and G' is obtained from G by contracting e, then $\chi''(G') = \chi''(G)$ (as the elementary contraction of identifying the end vertices of a bridge affects neither the number of regions in the planar graph nor the adjacency of any of the regions). We now prove that every bridgeless planar graph is 4-colourable if and only if every cubic bridgeless planar graph is 4-colourable. If every bridgeless planar graph is 4-colourable, then evidently every cubic bridgeless planar graph is 4-colourable. Conversely, let all cubic bridgeless planar graphs be 4-colourable. Let G be a bridgeless planar graph. We obtain G' from G by performing the following operations. In case G has a vertex v of degree two, let xv and yv be the edges incident with v. Subdivide xv at u and yv at w. Take two new vertices a and b and add the edges au, aw, bu, bw and ab. Now remove v (and uv and uw). In doing so, we have replaced v by a $K_4 - e$ and we see that each new vertex has degree three.

If G has a vertex v so that $d(v) \geq 4$, let $vx_1, vx_2, \ldots, vx_d$ be the edges incident with v. Subdivide each vx_i producing a new vertex v_i, for $1 \leq i \leq d$. We then remove v and add the new edges $v_1v_2, v_2v_3, \ldots, v_{d-1}v_d, v_dv_1$. Here we have replaced v by C_d and again the degree of each new vertex is three.

In both cases the graph G' formed is a bridgeless cubic graph. If these $K_4 - e$ and C_d introduced are contracted, we get the original graph G. Hence G is 4-colourable, since we have assumed that G' is 4-colourable. ■

Theorem 1.3.5 (Tait's Theorem)**.** *A cubic bridgeless, planar graph G is 4 region colorable if and only if, G is $3-$edge colorable.*

Proof. **Suppose G is 4-region colorable.** Let the 4-region coloring of G be given, where the colors are the four elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$. We now define an edge coloring of G. We define the color of an

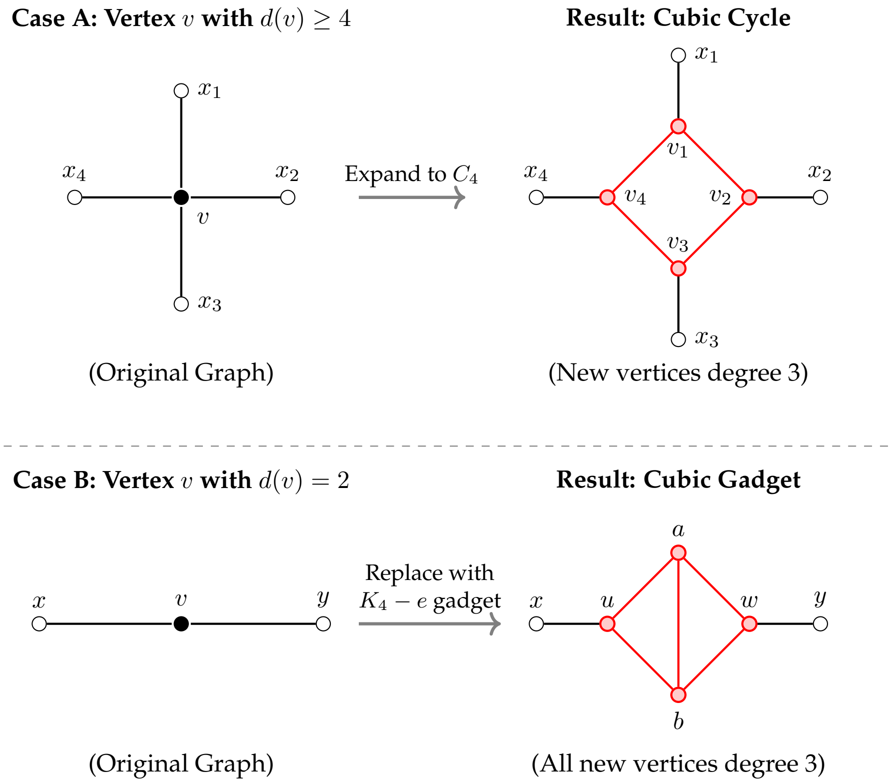


Figure 1.6: Visualisation of the operations described in Theorem 1.3.4. Top: Expanding a vertex of degree 4 into a cycle. Bottom: Replacing a vertex of degree 2 with a specialized gadget.

edge as the sum of colors of regions adjacent to it, noting that each edge lies on the boundary of at most two regions (because the given graph is cubic and bridgeless). Thus, we are done.

Conversely: [Refer to Figure above]. Suppose that G is 3-edge colorable. Let a 3-edge coloring of G be given using the colors $c_1 = (0,1), c_2 = (1,0), c_3 = (1,1)$. This produces a partition of $E(G)$ into three perfect matchings E_1, E_2, E_3 where E_i is the set of edges colored c_i. Let G_1 be a spanning subgraph of G with edge set $E(G_1) = E_1 \cup E_2$ and G_2 be the spanning subgraph of G with edge set $E(G_2) = E_2 \cup E_3$. Then, both G_1 and G_2 are 2-regular spanning subgraphs of G so each of G_1 and G_2 is the disjoint union of even cycles. For $i = 1, 2$ each region of G_i is the union of regions of G. For each cycle C in the graph G_i, every region of G either lies interior or exterior to C. Furthermore, each edge of G belongs to a cycle C' of G_i and is on the boundary of two distinct regions of G_i, one of which lies interior to C' and the other exterior to C'.

We now define a 4-region coloring of G. We assign to region R of G the color (a_1, a_2) with $a_i \in \{0,1\}$ for $i = 1, 2$, where $a_i = 0$ if R lies interior to even number of cycles in G_i and $a_i = 1$ if R lies interior to odd number of cycles in G_i.

Next, we shall show that this 4-region coloring of G is proper. Let R_1 and R_2 be two adjacent regions with common edge e. Then clearly either e is in G_1 or in G_2 or in both. Thus, e lies on a cycle in G_1 or a cycle in G_2 or both. Let C be a cycle in G_1 containing the edge e. Exactly, one of R_1 and R_2 lies interior to C, while for every other cycle C', either both lie interior to C' or both exterior to C. Hence, the first coordinate of the colours of R_1 and R_2 are different. Therefore, the colors of every two adjacent regions of G differ in the first coordinate or the second coordinate or both. Hence, the 4-region coloring of G is proper. ■

1.4 Heawood Map-Colouring Theorem

Let S_p be the orientable surface of genus p, so that S_p is topologically equivalent to a sphere with p handles. The chromatic number of S_p, denoted by $\chi(S_p)$, is the maximum chromatic number among

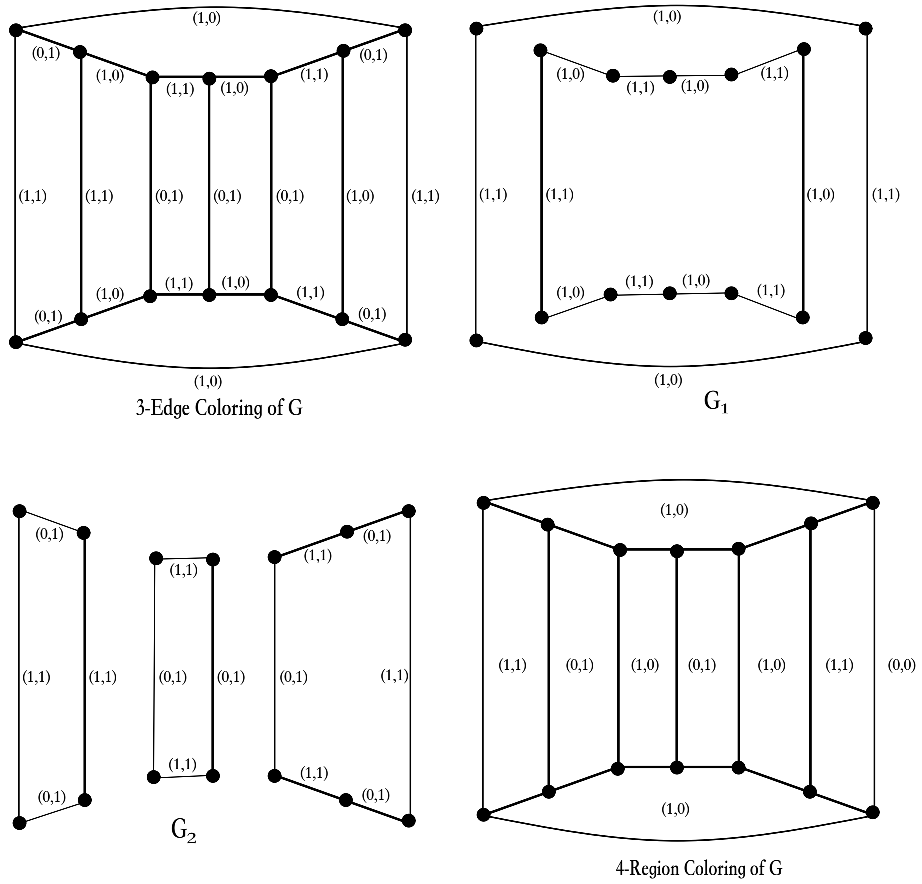


Figure 1.7: Tait's Theorem: $\chi'(G) = 3$ implies G is 4-region colorable.

all graphs which can be embedded on S_p. The surface S_0 is clearly the sphere. The four colour problem states that $\chi(S_0) = 4$.

Definition: The genus $\gamma(G)$ of a graph is the minimum number of handles which must be added to a sphere so that G can be embedded on the resulting surface. Clearly, $\gamma(G) = 0$ if and only if G is planar. Further, for a polyhedron, $n - m + f = 2 - 2\gamma$. For $n \geq 3$, the genus of the complete graph K_n is

$$\gamma(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

The following inequality is due to Heawood [113].

Theorem 1.4.1. *The chromatic number of the orientable surface of positive genus p has the upper bound*

$$\chi(S_p) \leq \frac{7 + \sqrt{1 + 48p}}{2}, \quad p > 0.$$

Proof. Let G be an (n, m) graph embedded on S_p. Since any graph can be augmented to a triangulation of the same genus by adding edges without reducing χ, we assume G to be a triangulation. Let d' be the average degree of the vertices of G. Then n, m and f (the number of regions) are related by the equations

$$d'n = 2m = 3f.$$

Solving for m and f in terms of n and using Euler's equation $n - m + f = 2 - 2\gamma$, we get

$$d' = \frac{12(p-1)}{n} + 6. \tag{1.2}$$

As $d' \leq n - 1$, this gives $n - 1 \geq \frac{12(p-1)}{n} + 6$. Solving for n and taking the positive square root, we obtain $n \geq \frac{7+\sqrt{1+48p}}{2}$.

Let $H(p) = \frac{7+\sqrt{1+48p}}{2}$. Then we show that $H(p)$ colours are sufficient to colour the vertices of G. If $n = H(p)$, obviously we have sufficient colours. In case $n > H(p)$, we have $\frac{1}{n} < \frac{1}{H(p)}$ then equation (1.2) gives obtain

$$d' < \frac{12(p-1)}{H(p)} + 6 \tag{1.3}$$

Now,

$$\begin{aligned}
2H(p) &= 7 + \sqrt{1+48p} \\
\text{or } (2H(p)-7)^2 &= 1 + 48p \\
\text{or } 4H^2(p) - 28H(p) + 49 &= 1 + 48p \\
\text{or } H(p)\,[H(p)-7] &= 12(p-1) \\
\text{or } H(p) - 7 &= \frac{12(p-1)}{H(p)} \\
\text{i.e. } H(p) - 1 &= \frac{12(p-1)}{H(p)} + 6
\end{aligned}$$

Hence, equation 1.3 gives, $d' < H(p) - 1$. Therefore, when $n > H(p)$, there is a vertex v of degree at most $H(p) - 2$. Identify v and any adjacent vertex by an elementary contraction to obtain a new graph G'. If $n' = n - 1 = H(p)$, then G' can be coloured in $H(p)$ colours. If $n' > H(p)$, repeat the argument and evidently we get an $H(p)$ colourable graph. It is then easy to see that the colouring of this graph induces a colouring of the preceding one in $H(p)$ colours, and so forth, so that G itself is $H(p)$-colourable. Hence the result follows.

■

Theorem 1.4.2. *(Heawood map-colouring theorem) For every positive integer p, the chromatic number of the orientable surface of genus p is given by*

$$\chi(S_p) = \frac{7+\sqrt{1+48p}}{2}, \quad p > 0.$$

Proof. Let G be an (n, m) graph embedded on S_p. Then,

$$\chi(S_p) \le \frac{7+\sqrt{1+48p}}{2}, \quad p > 0.$$

Now, if the complete graph K_n, is embedded in S_p, then

$$p \ge \gamma(K_n) = \frac{(n-3)(n-4)}{12}. \tag{1.4}$$

Setting n to be the largest integer satisfying equation 1.4, we have

$$\frac{(n-3)(n-4)}{12} \le p < \left\lfloor \frac{(n-2)(n-3)}{12} \right\rfloor - 1 < \frac{(n-2)(n-3)}{12}.$$

Solving for n, we get $n^2 - 7n + 12 \le 12p < n^2 - 5n + 12$. This gives, $\frac{5+\sqrt{1+48p}}{2} < n \le \frac{7+\sqrt{1+48p}}{2}$, which is equivalent to $\frac{5+\sqrt{1+48p}}{2} + 1 \le n \le \frac{7+\sqrt{1+48p}}{2}$. Thus, $n = \frac{7+\sqrt{1+48p}}{2}$.

Since $\chi(K_n) = n$, we have found a graph with genus p and chromatic number equal to $H(p)$. This shows that $H(p)$ is a lower bound for $\chi(S_p)$, completing the proof.

■

Theorem 1.4.3 (Heawood Map Coloring Theorem)**.** *For any orientable surface Σ_g of genus $g \ge 1$, the maximum number of colors required to color any map on that surface is given by the Heawood number $H(g)$:*

$$\chi(g) = \left\lfloor \frac{7+\sqrt{1+48g}}{2} \right\rfloor$$

where $\chi(g)$ is the chromatic number of the surface.

The Upper Bound (Heawood, 1890)

Heawood demonstrated that $H(g)$ colors are *sufficient* to color any map on a surface of genus $g \geq 1$.

1. **Graph Conversion:** A map coloring problem is equivalent to coloring the vertices of its dual graph, where vertices represent regions and edges represent shared boundaries. This dual graph can be embedded on the same surface.
2. **Euler's Formula and Degree Bounds:** For any graph G embedded on an orientable surface Σ_g, Euler's formula states $V - E + F = 2 - 2g$. If we consider a maximal (triangulated) embedding, $3F \leq 2E$. This leads to a bound on the average degree of such graphs. Specifically, if a graph requires k colors, it must contain a vertex of degree at least $k-1$. By relating V, E, F, and g, it can be shown that any graph embeddable on Σ_g must contain a vertex of degree at most $H(g) - 1$.
3. **Inductive Argument:** A proof by induction on the number of vertices, similar to the proof of Brooks' Theorem, can then establish that $H(g)$ colors are sufficient. If a vertex v has degree $\leq H(g) - 1$, we can remove v, color the remaining graph inductively, and then color v with one of the $H(g)$ colors, as it has at most $H(g) - 1$ neighbors.

The Lower Bound (Ringel and Youngs, 1960-1968)

This part, completing the theorem, proves that for every $g \geq 1$, there exists a map on Σ_g that *requires* $H(g)$ colors. This is achieved by showing that the complete graph $K_{H(g)}$ can be embedded on Σ_g.

1. **Complete Graphs and Coloring:** A complete graph K_n requires n colors. If K_n can be embedded on Σ_g, then $\chi(g) \geq n$.
2. **Genus of Complete Graphs:** The critical step was proving the exact genus of the complete graph K_n, denoted $\gamma(K_n)$. Ringel and Youngs showed:

$$\gamma(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil$$

   This formula gives the minimum genus of an orientable surface on which K_n can be embedded.
3. **Equating for Tightness:** To find the maximum n such that K_n can be embedded on a surface of genus g, we set $\gamma(K_n) = g$. Solving the equation $g = \frac{(n-3)(n-4)}{12}$ for n yields:

$$n = \frac{7 + \sqrt{1 + 48g}}{2}$$

   Taking the floor of this expression gives the largest integer n for which K_n is embeddable on Σ_g, which is precisely $H(g)$.
4. **Constructive Embeddings:** The final step involved constructing explicit embeddings of $K_{H(g)}$ on Σ_g for all $g \geq 1$. This monumental task was completed by Ringel and Youngs over several years, cementing the lower bound and thus proving the theorem for all $g \geq 1$.

1.5 Uniquely Colorable Graphs

Definition 1.5.1. A graph $G(V, E)$ is said to be uniquely k-vertex-colourable (or uniquely k-colourable) if there is a unique k-part partition of the vertex set V into independent subsets. That is, in the uniquely k-colouring of a graph G, every k-colouring of G induces the same partition of V.

Theorem 1.5.2. *A uniquely k-colorable graph is $(k-1)$-connected.*

Proof. Let G be a uniquely k-colorable graph. If G is K_n, then G is $(k-1)$-connected. Now, assume G is not complete. We will show G is $(k-1)$-connected. Suppose, for contradiction, that G is not $(k-1)$-connected. Then there exists a set X of $(k-2)$ vertices whose removal from G disconnects G. In any k-coloring of G, there are at least two colors, say c_1 and c_2, that are not assigned to any vertex in X. A vertex colored c_1 is connected to any vertex colored c_2 by a path whose vertices are colored c_1 or c_2. Therefore, the set of vertices in G colored c_1 or c_2 lies within the same component of $G-X$, say G_1. Another k-coloring of G may thus be obtained by taking any vertex of $G-X$ which is not in G_1 and recoloring it with either c_1 or c_2. This contradicts the hypothesis that G is uniquely k-colorable. Thus, G must be $(k-1)$-connected. ■

Theorem 1.5.3. *Each vertex in a uniquely k-colorable graph has degree $\geq k-1$.*

Proof. Let G be a uniquely k-colorable graph and c be its unique k-coloring. Assume, for contradiction, that there exists a vertex $v \in V(G)$ with $\deg(v) < k-1$. This implies that v has at most $k-2$ neighbors.

Since there are k available colors and at most $k-2$ colors are used by the neighbors of v, there must be at least two colors, say j_1 and j_2, that are not used by any neighbor of v.

If $c(v)$ is distinct from both j_1 and j_2, we can recolor v with j_1. This produces a new proper k-coloring, contradicting uniqueness. If $c(v)$ is one of j_1 or j_2 (say $c(v) = j_1$), we can recolor v with j_2. This also produces a new proper k-coloring (since $j_2 \neq j_1$), again contradicting uniqueness.

In both cases, we find a distinct proper k-coloring, which contradicts the assumption that G is uniquely k-colorable. Therefore, every vertex in a uniquely k-colorable graph must have degree at least $k-1$. ■

Theorem 1.5.4. *In a uniquely k-colorable graph G, the subgraph induced by the union of any two color classes in a k-coloring is connected.*

Proof. Let G be a uniquely k-colorable graph with a k-coloring $c : V(G) \to \{1, \ldots, k\}$. Consider any two distinct color classes C_i and C_j. Let $G_{i,j}$ be the subgraph induced by $V(G_{i,j}) = C_i \cup C_j$.

Assume, for contradiction, that $G_{i,j}$ is disconnected. Then $G_{i,j}$ has at least two connected components. Let X be one such component. Define a new coloring c' for G as follows:

- For $v \in X$: Swap $c(v)$ if it's i or j. That is, if $c(v) = i$, set $c'(v) = j$; if $c(v) = j$, set $c'(v) = i$.
- For $v \notin X$: $c'(v) = c(v)$.

We verify that c' is a proper k-coloring.

- Edges (u, w) not in $G_{i,j}$, or with only one endpoint in $G_{i,j}$, maintain proper coloring since colors outside $\{i, j\}$ are untouched, and colors i, j are only assigned to neighbors whose original color was not i or j.
- For edges (u, w) within $G_{i,j}$:
  - If $u, w \in X$: $c(u)$ and $c(w)$ are different (one i, one j). $c'(u)$ and $c'(w)$ are also different (one j, one i).
  - If $u, w \notin X$ (i.e., in a component of $G_{i,j}$ other than X): $c'(u) = c(u)$ and $c'(w) = c(w)$, so they remain different.
  - If $u \in X$ and $w \notin X$: This is impossible, as X is a connected component of $G_{i,j}$, implying no edges connect X to any other part of $G_{i,j}$.

Thus, c' is a proper k-coloring.

Since X is a non-empty component, c' is distinct from c (e.g., any vertex $v \in X$ with color i now has color j). This contradicts the assumption that G is uniquely k-colorable. Therefore, $G_{i,j}$ must be connected. ■

Theorem 1.5.5. *A uniquely k-colorable graph is $(k-1)$-connected.*

Proof. Let G be a uniquely k-colorable graph. Assume, for contradiction, that G is not $(k-1)$-connected. Then there exists a vertex cut X of size $|X| \leq k-2$ such that $G - X$ is disconnected. Let c be a unique k-coloring of G.

Since $|X| \leq k-2$, there are at least two colors, say c_1 and c_2, not used by any vertex in X. Consider the subgraph G_{c_1,c_2} induced by the union of color classes C_{c_1} and C_{c_2}. It is a known property of uniquely k-colorable graphs that the subgraph induced by the union of any two color classes is connected. Thus, G_{c_1,c_2} is connected.

As X does not contain any vertex colored c_1 or c_2, all vertices in $C_{c_1} \cup C_{c_2}$ must belong to $G - X$. Since G_{c_1,c_2} is connected, all vertices colored c_1 or c_2 must lie within a single connected component of $G - X$.

Let D_1 and D_2 be two distinct connected components of $G - X$. Assume all vertices of $C_{c_1} \cup C_{c_2}$ are contained in D_1. This means D_2 contains no vertices colored c_1 or c_2. Its vertices are colored from the remaining $k-2$ colors. Pick an arbitrary vertex $v \in D_2$. Let $c(v)$ be its color. Since $v \in D_2$, $c(v) \notin \{c_1, c_2\}$. Now, construct a new coloring c' by swapping color $c(v)$ with c_1 in the Kempe chain containing v within the subgraph induced by $C_{c(v)} \cup C_{c_1}$. This Kempe chain must be entirely contained within D_2 because c_1 is not present in D_2 initially (so C_{c_1} has no vertices in D_2), and swapping only affects vertices within D_2. This operation produces a new proper k-coloring c' that differs from c for at least one vertex (namely, v). This contradicts the uniqueness of the k-coloring.

Therefore, our initial assumption that G is not $(k-1)$-connected must be false. Hence, a uniquely k-colorable graph is $(k-1)$-connected. ∎

Theorem 1.5.6. *A graph is four colorable if and only if every Hamiltonian planar graph is four colorable.*

Proof. Let P be the statement "Every planar graph is four colorable" and H be the statement "Every Hamiltonian planar graph is four colorable." We need to prove $P \iff H$.

(⇒) Assume that every planar graph is four colorable. A Hamiltonian planar graph is a specific type of planar graph (one that contains a Hamiltonian cycle). If all planar graphs are 4-colorable, then it logically follows that any subset of planar graphs, including Hamiltonian planar graphs, must also be 4-colorable. Thus, $P \implies H$.

(⇐) Assume that every Hamiltonian planar graph is four colorable. We need to show that this implies every planar graph is 4-colorable. This direction is less straightforward. It relies on a known result in graph theory:

Any planar graph G can be embedded as a subgraph of a Hamiltonian planar graph G'.

To construct such a G', one can take the original planar graph G and add edges in a planar way to make it maximal planar (triangulated). Then, it is known that every maximal planar graph with $n \geq 3$ vertices is Hamiltonian if its minimum degree is at least 3. If a planar graph is not 3-connected, it can be extended to a 3-connected maximal planar graph by adding edges. A maximal planar graph that is 3-connected is Hamiltonian.

Specifically, it is known that for any planar graph G, we can construct a supergraph G' such that G' is a Hamiltonian planar graph and any proper coloring of G' induces a proper coloring of G. If G' is 4-colorable (by our assumption that every Hamiltonian planar graph is 4-colorable), then G is also 4-colorable.

Alternatively, and more directly, a key result known as **Tait's Theorem** (for 3-edge-connected cubic planar graphs) connects 4-coloring of faces to 3-edge-coloring, and ultimately to Hamiltonian cycles in dual graphs. However, a more general argument is needed here:

It is a well-established result that if the Four Color Theorem holds (i.e., every planar graph is 4-colorable), then it implies properties for various sub-classes. Conversely, proving the Four Color Theorem is often done by reducing it to a specific class of graphs. It turns out that any planar graph can be embedded within a Hamiltonian planar graph in such a way that if the larger graph is 4-colorable, the smaller one is too. This is due to the fact that adding edges to a graph can only make it harder to color (i.e., the chromatic number can increase or stay the same). If we take an arbitrary planar graph G and embed it into a Hamiltonian planar graph G', if G' is 4-colorable, then G must also be 4-colorable because G is a subgraph of G'.

Therefore, if every Hamiltonian planar graph is 4-colorable, then every planar graph is 4-colorable. Thus, $H \implies P$.

Since both directions are true, $P \iff H$. ■

Chapter 2

Matchings and Factors

Historical Note and Motivation

The theory of matchings began with **Julius Petersen's** 1891 study of regular graphs and evolved through **Philip Hall's** 1935 matching criteria for bipartite graphs. The field was revolutionized by **W.T. Tutte** in 1947, who characterized 1-factors in general graphs and later extended this to f-factors. These structural foundations were complemented by **Claude Berge's** algorithmic approach using augmenting paths and the **Erdős-Gallai** characterization of degree sequences. The study culminates in the factorization of K_n, a problem rooted in 19th-century tournament designs.
This chapter explores the fundamental tension between **local constraints** (vertex degrees) and **global structure** (factors). By mastering these topics, we learn how to determine when local demands can be universally satisfied, a concept essential for solving real-world assignment and scheduling problems. Theoretically, this section highlights the power of reduction, demonstrating how complex f-factor problems can be solved using the simpler mechanics of 1-factors.

2.1 Matchings

Definition 2.1.1 (Matching, Maximal, and Maximum)**.** A **matching** in a graph is a set of independent edges. That is, a subset M of the edge set E of a graph $G(V, E)$ is a matching if no two edges of M have a common vertex.

- A matching M is said to be **maximal** if there is no matching M' strictly containing M. That is, M is maximal if it cannot be enlarged (you cannot add another edge without breaking the matching property).
- A matching M is said to be **maximum** if it has the largest possible cardinality. That is, M is maximum if there is no matching M' such that $|M'| > |M|$.

Definition 2.1.2 (Saturated and Unsaturated Vertices)**.** Let M be a matching in a graph G.

- A vertex v in G is said to be M**-saturated** (or saturated by M) if there is an edge $e \in M$ incident with v.
- A vertex which is not incident with any edge of M is said to be M**-unsaturated**.

In other words, given a matching M in a graph G, the vertices belonging to the edges of M are M-saturated and the vertices not belonging to the edges of M are M-unsaturated.

Consider the graph shown in Figure 2.1 below. Clearly, $M = \{v_1v_2, v_3v_7, v_4v_5\}$ is a matching. The vertices $v_1, v_2, v_3, v_4, v_5, v_7$ are M-saturated, but v_6 is M-unsaturated.

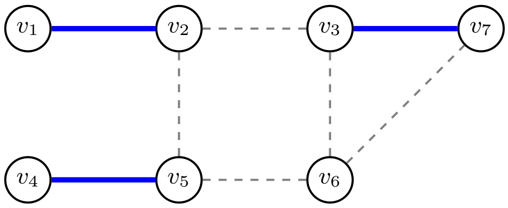


Figure 2.1: Visual representation of the matching M. The blue edges form the matching.

Definition 2.1.3 (Perfect Matching)**.** A matching M in a graph G is said to be a **perfect matching** if M saturates every vertex of G.

Definition 2.1.4 (Complete Matching in Bipartite Graphs)**.** If G is a bipartite graph with bipartition $V = V_1 \cup V_2$ and $|V_1| = n_1$, $|V_2| = n_2$, then a matching M of G saturating all the vertices in V_1 is called a **complete matching** (or V_1 matched into V_2).

Definition 2.1.5 (Alternating Path)**.** Let H be a subgraph of a graph G. An H**-alternating path (cycle)** in G is a path (cycle) whose edges are alternately in $E(G) \setminus E(H)$ and $E(H)$.

Specifically, if M is a matching in a graph G, then an M**-alternating path (cycle)** in G is a path (cycle) whose edges are alternately in $E(G) \setminus M$ and M. That is, in an M-alternating path, the edges alternate between M-edges and non-M-edges.

Definition 2.1.6 (Augmenting Path)**.** An M-alternating path whose end vertices are M-unsaturated is said to be an M**-augmenting path**.

Example 2.1.7. Examples to explain definitions given above.

- **Matching:** A set of edges without common vertices.
- **Maximal Matching:** A matching to which no more edges can be added.
- **Maximum Matching:** A matching with the largest possible number of edges.

Figure 2.2: **Maximal Matching**. Size = 1. We cannot add (A, B) or (C, D) because B and C are already busy.

Figure 2.3: **Maximum Matching**. Size = 2. This is the largest possible matching for this graph.

Saturated vs. Unsaturated Vertices

A vertex is **Saturated** if it touches a matched edge (blue). Otherwise, it is **Unsaturated**.

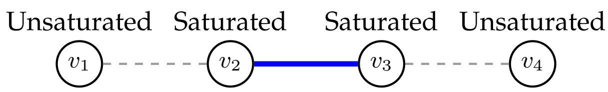


Figure 2.4: Vertices v_2, v_3 are saturated by the blue edge. v_1, v_4 are unsaturated.

Perfect Matching

A matching that saturates **every** vertex in the graph.

Complete Matching (Bipartite)

In a bipartite graph $G = (V_1 \cup V_2, E)$, a matching is **complete from** V_1 **to** V_2 if every vertex in V_1 is saturated.

Alternating and Augmenting Paths

- **Alternating Path:** Edges alternate between Non-Matching and Matching.
- **Augmenting Path:** An alternating path that starts and ends at unsaturated vertices.

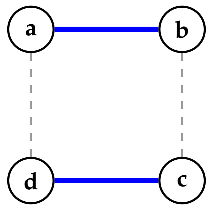


Figure 2.5: A Perfect Matching on C_4. All 4 vertices are covered.

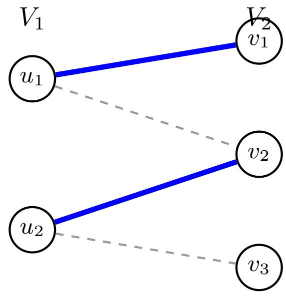


Figure 2.6: Complete Matching of V_1 into V_2. All vertices in V_1 are blue-connected. Vertex v_3 is left over, which is fine.

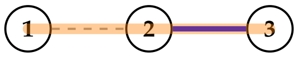


Figure 2.7: **Alternating Path** $(1-2-3)$.
Starts Unsat(1) $\to$ Sat(3).

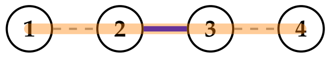


Figure 2.8: **Augmenting Path** $(1-2-3-4)$.
Starts Unsat(1) $\to$ Ends Unsat(4).

Theorem 2.1.8 (Berge). *A matching M of a graph G is maximum if and only if G contains no M-augmenting paths.*

Proof. Let M be a maximum matching in a graph G. Assume $P = v_1v_2\dots v_k$ is an M-augmenting path in G. Due to the alternating nature of an M-augmenting path, we observe that k is even (since the path starts and ends with edges not in M, the number of edges is odd, thus vertices are even). The edges $v_2v_3, v_4v_5, \dots, v_{k-2}v_{k-1}$ belong to M. Also, the edges $v_1v_2, v_3v_4, \dots, v_{k-1}v_k$ do not belong to M (Figure 2.9).

Now, let M_1 be the set of edges given by

$$M_1 = [M - \{v_2v_3, \dots, v_{k-2}v_{k-1}\}] \cup \{v_1v_2, \dots, v_{k-1}v_k\}.$$

Then M_1 is a matching and clearly M_1 contains one more edge than M. This contradicts the assumption that M is maximum. Thus G contains no M-augmenting paths.

Conversely, assume that G has no M-augmenting paths. Let M' be a matching such that $|M'| > |M|$. Let H be a subgraph of G with $V(H) = V(G)$ and $E(H)$ be the set of edges of G that appear in exactly one of M and M' (the symmetric difference $M \Delta M'$).

Since every vertex of G lies on at most one edge from M and at most one edge from M', the degree (in H) of each vertex of H is at most 2. This implies that each connected component of H is either a single vertex, or a path, or a cycle (Fig. 2.10).

If a component is a cycle, then it is an even cycle, because the edges alternate between M-edges and M'-edges. Since $|M'| > |M|$, there is at least one component in H that is a path which begins and ends with edges from M'. Clearly, this path is an M-augmenting path, contradicting the assumption. Thus no such matching M' can exist and hence M is maximum. ∎

Theorem 2.1.9 (Hall's Marriage Theorem). *A bipartite graph $G = (X \cup Y, E)$ has a matching that covers all vertices in X if and only if for every $S \subseteq X$, $|N(S)| \geq |S|$.*

Proof. Assume a matching M covers all vertices in X. For any $S \subseteq X$, each vertex in S is matched by M to a distinct vertex in Y. Let $M(S)$ be this set of matched vertices in Y. Then $|M(S)| = |S|$. Since every vertex in S is connected to its matched partner, $M(S) \subseteq N(S)$. Thus, $|N(S)| \geq |M(S)| = |S|$.

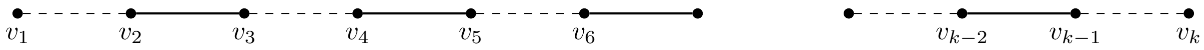


Figure 2.9:

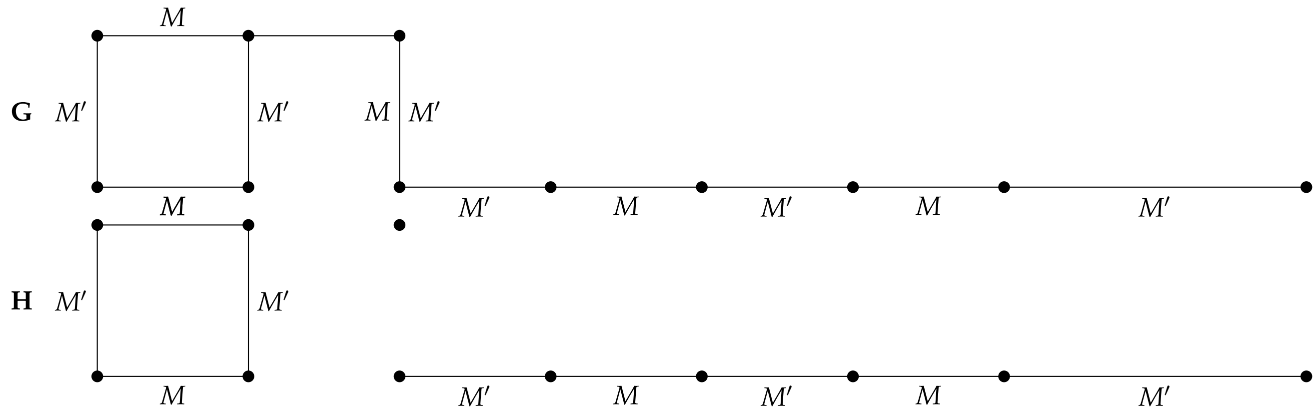


Figure 2.10:

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. We prove by contradiction that a matching covering X exists. Suppose no such matching exists. Let M be a maximum matching in G. Since M does not cover X, there must be at least one unmatched vertex $u \in X$. By Berge's Lemma, since M is a maximum matching, there are no M-augmenting paths in G. Let A be the set of all vertices reachable from u by an M-alternating path. Let $X' = A \cap X$ and $Y' = A \cap Y$. Since u is unmatched and starts an M-alternating path, $u \in X'$. For any $y \in Y'$, it must be connected to some $x \in X'$ by an edge not in M. M must match this x to some $y' \in Y'$, otherwise we could extend the path to x to form an M-augmenting path from u to x, a contradiction (since x would be unmatched). Specifically, any vertex $x \in X'$ (other than u) must be matched by M to some $y \in Y'$. If x were unmatched, then an M-alternating path from u to x would be an M-augmenting path, a contradiction. Similarly, any vertex $y \in Y'$ must be matched by M to some $x \in X'$. If y were matched to $x \notin X'$, then combining the path from u to y (ending with an M-edge) with the edge (y, x) (which is in M) would form an M-alternating path from u to x ending with an M-edge. This would imply $x \in X'$, a contradiction. Therefore, all edges of M that are incident to A must connect a vertex in $X' \setminus \{u\}$ to a vertex in Y'. This implies that M provides a perfect matching between $X' \setminus \{u\}$ and Y'. Thus, $|X'| - 1 = |Y'|$.

Now consider the set $S = X'$. By construction of A, all neighbors of X' must be in Y'. Thus $N(X') \subseteq Y'$. From our previous deduction, $|Y'| = |X'| - 1$. Therefore, $|N(X')| \leq |Y'| = |X'| - 1$. This means $|N(X')| < |X'|$, which contradicts the given condition that $|N(S)| \geq |S|$ for all $S \subseteq X$. Our initial assumption that no matching covering X exists must be false. Hence, such a matching exists. ■

2nd Proof using Augmenting Paths. Assume a matching M covers all vertices in X. For any $S \subseteq X$, each $x \in S$ is matched to a distinct $y_x \in Y$. These y_x form the set $M(S)$. Clearly, $|M(S)| = |S|$ and $M(S) \subseteq N(S)$. Thus, $|N(S)| \geq |S|$.

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. We prove by contradiction. Suppose no matching covers X. Let M be a maximum matching. Since it doesn't cover X, there's an unmatched $u \in X$. By the given theorem (Berge's Lemma), since M is maximum, there are no M-augmenting paths.

Let A be the set of vertices reachable from u by M-alternating paths. Let $X_A = A \cap X$ and $Y_A = A \cap Y$. (See Figure 2.11). Since there are no $M-$augmenting paths, u is the only vertex in A that is $M-$unsaturated. Furthermore, if $y \in Y_A$ is matched by M, its matched partner x' must be in X_A. (If $x' \notin X_A$, then extending the M-alternating path from u to y with the edge $(y, x') \in M$ would make x' reachable, implying $x' \in X_A$, a contradiction). Thus, M perfectly matches Y_A with $X_A \setminus \{u\}$. Therefore, $|Y_A| = |X_A| - 1$.

Now, let $S = X_A$. We claim $N(S) \subseteq Y_A$. Take any $x \in S$ and its neighbor $y \in N(x)$. If $y \notin Y_A$,

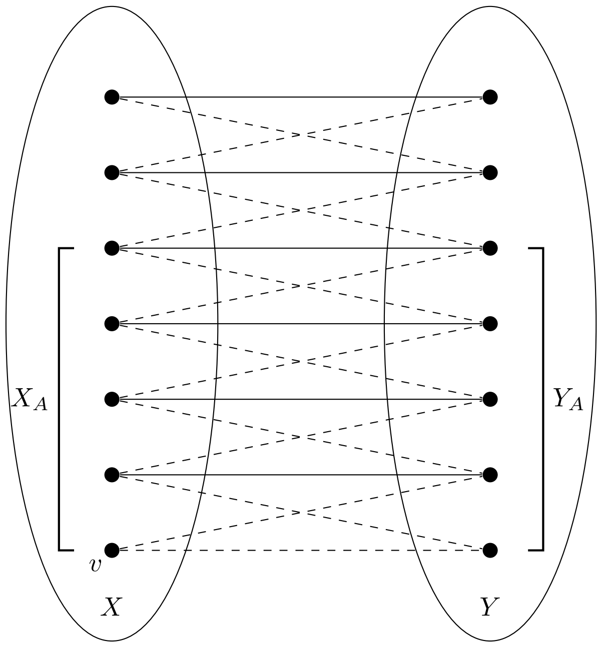


Figure 2.11: Bipartite graph illustrating sets $X_A \subseteq X$ and $Y_A \subseteq Y$ relative to a matching.

then the edge (x, y) cannot be in M (because if it were, then x would be matched to y, but $x \in X_A$ and $y \notin Y_A$ implies x is not matched within A, a contradiction to M matching $X_A \setminus \{u\}$ to Y_A). If $(x, y) \notin M$, then combining the M-alternating path from u to x with (x, y) would make y reachable, implying $y \in Y_A$, a contradiction. Thus, $N(S) \subseteq Y_A$.

Combining these, $|N(S)| \leq |Y_A| = |X_A| - 1 = |S| - 1$. So, $|N(S)| < |S|$ for $S = X_A$. This contradicts our initial assumption. Therefore, a matching covering X must exist. ■

Theorem 2.1.10 (Hall's SDR Theorem). *A family of finite nonempty sets $A = \{A_i : 1 \leq i \leq r\}$ has an SDR (System of Distinct Representatives) if and only if for every k, $1 \leq k \leq r$, the union of any k of these sets contains at least k elements.*

Proof. We prove this theorem by translating the problem of finding an SDR into a problem of finding a matching in a specific bipartite graph.

Construction of the Bipartite Graph: Let $G = (X, Y, E)$ be a bipartite graph constructed as follows:

- Let $X = \{1, 2, \dots, r\}$ be the set of indices representing the sets $A_1, \dots, A_r$.
- Let $Y = \bigcup_{i=1}^{r} A_i$ be the set containing all elements present in the family A.
- We draw an edge from vertex $i \in X$ to vertex $y \in Y$ if and only if the element y is contained in the set A_i (i.e., $y \in A_i$).

Relating SDRs to Matchings: An SDR is defined as a set of distinct elements $\{x_1, x_2, \dots, x_r\}$ such that $x_i \in A_i$ for all i, and $x_i \neq x_j$ for $i \neq j$. In our graph G, choosing an element $x_i \in A_i$ corresponds to selecting an edge (i, x_i). Therefore, finding a System of Distinct Representatives is equivalent to finding a matching in G that saturates every vertex in X.

Applying Hall's Marriage Theorem: By Theorem 8.2 (Hall's Marriage Theorem), a matching saturating X exists if and only if for every subset $S \subseteq X$,

$$|N(S)| \geq |S|$$

where $N(S)$ is the set of neighbors of S in G.

In our constructed graph, the set of neighbors $N(S)$ for a subset of indices S consists of all elements $y \in Y$ that are connected to at least one index in S. By definition of our edges, this is exactly the union of the sets indexed by S:

$$N(S) = \bigcup_{i \in S} A_i$$

Thus, the condition $|N(S)| \geq |S|$ translates directly to:

$$\left| \bigcup_{i \in S} A_i \right| \geq |S|$$

Letting $k = |S|$, this means that the union of any k sets must contain at least k elements.

Conclusion: Since the existence of a saturating matching is equivalent to the existence of an SDR, the family A has an SDR if and only if the union of any k sets contains at least k elements. ∎

Theorem 2.1.11. *Let G be a bipartite graph with bipartite sets V_1 and V_2 having orders n_1 and n_2 where $n_1 \leq n_2$. Suppose G satisfies Hall's condition $|S| \leq |N(S)|$ for all subsets $S \subseteq V_1$. Then G has at least $r(\delta, n_1) = \prod_{i=1}^{\min(\delta, n_1)} (\delta - i + 1)$ complete matchings, where $\delta = \min\{\deg(v) : v \in V_1\}$.*

Proof. The theorem can be proved by induction on n_1. Base case: $n_1 = 1$. $V_1 = \{v_1\}$. $\deg(v_1) = \delta$. The number of complete matchings is δ. $r(\delta, 1) = (\delta - 1 + 1) = \delta$. Holds.

Inductive step: Assume the theorem holds for $n_1 - 1$. Let $v_1 \in V_1$. $\deg(v_1) \geq \delta$. There are at least δ choices for $u_1 \in N(v_1)$. For each such choice (v_1, u_1), consider the graph $G' = G - \{v_1, u_1\}$. The minimum degree of vertices in $V_1' = V_1 \setminus \{v_1\}$ in G' is at least $\delta - 1$. (A vertex $v \in V_1'$ could lose an edge to u_1, but its degree in G was $\geq \delta$, so in G' it's $\geq \delta - 1$). The Hall's condition for G': $|S'| \leq |N_{G'}(S')|$ for $S' \subseteq V_1'$. This needs to be carefully established. If G satisfies Hall's condition, then G' also satisfies Hall's condition. (Proof: Assume $|S'| > |N_{G'}(S')|$ for some $S' \subseteq V_1'$. Then u_1 must be in $N_G(S')$. So $|N_G(S')| = |N_{G'}(S')| + 1 < |S'| + 1$. This doesn't directly contradict Hall for G. A more rigorous argument involves assuming violation in G' and showing it implies violation in G.)

Assuming Hall's condition holds for G', by induction hypothesis, G' has at least $r(\delta - 1, n_1 - 1)$ complete matchings. So the total number of complete matchings in G is at least $\delta \cdot r(\delta - 1, n_1 - 1)$= $\delta \cdot \prod_{i=1}^{\min(\delta-1, n_1-1)} (\delta - 1 - i + 1) = \delta \cdot \prod_{i=1}^{\min(\delta-1, n_1-1)} (\delta - i)$.

Case 1: $\delta \geq n_1$. Then $\delta - 1 \geq n_1 - 1$. $r(\delta, n_1) = \delta \cdot \prod_{i=1}^{n_1-1} (\delta - i) = \delta(\delta - 1) \cdots (\delta - n_1 + 1)$. This matches the product $\delta \cdot r(\delta - 1, n_1 - 1)$ where $\min(\delta - 1, n_1 - 1) = n_1 - 1$.

Case 2: $\delta < n_1$. Then

$$\begin{aligned}
\delta \cdot \prod_{i=1}^{\min(\delta-1, n_1-1)} (\delta - i) &= \delta \cdot \prod_{i=1}^{\delta-2} (\delta - i) \\
&= \delta \left((\delta - 1)(\delta - 2) \cdots (\delta - (\delta - 2)) \right) \\
&= \delta(\delta - 1)! = \delta! \\
&= \prod_{i=1}^{\delta} (\delta - i + 1).
\end{aligned}$$

∎

Theorem 2.1.12. *Every k-regular bipartite graph with $k > 0$ has a perfect matching.*

Proof. Let $G = (X \cup Y, E)$ be a k-regular bipartite graph. Summing degrees in X: $\sum_{x \in X} \deg(x) = |X|k$. Summing degrees in Y: $\sum_{y \in Y} \deg(y) = |Y|k$. Both sums equal $|E|$, so $|X|k = |Y|k \implies |X| = |Y|$ (since $k > 0$).

For any $S \subseteq X$, let $E(S, N(S))$ be the set of edges between S and $N(S)$. The sum of degrees in S is $\sum_{x \in S} \deg(x) = |S|k = |E(S, N(S))|$. Since each vertex in $N(S)$ has degree k, $|E(S, N(S))| \leq \sum_{y \in N(S)} \deg(y) = |N(S)|k$. Thus, $|S|k \leq |N(S)|k$, which implies $|S| \leq |N(S)|$ (since $k > 0$).

By Hall's Marriage Theorem, a matching exists that covers all vertices in X. Since $|X| = |Y|$, this matching is perfect.

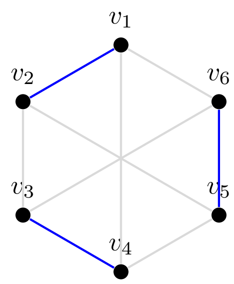
■

2.2 Factors of Graphs

Definition 2.2.1 (Factor)**.** Let G be a graph. A **factor** F of G is a spanning subgraph of G. That is, $V(F) = V(G)$ and $E(F) \subseteq E(G)$.

Definition 2.2.2 ($\mathcal{F}$-Factor and H-Factor)**.** Let $\mathcal{F} = \{H_1, H_2, \ldots, H_k\}$ be a collection of graphs. A factor F is called an **$\mathcal{F}$-factor** if every connected component of F is isomorphic to some element of $\mathcal{F}$.

In the specific case where $\mathcal{F} = \{H\}$ (a singleton set), the factor F is called an **H-factor**. In this case, every component of the spanning subgraph is isomorphic to H.



K_2-Factor
The blue edges form a spanning subgraph
where every component is isomorphic to K_2.

Figure 2.12: Visualisation of an H-factor where $H = K_2$.

Degree Constraints: f-Factors and k-Factors

Example 2.2.3. Consider the graph $G = K_3 \square K_3$ (also known as the Rook's graph on a 3×3 grid). This graph has 9 vertices and is 4-regular. It admits a K_3-factorization consisting of two factors, F_1 and F_2.

- **Factor 1 (Red):** Three disjoint triangles formed horizontally.
- **Factor 2 (Blue):** Three disjoint triangles formed vertically.

Since $E(G) = E(F_1) \cup E(F_2)$ and $E(F_1) \cap E(F_2) = \emptyset$, this constitutes a factorization.

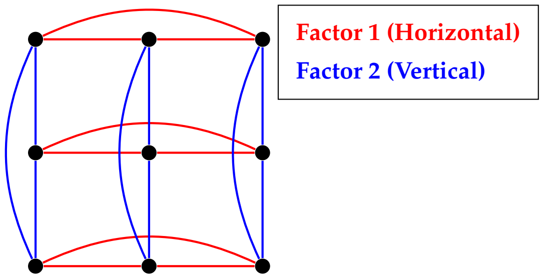
Definition 2.2.4 (Vertex Function)**.** A **vertex function** of a graph $G(V, E)$ is a mapping $f : V \to \mathbb{N}_0$, where $\mathbb{N}_0$ denotes the set of non-negative integers.

Definition 2.2.5 (f-Factor)**.** A spanning subgraph F of a graph G is called an **f-factor** if the degree of every vertex v in F corresponds to the value assigned by the vertex function. That is:

$$\deg_F(v) = f(v) \quad \forall v \in V.$$

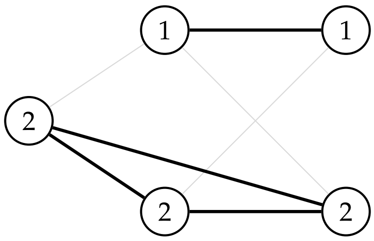
Definition 2.2.6 (k-Factor)**.** A **k-factor** is a special case of an f-factor where $f(v) = k$ for all $v \in V$. Thus, a k-factor is a k-regular spanning subgraph.

- A **1-factor** is a perfect matching (vertices have degree 1).
- A **2-factor** is a disjoint union of cycles covering all vertices.



Graph $G = F_1 \cup F_2$

Figure 2.13: Visualisation of G decomposed into two K_3-factors.



Example of an f-factor
Numbers inside nodes represent the target degree $f(v)$. The thick edges form the subgraph satisfying these constraints.

Figure 2.14: An f-factor where some vertices require degree 1 and others degree 2.

Factorisation

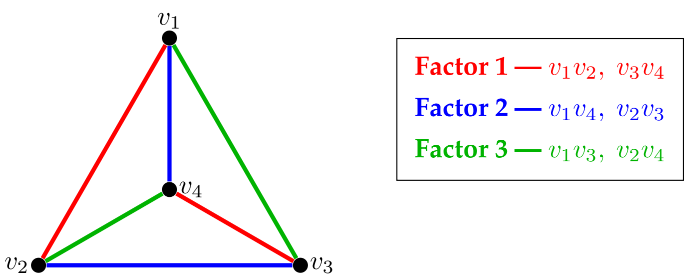
Definition 2.2.7 (Factorisation / Decomposition)**.** Let G be a graph and let $\{F_1, F_2, \dots, F_q\}$ be a set of factors of G. These factors constitute a **factorisation** (or decomposition) of G if their edge sets form a partition of $E(G)$. We write:

$$G = F_1 \oplus F_2 \oplus \cdots \oplus F_q$$

This means every edge of G belongs to exactly one factor F_i.

Definition 2.2.8 (Isomorphic Factorisation)**.** If each factor F_i in a factorisation is isomorphic to the same graph F, the factorisation is called an **isomorphic factorisation** or an F**-decomposition** of G.

Definition 2.2.9 (k-Factorisation)**.** A factorisation is called a k**-factorisation** if every factor F_i is a k-factor. A graph admitting such a decomposition is said to be k**-factorable**.



1-Factorisation of K_4

Figure 2.15: A 1-factorisation of the complete graph K_4. The edges are partitioned into three distinct perfect matchings (Red, Blue, Green).

Definition 2.2.10. A component of a graph is odd or even according to whether it has an odd or even number of vertices. The number of odd components of a graph G is denoted by $k_0(G)$.

Theorem 2.2.11 ((Tutte's One-Factor Theorem). *A graph $G = (V, E)$ has a perfect matching if and only if for every subset $S \subseteq V$, the number of odd components in $G - S$ is less than or equal to $|S|$. That is, $o(G - S) \leq |S|$ for all $S \subseteq V$.*

Proof. See proof from Introduction to Graph Theory by S. Pirzada

Fully explained Video lecture available at
https://youtu.be/EW4oA9HZzgg?si=8QGKe7Sfpdem9Rg5
or visit YouTube via **https://aijaz.vercel.app/** ■

Theorem 2.2.12 (Peterson). *Every cubic graph without cut edges has a 1-factor*

Proof. Let G be a cubic graph with vertex set $V(G)$. Let G have no cut edges and let $S \subseteq V$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S. Let $O_1, O_2, \ldots, O_r$ be the odd components in $G - S$. Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S. Then we have;

$$\sum_{v \in V(O_i)} \deg(v) = 3n_i = 2m_i + k_i.$$

This gives, $k_i = 3n_i - 2m_i,\ 1 \leq i \leq r$. As n_i is odd, therefore, k_i must be odd. Moreover, since there are no cut edges in the graph, we have $k_i \neq 1$, that is, $k_i \geq 3$, that is, there are at least 3 edges from each O_i to S. Hence, we have

$$3r \leq \sum_{s \in S} \deg(s) = 3|S|.$$

Therefore,

$$k_o(G - S) = r \leq |S|$$

for each $S \subseteq V(G)$. The proof then follows by Tutte's theorem. ■

Theorem 2.2.13 (Generalization of Theorem 2.2.12). *Let G be a $k-$regular graph having even number of vertices and is $(k-1)-$edge connected. Then G has a $1-$factor.*

Proof. Let G be a k-regular, $(k-1)$-edge-connected graph with even order and let S be any subset of V. Let $O_1, O_2, \ldots, O_r$ be the odd components of $G - S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S. Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S for $1 \leq i \leq r$. Then as G is $(k-1)-$connected, we have

$$k_i \geq k - 1, \tag{2.1}$$

Now, G is $k-$regular and vertices of any O_i are either adjacent within O_i or adjacent to some vertex of S, therefore,

$$\sum_{v \in V(O_i)} d(v) = kn_i = 2m_i + k_i \tag{2.2}$$

and vertices in S can be adjacent within S or adjacent to some vertex of $O_i,\ 1 \leq i \leq r$ or may be adjacent to some other vetex of G, so

$$\sum_{v \in S} d(v) = k|S| \geq \sum_{i=1}^{r} k_i + 2|E(S)|. \tag{2.3}$$

Now, (2.2) gives $k_i = kn_i - 2m_i$, so that if k and $k)i$ have the same parity, therefore, $k_i \geq k-1$ implies $k_i \geq k$. Summing this over r odd components, we obtain by using (2.3)

$$rk \leq \sum_{i=1}^{r} k_i \leq k|S| + 2|E(S)| \leq k|S|.$$

Therefore, $k_0(G - S) = r \leq |S|$ and by Tutte's theorem, G has a 1-factor. ■

Definition 2.2.14 (Anti-factors)**.** A subset $S \subseteq V$ of a graph $G(V, E)$ such that $k_0(G - S) > |S|$ is called an antifactor set. An antifactor set S such that no proper subset of S is an antifactor set is called a minimal antifactor set.

Theorem 2.2.15. *Let G be a $k-$regular graph of even order that is also $(k-1)-$edge connected. If G_1 is obtianed from G by deleting any $k-1$ edges, then G_1 has a $1-$factor.*

Proof. Let G be a k-regular, $(k-1)$-edge-connected graph with even order. Let $K \subseteq V(G)$ and let $V(G_1) = V(G) - K$ and let S be any subset of V. Let $O_1, O_2, \ldots, O_r$ be the odd components of $G - S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S. Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S for $1 \leq i \leq r$. Then as G is $(k-1)-$connected, we have $a_i =$number of edges from K to S
$b_i =$number of edges within O_i
$m_i =$number of edges from O_i to S

$$a_i + m_i \geq k - 1, \text{ then } \sum_{i=1}^{r} a_i + \sum_{i=1}^{r} m_i \geq k$$

Morevover, $\sum_{i=1}^{r} a_i + \frac{1}{2} \sum_{i=1}^{r} b_i \leq k - 1$, which gives, $2 \sum_{i=1}^{r} a_i + \sum_{i=1}^{r} b_i \leq 2(k-1)$.

Also, $\sum_{i=1}^{r} m_i \leq k|S|$. Adding, we get,

$$2\sum_{i=1}^{r} a_i + \sum_{i=1}^{r} b_i + \sum_{i=1}^{r} m_i \leq 2(k-1) + k|S| = k(|S| + 2) - 2.$$

Finally,

$$\begin{aligned}
kr &\leq \sum_{i=1}^{r} a_i + \sum_{i=1}^{r} m_i \\
&\leq 2\sum_{i=1}^{r} a_i + \sum_{i=1}^{r} b_i + \sum_{i=1}^{r} m_i \\
&\leq k(|S| + 2) - 2 \\
&< k(|S| + 2).
\end{aligned}$$

Thus, $r < |S| + 2$, a contradiction. ∎

Corollary 2.2.16. *A $(k-1)$-regular simple graph on $2k$ vertices has a 1-factor.*

Proof. Assume G is a $(k-1)$-regular simple graph with $2k$ vertices having no 1-factor. Then G has an antifactor set S and by Lemma 8.4, $k_0(G - S) \geq |S| + 2$. Therefore,

$$|S| + (|S| + 2) \leq 2k \implies |S| \leq k - 1.$$

Let $|S| = k - r$. Then $r \neq 1$. For if $r = 1$, then $|S| = k - 1$ and therefore $k_0(G - S) = k + 1$. Thus each odd component of $G - S$ is a singleton and therefore each such vertex is adjacent to all the $k-1$ vertices of S, as G is $(k-1)$-regular. This implies that every vertex of S is of degree at least $k + 1$, a contradiction.

Thus, $|S| = k - r$, with $2 \leq r \leq k - 1$.

If G_1 is any component of $G - S$ and $v \in V(G_1)$, then v can be adjacent to at most $|S|$ vertices of S. Since G is $(k-1)$-regular, v is adjacent to at least $(k-1) - (k-r) = r - 1$ vertices of G_1. So, $|V(G_1)| \geq r$.

Counting the vertices of S and the components, we obtain:

$$(|S| + 2)r + |S| \leq 2k$$

Substituting $|S| = k - r$:

$$(k - r + 2)r + (k - r) \leq 2k$$

Expanding and simplifying:

$$kr - r^2 + 2r + k - r \leq 2k$$
$$kr - r^2 + r - k \leq 0$$

Multiplying by -1 and factoring:

$$r^2 - r(k + 1) + k \geq 0$$
$$(r - 1)(r - k) \geq 0$$

Since $2 \leq r \leq k - 1$, the term $(r - 1)$ is positive and $(r - k)$ is negative. Their product is negative, which violates the condition $(r - 1)(r - k) \geq 0$. ■

Definition 2.2.17. A *vertex function* for a general graph $G(V, E)$ is a function $f : V \to \mathbb{N}_0$ such that $0 \leq f(v) \leq d(v)$, for every $v \in V$.

For any function $f : V \to \mathbb{N}_0$ and for any $U \subseteq V$, we set

$$f(U) = \sum_{v \in U} f(v).$$

Definition 2.2.18. Three pair-wise disjoint (possibly empty) subsets S, T and U of the vertex set V of a general graph G, such that $V = S \cup T \cup U$, is called a *decomposition* $D = (S, T, U)$ of G. The components, say $C_1, C_2, \ldots, C_k$, of the vertex-induced subgraph $\langle U \rangle$ of G are called the components of U. For each C_i, let

$$h_i = f(C_i) + q_G[T, C_i],$$

where f is a given vertex function of G, and $q_G[T, C_i]$ denotes the number of edges of G, whose one end is in T and the other end is in C_i. Then C_i is said to be an odd or even component of U, depending upon h_i being odd or even, and the number of odd components of U is denoted by $k_0'(D, f)$.

The *f-deficiency* of D for the given G is defined as

$$\delta(D, f) = k_0'(D, f) - f(S) + f(T) - d(T) + q[S, T].$$

Construction 1 (Construction of Graph G' for $f-$factor Theorem)**.** Let G be a general graph with the vertex function f. This defines for each vertex v_i, its degree $d_i = d(v_i|G)$, $f_i = f(v_i)$ and $k_i = d_i - f_i$. For each vertex v_i of G, take two subsets X_i and Y_i of vertices of G', where $|X_i| = d_i$ and $|Y_i| = k_i$. Let the vertices of Y_i be labelled y^i_j, $1 \leq j \leq k_i$, arbitrarily. Now, label the vertices of X_i in the following manner.

Label the edges of G arbitrarily as $1, 2, \ldots, m$. Since $|X_i|$ = number of link edges l_i incident with v_i plus twice the number of loops ℓ^0_i incident with v_i, a vertex $x^i_j \in X_i$ is taken corresponding to each link edge j incident with v_i and two vertices $x^i_h, x^i_{h'}$ are taken corresponding to each loop h incident at v_i. As for edges, if link j joins v_i and v_k, then $x^i_j x^k_j$ is made an edge. Corresponding to loop h incident with x^i_h, the edge $x^i_h x^i_{h'}$ is taken. In addition, all the edges of the complete bipartite subgraph with vertex partition $\{X_i, Y_i\}$ are taken for each i. These constitute the graph G'. The induced subgraph $< X_i \cup Y_i >$ of G' is called the *star graph* at i and is denoted by $St(i)$. For multi graph G, it is a complete bipartite graph, but for general graphs it will contain edges in $< X_i >$ which we call *loop edges*. The other edges of $St(i)$ are called *star edges* and the remaining edges of G' are called as *link edges*.

Illustration: This construction is illustrated in Figure 2.16. In the general graph G, $d(v_1) = 5$, $d(v_2) = 3$ and $d(v_3) = 2$. Let $f = (3, 2, 1)$ be the vertex function, so that $f(v_1) = 3$, $f(v_2) = 2$ and $f(v_3) = 1$. Therefore, $k_1 = 2$, $k_2 = 1$ and $k_3 = 1$. Here $|X_1| = 5$, $|X_2| = 3$ and $|X_3| = 2$, and $|Y_1| = 2$, $|Y_2| = 1$ and $|Y_3| = 1$.

Label the edges of G as 1, 2, 3, 4 and 5 as shown. Then $Y_1 = \{y^1_1, y^1_2\}$, $Y_2 = \{y^2_1\}$ and $Y_3 = \{y^3_1\}$, and we have $X_1 = \{x^1_1, x^1_{1'}, x^1_2, x^1_3, x^1_4\}$, $X_2 = \{x^2_2, x^2_3, x^2_5\}$ and $X_3 = \{x^3_4, x^3_5\}$. The vertex set of G' is $X_i \cup Y_i$, $1 \leq i \leq 3$. The edges of G' are $x^1_1 x^1_{1'}$, $x^1_2 x^2_2$, $x^1_3 x^2_3$, $x^1_4 x^3_4$, $x^2_5 x^3_5$ (shown by dotted lines) and the star edges (shown by bold lines). The dotted lines in G form an f-factor with $f = (3, 2, 1)$.

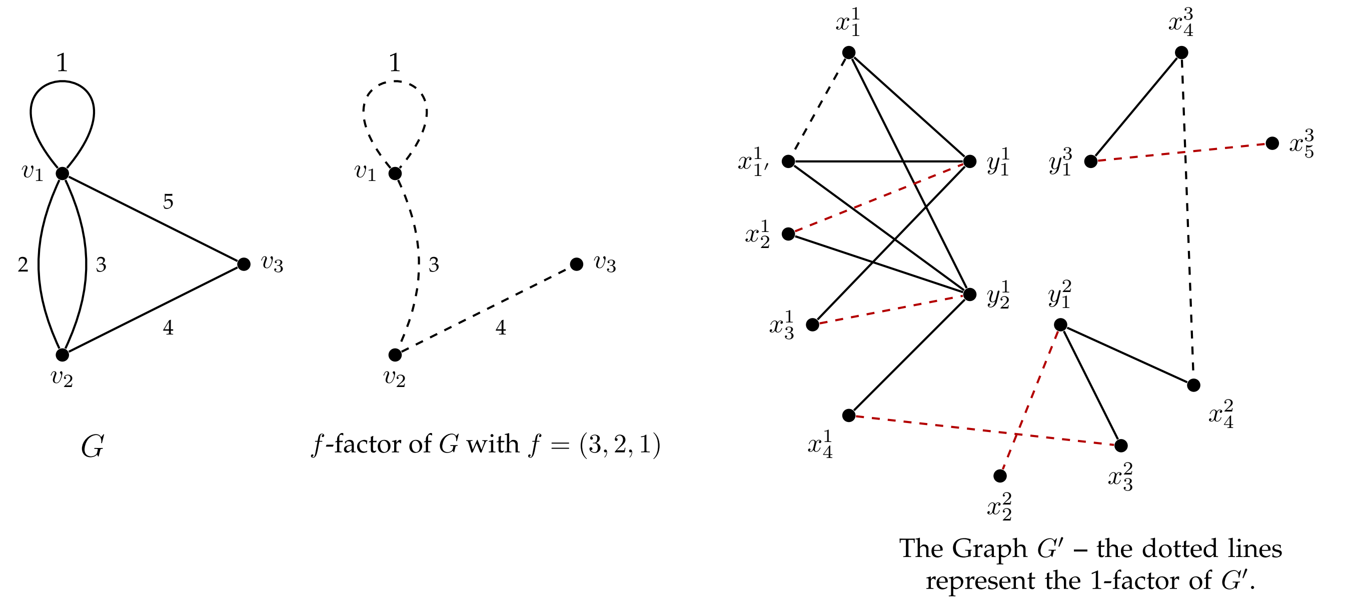


Figure 2.16: Dark dotted edges in G' represent edges induced by $f-$factor of G for a $1-$factor of G' and red dotted edges are additional edges required to produce a $1-$factor of G'.

Example 2.2.19 (Construction of G' for a Cycle C_4)**.** Let G be a cycle C_4 with vertices v_1, v_2, v_3, v_4. Let $f(v_i) = 1$ for all i. Thus, for every vertex v_i:

$$|X_i| = d(v_i) = 2 \quad \text{and} \quad |Y_i| = d(v_i) - f(v_i) = 1.$$

The construction is visualized below (Figure 2.17)The graph G' consists of 4 "corner" gadgets. Finding a 1-factor in G' corresponds to finding a 1-factor in G.

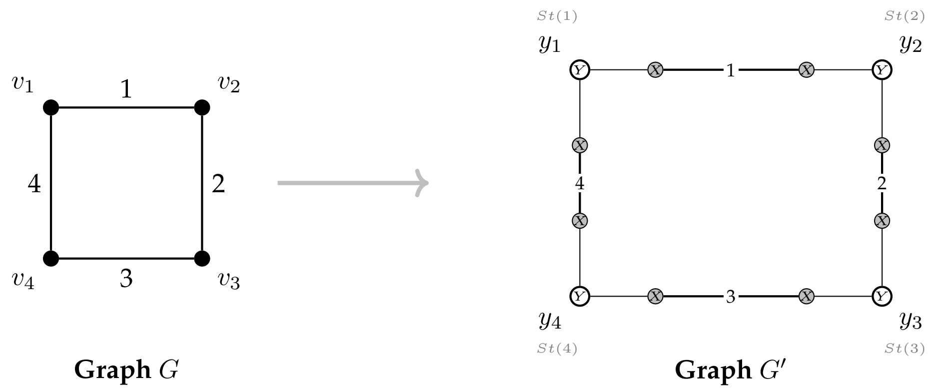


Figure 2.17: Transformation of C_4 into G' with $f(v) = 1$.

Theorem 2.2.20. *A general graph G has an f-factor if and only if G' has a 1-factor*

Proof. We prove the necessary and sufficient conditions separately.

(⇒) Necessity Assume G has an f-factor F with link edges L and loop edges L^0. Let g_i denote the number of link edges and f_i^0 the number of loop edges of F incident at vertex v_i. Consequently, the degree in the factor is $f_i = g_i + 2f_i^0$.

Recall that for the constructed graph G', the size of the sets X_i and Y_i are defined as:

$$|X_i| = d_i = l_i + 2l_i^0 \quad \text{and} \quad |Y_i| = k_i = d_i - f_i,$$

where l_i is the number of link edges incident with v_i and l_i^0 is the number of loops incident with v_i in the original graph G. We can expand the expression for $|Y_i|$ as follows:

$$\begin{aligned}|Y_i| &= (l_i + 2l_i^0) - (g_i + 2f_i^0)\\ &= (l_i - g_i) + 2(l_i^0 - f_i^0).\end{aligned}$$

Let L' and $(L^0)'$ be the set of link and loop edges in G' corresponding to L and L^0. The matching consisting of these edges keeps all vertices of Y_i unsaturated. Furthermore, it leaves unsaturated a set of vertices in X_i with cardinality:

$$(l_i - g_i) + 2(l_i^0 - f_i^0) = k_i = |Y_i|.$$

We can now match the vertices of Y_i with these remaining unsaturated vertices of X_i using the star edges $St(i)$. Let S' be this set of edges. Clearly, the union $L' \cup (L^0)' \cup S'$ constitutes a 1-factor in G'.

(⇐) Sufficiency Conversely, assume G' has a 1-factor H. Since the vertices in Y_i can only be matched to vertices in X_i (by the construction of the star graphs), there are exactly

$$d_i - |Y_i| = d_i - (d_i - f_i) = f_i$$

vertices of X_i that are *not* matched by H to vertices in Y_i. Let us denote this set of vertices as X_i^*. Note that $f_i = g_i + 2f_i^0$.

Clearly, $2f_i^0$ vertices from X_i^* correspond to loop edges of the induced subgraph $\langle X_i \rangle$ and are thus matched in pairs amongst themselves. The remaining g_i vertices in X_i^* are matched by the link edges of H to vertices in X_j belonging to other star graphs $St(j)$ of G'.

If F is the set of edges in G corresponding to these loop edges and link edges selected by H, then the degree of v_i in the subgraph induced by F is:

$$d_{<F>}(v_i) = g_i + 2f_i^0 = f_i.$$

Thus, F is an f-factor of G. ■

Theorem 2.2.21 ($f-$factor theorem)**.** *A general graph G has an $f-$factor if and only if it has no decomposition D with $\delta(D, f) > 0$.*

OR

If f is a vertex function for a graph G, then either G has an $f-$factor or an $f-$barrier but not both.

Proof. Refer to "Introduction to Graph Theory" by S Pirzada. ■

Definition 2.2.22. Suppose $D = (S, T, U)$ is a decomposition of V and $x \in S$, $y \in T$. The decomposition $D_1 = (S \setminus \{x\}, T, U \cup \{x\})$ is said to be obtained by an **x-transfer from** S, and the decomposition $D_2 = (S, T \setminus \{y\}, U \cup \{y\})$ is said to be obtained by a **y-transfer from** T.

Lemma 2.2.23. *An x-transfer from S does not reduce the deficiency of D if $f(x) = d(x)$ or $d(x) - 1$, and a y-transfer does not reduce the deficiency of D if $f(y) = 0$ or 1.*

Theorem 2.2.24. *The f-factor Theorem implies the 1-factor Theorem.*

Proof. Clearly, by the f-factor Theorem with $f(v) = 1$ for all $v \in V$, G has a 1-factor if and only if it has a 1-barrier $D = (S, T, U)$. But by Lemma 2.2.23, $D_1 = (S, \emptyset, V \setminus S)$ has no less deficiency than D (by y-transfer from T). Therefore, G has no 1-factor if and only if there is a 1-barrier of the form D_1.

The deficiency is given by:

$$\delta(D_1, f) = k_0(G - S) - f(S) - d(T) + f(T) + q[S, T].$$

Since $T = \emptyset$, this simplifies to:

$$\delta(D_1, f) = k_0(G - S) - |S|.$$

Thus, D_1 is a 1-barrier if and only if $\delta(D_1, f) > 0$, which implies $k_0(G - S) > |S|$. Hence, the 1-factor Theorem holds. ■

Corollary 2.2.25. *The Erdős-Gallai Theorem on degree sequences follows from the f-factor Theorem.*

Proof. Erdős-Gallai Theorem gives a necessary and sufficient condition for a non-increasing sequence $d = [d_i]_1^n$ of non-negative integers to be the degree sequence of a simple graph. In order to get this condition, we observe that d is graphic if and only if the complete graph K_n has a d-factor, where $d(v_i) = d_i$, given $\sum_{i=1}^n d_i = 2m$.

By the f-factor Theorem, d fails to be graphic if and only if K_n has a d-barrier $D = (S, T, U)$. Since for K_n, U can have at most one component, $k_0(D, d) = 0$ or 1. Furthermore, $\delta(D, d) > 0$ is even, as $d(K_n) = \sum d_i = 2m$ is even. Therefore, the condition $\delta(D, d) > 0$ becomes:

$$q_{K_n}[S, T] - d(S) - (n-1)|T| + d(T) > 0. \tag{2.4}$$

Now, transfer of elements t from T to U, or elements s from S to U does not decrease the deficiency $\delta(D, d)$. Specifically:

- Elements $t \in T$ with $d(t) < |T| + |U|$ can be transferred to U without decreasing the deficiency.
- Elements $s \in S$ with $d(s) > |T|$ can be transferred to U without decreasing the deficiency.
- Elements $u \in U$ with $d(u) \geq |T| + |U|$ can be transferred to S without decreasing the deficiency.

Therefore, the f-barrier can be made such that:

$$\begin{gathered} d(t) > |T| + |U| \quad \text{for all } t \in T, \\ |T| < d(u) < |T| + |U| \quad \text{for all } u \in U, \\ d(s) \leq |T| \quad \text{for all } s \in S. \end{gathered}$$

Consequently, d is non-graphical if and only if K_n has such a d-barrier D. But for such a D, $q[S, T] = r(n - r - |U|)$, where $|T| = r$ and $d(T) = \sum_1^r d_i$. Thus condition (2.4) becomes:

$$r(n - r - |U|) - d(S) - r(n-1) + d(T) > 0.$$

Rearranging for $d(T)$ (which corresponds to $\sum_{i=1}^r d_i$):

$$\begin{aligned} \sum_{i=1}^r d_i &> r(n-1) + d(S) - r(n - r - |U|) \\ &= r(r-1) + r(n-r) + d(S) - r(n - r - |U|) \\ &= r(r-1) + r|U| + d(S). \end{aligned}$$

Observing that vertices in S have degree $\leq r$ and vertices in U have degree $> r$, we can relate $r|U| + d(S)$ to the minimum function used in the Erdős-Gallai condition:

$$\sum_{i=1}^r d_i > r(r-1) + \sum_{i=r+1}^n \min\{r, d_i\},$$

which is the Erdős-Gallai condition. ■

Theorem 2.2.26 (k-factor Theorem)**.** *Let $d = [d_i]_1^n$ be a graphical sequence. If the sequence $d' = [d_i - k]_1^n$ is also graphical, then there exists a realization of d that contains a k-factor.*

Proof. Let G_1 and G_2 be two graphs on the vertex set $V = \{v_1, v_2, \ldots, v_n\}$. The degrees are constrained such that for every vertex v_i, $d(v_i|G_1) = d_i$ and $d(v_i|G_2) = d_i - k$. To proceed, we assume that G_1 and G_2 are chosen specifically such that they share the **maximum possible number of edges**. We aim to prove that $E(G_2) \subseteq E(G_1)$. We assume the contrary—that G_2 contains edges not in G_1—and derive a contradiction. Let $E_1 = E(G_1)$ and $E_2 = E(G_2)$.

The Switching Lemma (The Claim)

The proof utilizes an "Edge Disjoint Transfer" (EDT), or degree-preserving switch. We establish the following claim: *If $v_iv_j \in E_2 \setminus E_1$ and $v_iv_k, v_jv_r \in E_1 \setminus E_2$ (where indices are distinct), then the edge v_kv_r must belong to $E_1 \setminus E_2$.*

To see why, consider the alternatives. If v_kv_r were in E_2, we could switch the pair (v_iv_j, v_kv_r) in G_2 to (v_iv_k, v_jv_r). Since the new edges are in G_1, this would increase the overlap between G_1 and G_2, contradicting our maximality assumption. Conversely, if v_kv_r were not in E_1, we could switch (v_iv_k, v_jv_r) in G_1 to (v_iv_j, v_kv_r). Since $v_iv_j \in E_2$, this would also increase the overlap. Thus, v_kv_r must be in $E_1 \setminus E_2$.

The Main Argument

Assume $E_2 \not\subseteq E_1$. We select a vertex v_1 that has the **maximum number**, denoted by t, of edges in $E_2 \setminus E_1$ incident to it. By the degree condition ($d(v|G_1) - d(v|G_2) = k$), v_1 must be incident to exactly $t+k$ edges in $E_1 \setminus E_2$. Let these neighbors be $v_2, \ldots, v_{t+k+1}$.

Consider a path consisting of a "bad" edge $v_1v_k \in E_2 \setminus E_1$ and a "good" edge $v_kv \in E_1 \setminus E_2$. We analyze two cases for the vertex v:

Case 1: v is not one of the neighbors $v_2, \ldots, v_{t+k+1}$.

By applying the Switching Lemma, we infer that v must form an edge in $E_1 \setminus E_2$ with every vertex in $\{v_2, \ldots, v_{t+k+1}\}$. This implies v is incident to at least $t+k+1$ edges in $E_1 \setminus E_2$. By the degree constraints, this would require v to be incident to $t+1$ edges in $E_2 \setminus E_1$. This contradicts the choice of v_1 as the vertex with the maximum such edges (t).

Case 2: v is one of the neighbors (e.g., $v = v_2$).

Without loss of generality, let $v = v_2$. Applying the Switching Lemma again shows that v_2 must connect to $v_3, \ldots, v_{t+k+1}$ via edges in $E_1 \setminus E_2$. Additionally, v_2 is connected to v_1 and v_k by edges in $E_1 \setminus E_2$. Counting these reveals that v_2 is incident to $t+k+1$ edges in $E_1 \setminus E_2$. As in Case 1, this implies v_2 must have $t+1$ edges in $E_2 \setminus E_1$, contradicting the maximality of v_1.

Both cases lead to a contradiction. Therefore, the set $E_2 \setminus E_1$ must be empty, proving that $E(G_2) \subseteq E(G_1)$. The graph with edge set $E(G_1) \setminus E(G_2)$ is a k-factor of G_1. ■

Proof. Let G_1 and G_2 be realizations of the sequences d and d' respectively, defined on the same vertex set $V = \{v_1, v_2, \ldots, v_n\}$. That is, $\deg_{G_1}(v_i) = d_i$ and $\deg_{G_2}(v_i) = d_i - k$.

We choose the pair of graphs G_1 and G_2 such that the number of common edges, $|E(G_1) \cap E(G_2)|$, is maximized. Let $E_1 = E(G_1)$ and $E_2 = E(G_2)$.

Our goal is to show that $E_2 \subseteq E_1$. If this holds, then the graph $G = (V, E_1 \setminus E_2)$ has degrees $d_i - (d_i - k) = k$ for all i, making it the desired k-factor.

The Switching Lemma. Suppose there exist vertices v_i, v_j, v_k, v_r such that:

1. $v_iv_j \in E_2 \setminus E_1$, and
2. $v_iv_k \in E_1 \setminus E_2$ and $v_jv_r \in E_1 \setminus E_2$.

Then, it must be that $v_kv_r \in E_1 \setminus E_2$.

Proof of Lemma: Consider the edge v_kv_r.

- If $v_kv_r \in E_2$, we perform a 2-switch in G_2: replace edges v_iv_j, v_kv_r with v_iv_k, v_jv_r. This operation preserves the degrees of G_2 but increases the number of common edges with G_1 (since $v_iv_k, v_jv_r \in E_1$), contradicting the maximality of the intersection.
- If $v_kv_r \notin E_1$, we perform a 2-switch in G_1: replace edges v_iv_k, v_jv_r with v_iv_j, v_kv_r. This preserves degrees in G_1 but adds v_iv_j (which is in E_2) to G_1, increasing the intersection, again a contradiction.

Thus, v_kv_r must exist in E_1 and must not exist in E_2. ◇

Assume for the sake of contradiction that $E_2 \not\subseteq E_1$. Then $E_2 \setminus E_1$ is non-empty.

For any vertex u, observe that:

$$\deg_{E_1 \setminus E_2}(u) - \deg_{E_2 \setminus E_1}(u) = \deg_{G_1}(u) - \deg_{G_2}(u) = d_i - (d_i - k) = k.$$

Let $t(u) = \deg_{E_2 \setminus E_1}(u)$. Then $\deg_{E_1 \setminus E_2}(u) = t(u) + k$.

Choose a vertex v_1 that maximizes $t(v_1)$. Let $t = t(v_1) > 0$. Since $t > 0$, there exists a neighbor v_k such that $v_1v_k \in E_2 \setminus E_1$. Since $\deg_{E_1\setminus E_2}(v_1) = t + k$, let the neighbors of v_1 in $E_1 \setminus E_2$ be $\{v_2, v_3, \ldots, v_{t+k+1}\}$.

Consider the vertex v_k. Since $v_1v_k \in E_2 \setminus E_1$, $t(v_k) \geq 1$. Consequently, $\deg_{E_1\setminus E_2}(v_k) \geq k + 1 \geq 1$. Thus, there exists a vertex v such that $v_kv \in E_1 \setminus E_2$.

We analyze two cases for v:

Case 1: $v \notin \{v_2, \ldots, v_{t+k+1}\}$. We have the configuration: $v_1v_k \in E_2 \setminus E_1$, while v_kv and v_1v_i (for all $i \in \{2, \ldots, t + k + 1\}$) are in $E_1 \setminus E_2$. By applying the Switching Lemma with the pairs (v_1v_k) and (v_1v_i, v_kv), we deduce that $vv_i \in E_1 \setminus E_2$ for all $i \in \{2, \ldots, t + k + 1\}$. Additionally, we know $vv_k \in E_1 \setminus E_2$. Thus, vertex v is incident to at least $(t + k) + 1$ edges in $E_1 \setminus E_2$. Using the degree formula:

$$t(v) = \deg_{E_1\setminus E_2}(v) - k \geq (t + k + 1) - k = t + 1.$$

This contradicts the choice of v_1 as the vertex maximizing t.

Case 2: $v \in \{v_2, \ldots, v_{t+k+1}\}$. Without loss of generality, let $v = v_2$. We have $v_1v_k \in E_2 \setminus E_1$. Also, $v_kv_2 \in E_1 \setminus E_2$ and $v_1v_i \in E_1 \setminus E_2$ for $i \in \{3, \ldots, t + k + 1\}$. Applying the Switching Lemma to the pairs (v_1v_k) and (v_1v_i, v_kv_2), we deduce that $v_2v_i \in E_1 \setminus E_2$ for all $i \in \{3, \ldots, t + k + 1\}$.

Counting the edges incident to v_2 in $E_1 \setminus E_2$:

1. v_2v_1 (since v_2 is in the neighbor set of v_1),
2. v_2v_k (by assumption of this case),
3. v_2v_i for $i \in \{3, \ldots, t + k + 1\}$ (by the Lemma).

Total edges $= 1 + 1 + (t + k - 1) = t + k + 1$. Again, this implies $t(v_2) = (t + k + 1) - k = t + 1$, contradicting the maximality of t.

Since both cases lead to a contradiction, our assumption that $E_2 \setminus E_1 \neq \emptyset$ must be false. Therefore, $E_2 \subseteq E_1$, and G_1 contains the edges of G_2. The graph with edge set $E(G_1) \setminus E(G_2)$ is a k-factor of G_1. ■

2.3 Factorization of Complete Graphs

Definition 2.3.1. A **factorization** of a graph G is a decomposition of the edges of G into edge-disjoint spanning subgraphs, known as factors. If the graph is decomposed into k-factors (where every vertex has degree k), it is called a k**-factorization**.

For the complete graph K_n, the type of factorization possible depends on the parity of n.

Case 1: n **is Even (1-Factorization)**
Consider a complete graph K_{2n}. It has $2n$ vertices and is $(2n - 1)$-regular.

- A 1-factor is a perfect matching (regular of degree 1).
- Since the total degree of each vertex is $2n-1$, K_{2n} can be decomposed into exactly $2n-1$ perfect matchings.

Case 2: n **is Odd (2-Factorization)**
Consider a complete graph K_{2n+1}. It has $2n + 1$ vertices and is $2n$-regular.

- A 2-factor is a spanning subgraph where every vertex has degree 2 (a disjoint union of cycles).
- In the context of K_n, we are often interested in decomposing it into Hamiltonian cycles.
- Since the total degree is $2n$, K_{2n+1} can be decomposed into n Hamiltonian cycles.

Theorem 2.3.2. *For every positive integer n, the complete graph K_{2n} is 1-factorable.*

Proof. Let the vertex set be $V = \{0, 1, \dots, 2n-2, 2n-1\}$. We designate the vertex $2n-1$ as the fixed center and treat the remaining vertices $\{0, 1, \dots, 2n-2\}$ as elements of $\mathbb{Z}_{2n-1}$ (the integers modulo $2n-1$).

We construct $2n-1$ factors, denoted $F_0, F_1, \dots, F_{2n-2}$. For each $i \in \{0, \dots, 2n-2\}$, define the set of edges F_i as:

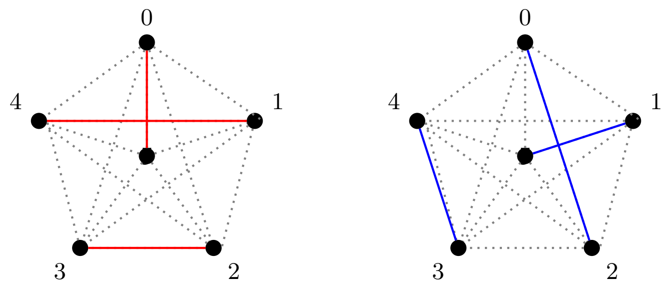
$$F_i = \{\{2n-1, i\}\} \cup \{\{i-k, i+k\} \pmod{2n-1} \mid k = 1, \dots, n-1\}.$$

1. F_i is a 1-factor: The fixed vertex $2n-1$ is matched to vertex i. The remaining $2n-2$ vertices form the polygon and are paired distinctly. The relation $i - k \equiv i + k \pmod{2n-1}$ implies $2k \equiv 0$, which is impossible for $1 \leq k \leq n-1$ since $2n-1$ is odd. Thus, every vertex has degree exactly 1 in F_i.

2. Decomposition:

- Every edge incident to the center $2n-1$ is of the form $\{2n-1, x\}$ and appears uniquely in factor F_x.
- For any edge $\{u, v\}$ not incident to $2n-1$, the sum of the indices modulo $2n-1$ is invariant for a fixed factor. Specifically, $u + v \equiv (i-k) + (i+k) \equiv 2i \pmod{2n-1}$. Since $2n-1$ is odd, $\gcd(2, 2n-1) = 1$, meaning 2 is invertible. Thus, $i \equiv (u+v) \cdot 2^{-1} \pmod{2n-1}$ is uniquely determined.

Therefore, every edge of K_{2n} belongs to exactly one factor, constituting a 1-factorization. ■



Factor F_0 **Factor F_1**

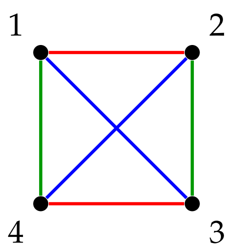
Figure 2.18: 1-Factorization of K_6 ($n = 3$). The solid colored lines represent the specific 1-factor, while the dotted lines represent the remaining edges of the complete graph.

Theorem 2.3.3. *For every positive integer n, the complete graph K_{2n+1} is 2-factorable into Hamiltonian cycles.*

Example 2.3.4 (1-Factorization of K_4)**.** Let $n = 4$. The graph K_4 has 4 vertices and degree 3. It can be decomposed into $4 - 1 = 3$ perfect matchings (1-factors).

The Factors:

- F_1 **(Red)**: Matches $\{(1,2), (3,4)\}$
- F_2 **(Blue)**: Matches $\{(1,3), (2,4)\}$
- F_3 **(Green)**: Matches $\{(1,4), (2,3)\}$

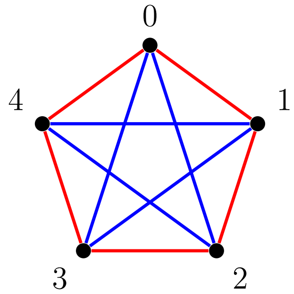


Decomposition of K_4

Example 2.3.5 (2-Factorization of K_5)**.** Let $n = 5$. The graph K_5 has 5 vertices and degree 4. It can be decomposed into $(5-1)/2 = 2$ Hamiltonian cycles.

The Factors:

- H_1 **(Red)**: The outer pentagon cycle $\{(0,1),(1,2),(2,3),(3,4),(4,0)\}$.
- H_2 **(Blue)**: The inner star cycle $\{(0,2),(2,4),(4,1),(1,3),(3,0)\}$.



Decomposition of K_5

Theorem 2.3.6. *For every positive integer n, the complete graph K_{2n+1} is 2-factorable into Hamiltonian cycles.*

Proof. Let the vertex set be $V = \{0,1,\ldots,2n\}$. We designate the vertex $2n$ as the fixed center and treat the remaining set $\{0,\ldots,2n-1\}$ as elements of $\mathbb{Z}_{2n}$.

We construct n Hamiltonian cycles $C_0, C_1, \ldots, C_{n-1}$ as follows:

1. The Base Cycle C_0: Define the cycle C_0 by the sequence of vertices:

$$C_0 = (2n,\ 0,\ 1,\ 2n-1,\ 2,\ 2n-2,\ \ldots,\ n,\ 2n).$$

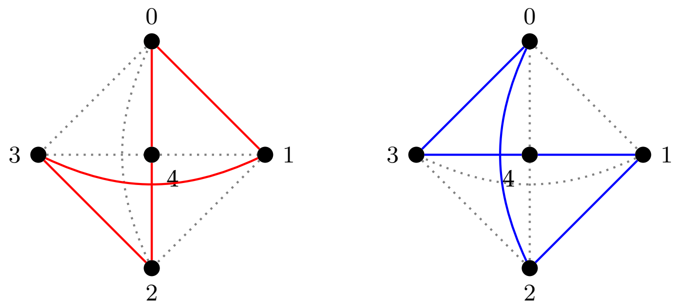
The path within $\mathbb{Z}_{2n}$ (excluding the fixed vertex $2n$) zigzags through the vertices. The lengths of the edges in this path correspond to the distances $1, 2, \ldots, n$ in the modular arithmetic of $\mathbb{Z}_{2n}$.

2. The General Cycles: For each $i \in \{0,\ldots,n-1\}$, let C_i be the cycle obtained by adding $i \pmod{2n}$ to every vertex in $\{0,\ldots,2n-1\}$, while keeping the vertex $2n$ fixed.

3. Verification:

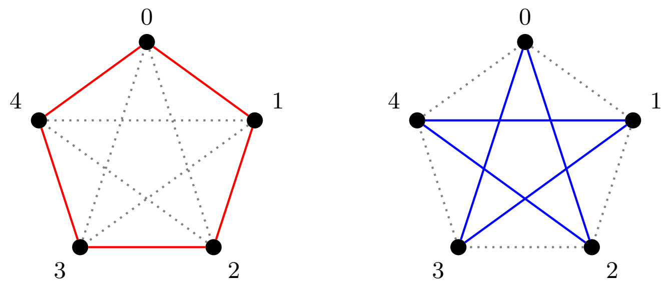
- **Edges incident to $2n$:** In cycle C_i, the vertex $2n$ connects to i and $n+i$. As i ranges from 0 to $n-1$, these pairs $\{i, n+i\}$ are disjoint and cover all vertices in $\mathbb{Z}_{2n}$. Thus, every edge incident to $2n$ appears exactly once.
- **Edges within $\mathbb{Z}_{2n}$:** The base path in C_0 contains exactly one edge of each "length" class $d \in \{1,\ldots,n\}$. By the properties of cyclic groups, rotating a set of edges that contains one representative of each length generates all possible edges within the complete graph induced by $\mathbb{Z}_{2n}$.

Thus, the collection $\{C_0,\ldots,C_{n-1}\}$ forms a 2-factorization of K_{2n+1}. ■



Cycle C_0 **Cycle** C_1

Figure 2.19: Decomposition of K_5. We curve the diagonal edges (0-2 and 1-3) so they do not overlap with the central vertex 4. All 10 edges of K_5 are present in both figures (either solid or dotted).



Factor 1: Pentagon **Factor 2: Star**

Figure 2.20: Decomposition of K_5 into 2 Hamiltonian Cycles. The dotted lines in each figure represent the edges belonging to the other factor.

Chapter 3

Edge Graphs

Please refer to Chapter 9 of the Book "An Introduction to Graph Theory" by S Pirzada.

Chapter 4

Spectral Theory of Graphs

Introduction

The algebraic study of graphs has two distinct but powerful roots: group theory and linear algebra. The study of symmetry began with **Arthur Cayley** (1878), who introduced graphical representations of groups. This connection was cemented by **Robert Frucht** in 1939, who proved the surprising result that any finite group is the automorphism group of some graph. Parallel to this, **Spectral Graph Theory** emerged from the intersection of mathematics and quantum chemistry. While early connections were made by Hückel in the 1930s regarding molecular orbitals, the mathematical foundations were solidified by Collatz and Sinogowitz (1957). The concept of **Graph Energy** was formally introduced by **Ivan Gutman** in 1978, bridging the gap between the sum of eigenvalues and the total π-electron energy of molecules. Why translate graph theory into the language of algebra? The answer lies in the power of invariants. **Automorphism groups** allow us to quantify the symmetry and indistinguishability of vertices, while **Cayley graphs** provide a geometric way to visualize abstract algebraic structures.
Simultaneously, the **Spectrum** (eigenvalues of the Adjacency or Laplacian matrix) acts as a "fingerprint" of the graph, revealing structural properties—such as connectivity, bipartiteness, and expansion—that are not immediately visible in a drawing. By studying invariants like **Graph Energy**, we gain precise tools to compare graphs, measure their structural complexity, and predict their behavior in physical and chemical systems.

This chapter explores the intersection of combinatorics and algebra, using group theory to analyze graph symmetries through automorphisms and Cayley digraphs. We examine fundamental connections between abstract groups and topological structures before transitioning to spectral graph theory, where linear algebra reveals hidden graph properties. By investigating the eigenvalues of adjacency and Laplacian matrices, we characterize standard graph classes and introduce the concept of graph energy. Finally, we demonstrate how these algebraic tools provide a quantitative measure of structural complexity and stability.

Video Resources

A detailed lecture series for this chapter is available at

https://www.youtube.com/@Prof.DrAijaz/playlists

or visit YouTube via https://aijaz.vercel.app/

4.1 Spectrum of Graphs

Let $G = (V, E)$ be a simple undirected graph with n vertices $\{v_1, \ldots, v_n\}$ and m edges.

Definition 4.1.1 (Adjacency Matrix)**.** The adjacency matrix $A(G)$ is an $n \times n$ matrix where:

$$A_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise} \end{cases}$$

For simple graphs, $A_{ii} = 0$ for all i (no self-loops).

Definition 4.1.2 (Spectrum)**.** The spectrum of G, denoted Spec(G), is the set of eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ of $A(G)$. These are the roots of the characteristic polynomial $\phi(G, \lambda) = \det(\lambda I - A)$. We typically order them as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$.

Theorem 4.1.3 (Real Eigenvalues)**.** *The eigenvalues of the adjacency matrix $A(G)$ are all real numbers.*

Proof. $A(G)$ is a real symmetric matrix (since the graph is undirected, $v_i \sim v_j \implies v_j \sim v_i$, so $A_{ij} = A_{ji}$). By the Spectral Theorem of linear algebra, all eigenvalues of a real symmetric matrix are real, and there exists an orthonormal basis of eigenvectors. ■

Theorem 4.1.4 (Sum of Eigenvalues)**.** *The sum of the eigenvalues of a simple graph is zero.*

$$\sum_{i=1}^{n} \lambda_i = 0$$

Proof. For any matrix, the sum of the eigenvalues equals the trace (the sum of diagonal elements).

$$\sum_{i=1}^{n} \lambda_i = \text{tr}(A) = \sum_{i=1}^{n} A_{ii}$$

Since G is simple, it has no self-loops, meaning $A_{ii} = 0$ for all i. Thus, the sum is 0. ■

Lemma 4.1.5. *Let A be the adjacency matrix of a graph G.*

(A) The sum of all 2×2 principal minors of A equals $-|E(G)|$.

(B) The sum of all 3×3 principal minors of A equals twice the number of triangles of G.

Proof. Let $A = [a_{ij}]$ be the adjacency matrix. Since G is simple, $a_{ii} = 0$ and $a_{ij} = a_{ji} \in \{0, 1\}$.
(A) A 2×2 principal minor for indices $i < j$ is $\det \begin{pmatrix} 0 & a_{ij} \\ a_{ji} & 0 \end{pmatrix} = -a_{ij}^2 = -a_{ij}$. This value is -1 if $\{i, j\} \in E(G)$ and 0 otherwise. Summing over all pairs $\{i, j\}$ yields $-|E(G)|$.
(B) A 3×3 principal minor for indices $i < j < k$ is

$$\det \begin{pmatrix} 0 & a_{ij} & a_{ik} \\ a_{ji} & 0 & a_{jk} \\ a_{ki} & a_{kj} & 0 \end{pmatrix} = a_{ij}a_{jk}a_{ki} + a_{ik}a_{ji}a_{kj} = 2a_{ij}a_{jk}a_{ki}.$$

This product is 1 if edges $\{i, j\}, \{j, k\}, \{k, i\}$ all exist (forming a triangle), and 0 otherwise. Thus, the minor equals 2 if the vertices form a triangle and 0 otherwise. Summing over all vertex triplets gives twice the number of triangles. ■

Theorem 4.1.6 (Number of Walks and Second Moment)**.** *Let N_k be the total number of closed walks of length k in G. Then $\sum_{i=1}^{n} \lambda_i^k = N_k$. Specifically for $k = 2$:*

$$\sum_{i=1}^{n} \lambda_i^2 = 2m$$

Proof. A standard result in graph theory states that the (i, j)-entry of the matrix A^k represents the number of walks of length k from vertex v_i to vertex v_j. A closed walk starts and ends at the same vertex. Thus, the number of closed walks is the sum of the diagonal entries of A^k:

$$N_k = \sum_{i=1}^{n} (A^k)_{ii} = \text{tr}(A^k)$$

If λ_i are eigenvalues of A, then λ_i^k are eigenvalues of A^k. The trace is the sum of eigenvalues:

$$\text{tr}(A^k) = \sum_{i=1}^{n} \lambda_i^k$$

For $k = 2$, a closed walk of length 2 starting at v_i must go to a neighbor v_j and return to v_i. This corresponds to traversing an edge back and forth. Each edge contributes 2 to the total count (once for each endpoint). Thus $N_2 = 2m$. ∎

Theorem 4.1.7 (Spectrum of Bipartite Graphs). *A graph G is bipartite if and only if its spectrum is symmetric with respect to 0. That is, if λ is an eigenvalue, then $-\lambda$ is also an eigenvalue with the same multiplicity.*

Proof. Let G be bipartite with parts V_1 and V_2. We can index vertices such that the adjacency matrix has the block form:

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}$$

Let λ be an eigenvalue with eigenvector $\mathbf{x} = \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix}$, where $\mathbf{u}$ corresponds to V_1 and $\mathbf{v}$ to V_2. The equation $A\mathbf{x} = \lambda\mathbf{x}$ yields:

$$B\mathbf{v} = \lambda\mathbf{u} \quad \text{and} \quad B^T\mathbf{u} = \lambda\mathbf{v}$$

Now consider the vector $\mathbf{y} = \begin{pmatrix} \mathbf{u} \\ -\mathbf{v} \end{pmatrix}$.

$$A\mathbf{y} = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ -\mathbf{v} \end{pmatrix} = \begin{pmatrix} -B\mathbf{v} \\ B^T\mathbf{u} \end{pmatrix} = \begin{pmatrix} -\lambda\mathbf{u} \\ \lambda\mathbf{v} \end{pmatrix} = -\lambda \begin{pmatrix} \mathbf{u} \\ -\mathbf{v} \end{pmatrix} = -\lambda\mathbf{y}$$

Thus, $-\lambda$ is also an eigenvalue. ∎

Theorem 4.1.8. *The adjacency spectrum of the complete graph K_n is $\{(n-1)^{(1)}, (-1)^{(n-1)}\}$.*

Proof. The adjacency matrix is $A = J - I$, where J is the all-ones matrix. The matrix J has rank 1 with a non-zero eigenvalue n (corresponding to eigenvector $\mathbf{1}$) and eigenvalue 0 with multiplicity $n - 1$. Since A is a polynomial of J, specifically $A = J - I$, its eigenvalues are obtained by shifting the eigenvalues of J by -1. Thus, the spectrum is $n - 1$ (multiplicity 1) and $0 - 1 = -1$ (multiplicity $n - 1$). ∎

Theorem 4.1.9. *The adjacency spectrum of the complete bipartite graph $K_{r,s}$ is $\{\sqrt{rs}^{(1)}, -\sqrt{rs}^{(1)}, 0^{(n-2)}\}$.*

Proof. The adjacency matrix of $K_{r,s}$ has only two linearly independent rows so its rank is 2. Consequently, 0 is an eigenvalue with multiplicity $n-2$. Since the trace is zero, implying the two non-zero eigenvalues are λ and $-\lambda$. Using the property that the sum of squares of eigenvalues equals twice the number of edges, we have $\lambda^2 + (-\lambda)^2 = 2rs$. This simplifies to $2\lambda^2 = 2rs$, yielding $\lambda = \sqrt{rs}$. ∎

Lemma 4.1.10. *For $n \geq 2$, the eigenvalues of Q_n are $1, \omega, \omega^2, \ldots, \omega^{n-1}$, where $\omega = e^{\frac{2\pi i}{n}}$, is the primitive nth root of unity.*

Proof. The characteristic polynomial of Q_n is $\det(Q_n - \lambda I) = (-1)^n(\lambda^n - 1)$. Clearly, the roots of the characteristic polynomial are the n roots of unity. □ ∎

Theorem 4.1.11. *For $n \geq 2$, the eigenvalues of C_n are $2\cos\frac{2\pi k}{n}$, $k = 1, \ldots, n$.*

Proof. Note that $A(C_n) = Q_n + Q_n' = Q_n + Q_n^{n-1}$ is a polynomial in Q_n. Thus, the eigenvalues of $A(C_n)$ are obtained by evaluating the same polynomial at each of the eigenvalues of Q_n. Thus, by Lemma 3.5, the eigenvalues of $A(C_n)$ are $\omega^k + \omega^{n-k}$, $k = 1, \ldots, n$. Note that

$$\begin{aligned}
\omega^k + \omega^{n-k} &= \omega^k + \omega^{-k} \\
&= e^{\frac{2\pi i k}{n}} + e^{-\frac{2\pi i k}{n}} \\
&= 2\cos\frac{2\pi k}{n},
\end{aligned}$$

$k = 1, \ldots, n$, and the proof is complete. ■

2nd Proof. Let the vertices of C_n be labeled $0, 1, \ldots, n-1$ such that vertex j is adjacent to $(j-1) \pmod n$ and $(j+1) \pmod n$. The adjacency matrix A is a circulant matrix where the rows are cyclic permutations of the vector $(0, 1, 0, \ldots, 0, 1)$.

Let $\omega = e^{i\frac{2\pi}{n}}$ be the primitive n-th root of unity. Consider the vector $\mathbf{x}_k = (1, \omega^k, \omega^{2k}, \ldots, \omega^{(n-1)k})^T$. The r-th entry of $A\mathbf{x}_k$ is the sum of the neighbors of vertex r:

$$(A\mathbf{x}_k)_r = (\mathbf{x}_k)_{r-1} + (\mathbf{x}_k)_{r+1} = \omega^{k(r-1)} + \omega^{k(r+1)}$$

Factoring out ω^{kr}, we get:

$$(A\mathbf{x}_k)_r = \omega^{kr}(\omega^{-k} + \omega^k) = (\mathbf{x}_k)_r \cdot (\omega^{-k} + \omega^k)$$

Thus, $\mathbf{x}_k$ is an eigenvector with eigenvalue $\lambda_k = \omega^k + \omega^{-k}$. Using Euler's formula ($e^{ix} + e^{-ix} = 2\cos x$), we conclude:

$$\lambda_k = 2\cos\left(\frac{2\pi k}{n}\right)$$

■

Lemma 4.1.12. *If $X = (x_1, x_2, \ldots, x_n)$ is an eigenvector of the adjacency matrix A of the path P_n corresponding to the eigenvalue λ, then λ is an eigenvalue of C_{2n+2} with*

$$Y = (x_1, \ldots, x_n, 0, -x_n, \ldots, -x_1, 0) \text{ and}$$

$$Z = (0, x_1, \ldots, x_n, 0, -x_n, \ldots, -x_1)$$

as the corresponding eigenvectors.

Proof. Given (λ, X) is an eigenpair of $A(P_n)$ where $X = (x_1, x_2, \ldots, x_n)$. This gives

$$A(P_n) = \lambda X.$$

Let $P_n = v_1 - v_2 - \cdots - v_{n-1} - v_n$. Then

$$A(P_n)\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

which gives

$$\begin{aligned}
x_2 &= \lambda x_1 \\
x_1 + x_3 &= \lambda x_2 \\
x_2 + x_4 &= \lambda x_3 \\
&\vdots \\
x_{n-2} + x_n &= \lambda x_{n-1} \\
x_{n-1} &= \lambda x_n
\end{aligned}$$

Now,

$$A(C_{2n+2})Y = A(C_{2n+2})\begin{bmatrix} x_1 \\ \vdots \\ x_n \\ 0 \\ -x_n \\ \vdots \\ -x_1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2+0 \\ x_1+x_3 \\ \vdots \\ x_{n-1}+0 \\ x_n+(-x_n) \\ 0-x_{n-1} \\ \vdots \\ -x_2+0 \\ -x_1+x_1 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_n \\ \lambda\cdot 0 \\ -\lambda x_n \\ \vdots \\ -\lambda x_1 \\ -\lambda\cdot 0 \end{bmatrix} = \lambda Y.$$

Therefore, Y is an eigenvector of $A(C_{2n+2})$ corresponding to the eigenvalue λ. On similar lines. we have

$$A(C_{2n+2})Z = A(C_{2n+2})\begin{bmatrix} 0 \\ x_1 \\ x_2 \\ \vdots \\ x_n \\ 0 \\ -x_n \\ \vdots \\ -x_1 \end{bmatrix} = \lambda Z$$

so that Z is an eigenvector of $A(C_{2n+2})$ corresponding to the eigenvalue λ. Thus, λ is an eigenvalue of C_{2n+2} having two linearly independent eigenvectors corresponding to it. ■

Theorem 4.1.13. *The eigenvalues of the path P_n are $2\cos\left(\frac{\pi k}{n+1}\right)$, $k = 1, 2\ldots, n$.*

Proof. Let λ be an eigenvalue of $A(P_n)$ and let $X = (x_1, x_2, \ldots, x_n)$ be the corresponding eigenvector. Then by previous theorem, we have

$$Y = (x_1, x_2, \ldots, x_n, 0, -x_n, x_{n-1}, \ldots, -x_1, 0) \text{ and}$$
$$Z = (0, x_1, x_2, \ldots, x_n, 0, -x_n, -x_{n-1}, \ldots, -x_1)$$

are the eigenvectors of C_{2n+2} corresponding to λ. Clearly Y and Z are linearly independent eigenvectors of C_{2n+2} corresponding to eigenvector λ. This means each eigenvalue λ of P_n is also an eigenvalue of C_{2n+2} with geometric multiplicity 2. But as adjacency matrix is symmetric (hence diagonalizable), thus arithmetic multiplicity of λ as an eigenvalue of C_{2n+2} is also 2. Hence the eigenvalues of path P_n are precisely those eigenvalues of C_{2n+2} whose arithmetic multiplicity is 2.

Now, the eigenvalues of C_{2n+2} are $2\cos\left(\frac{2\pi k}{2n+2}\right) = 2\cos\left(\frac{\pi k}{n+1}\right)$; $k = 1, 2, \ldots, 2n+2$. Using values of $k = 1, 2, \ldots, 2n+2$, these can be written as

$$2\cos\left(\frac{\pi}{n+1}\right), 2\cos\left(\frac{2\pi}{n+1}\right), \ldots, 2\cos\left(\frac{n\pi}{n+1}\right), 2\cos\left(\frac{(n+1)\pi}{n+1}\right)$$

for $k = 1, 2, \ldots, 2n+1$ and for $k = 2n+1, 2n, 2n-1, \ldots, n+2$ and $2n+2$.

$$2\cos\left(\frac{(2n+1)\pi}{n+1}\right), 2\cos\left(\frac{2n\pi}{n+1}\right), \ldots, 2\cos\left(\frac{(n+2)\pi}{n+1}\right), 2\cos\left(\frac{(2n+2)\pi}{n+1}\right)$$

These two lists each containing $n+1$ eigenvalues give the $2n+2$ eigenvalue of C_{2n+2}. Now, we have $\cos\left(\frac{(2n+2-p)\pi}{n+1}\right) = \cos\left(2\pi - \frac{p\pi}{n+1}\right) = \cos\left(\frac{p\pi}{n+1}\right)$. Therefore, for different values of p we see that the terms in the above lists are identical except the last term in each list. Thus the eigenvalues of P_n are

$$2\cos\left(\frac{k\pi}{n+1}\right); \quad k = 1, 2, \ldots, n.$$

■

Definition 4.1.14. Determinant of a matrix $A = [a_{ij}]_{n\times n}$ is defined as $|A| = \sum\limits_{\sigma\in S_n} \text{sgn}(\sigma) \cdot \prod\limits_{i=1}^{n} a_{i\sigma(i)}$ where

$$\text{sgn}(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is even permutation} \\ -1 & \text{if } \sigma \text{ is an odd permutation} \end{cases}$$

Note: If a permutation can be decomposed into d-disjoint cycles of length ≥ 2, then

$$\text{sgn}(\sigma) = (-1)^{n-d}.$$

Example 4.1.15. Evaluate $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$ using the above definition.

Solution. $|A| = \sum\limits_{\sigma\in S_n} \text{sgn}(\sigma) \prod\limits_{i=1}^{n} a_{i\sigma(i)}$. Here $S_2 = \{\sigma_1 = I, \sigma_2 = (12)\}$. Therefore,

$$\begin{aligned} |A| &= \text{sgn}(\sigma_1) \prod_{i=1}^{2} a_{i\sigma_1(i)} + \text{sgn}(\sigma_2) \prod_{i=1}^{2} a_{i\sigma_2(i)} \\ &= \text{sgn}(\sigma_1) a_{1\sigma_1(1)} a_{2\sigma_1(2)} + \text{sgn}(\sigma_2) a_{1\sigma_2(1)} a_{2\sigma_2(2)} \\ &= 1 \cdot a_{11}a_{22} + (-1)a_{12}a_{21} \\ &= a_{11}a_{22} - a_{12}a_{21}. \end{aligned}$$

■

Definition 4.1.16 (Elementary Spanning Subgraph)**.** A spanning subgraph $H \subseteq G$ is *elementary* if every component of H is either:

1. An edge (K_2), or
2. A cycle (C_k with $k \geq 3$).

The relationship between graph structure and the characteristic polynomial $P(G, \lambda) = \det(\lambda I - A)$ is given by the following theorem, often attributed to Harary and Sachs.

Theorem 4.1.17 (Determinant Expansion). *Let A be the adjacency matrix of a simple graph G. Then:*

$$\det(A) = \sum_{H} (-1)^{n-c_1(H)-c(H)} 2^{c(H)}$$

where the sum runs over all elementary spanning subgraphs H. Here, $c_1(H)$ is the number of K_2 components and $c(H)$ is the number of cycle components ($C_k, k \geq 3$).

Example 4.1.18. Find the determinant of the adjacency matrix of the graph given in figure below.
Sol: We have

$$|A| = \sum_{H\in\mathcal{H}_G} (-1)^{n-c_1(H)-c(H)} \cdot 2^{c(H)}.$$

Here we have

$$\begin{aligned} &H_1 = \{(1,2,3,4,5)\}, \quad H_2 = \{(1,2),(3,4,5)\}, \\ &H_3 = \{(1,2,3),(4,5)\} \end{aligned}$$

Hence,

$$|A| = (-1)^{5-0-1} \cdot 2^1 + (-1)^{5-1-1} \cdot 2^1 + (-1)^{5-1-1} \cdot 2^1.$$

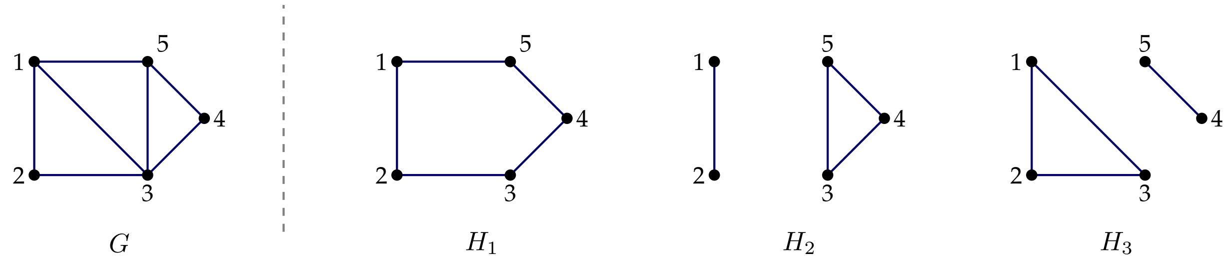


Figure 4.1: Elementary spanning subgraphs of G.

Proof of Theorem 4.1.17. Let $A = [a_{ij}]$ be the adjacency matrix of G. By the definition of the determinant:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^{n} a_{i\sigma(i)}$$

Consider a permutation $\sigma \in S_n$. Any permutation can be written as a product of disjoint cycles. We are only interested in permutations σ that contribute a non-zero quantity to the sum. Since G is a simple graph, $a_{ii} = 0$ for all i. Thus, for the product $\prod a_{i\sigma(i)}$ to be non-zero, $\sigma(i) \neq i$ for all i. Consequently, the disjoint cycle decomposition of such a σ contains no cycles of length 1. Every cycle must have length ≥ 2.

These non-zero contributions correspond to spanning subgraphs H where every vertex has degree exactly 1 (if part of a transposition) or degree 2 (if part of a longer cycle). Specifically:

- A cycle of length 2 in σ corresponds to a K_2 component (an edge) in H.
- A cycle of length $k \geq 3$ in σ corresponds to a cycle subgraph C_k in H.

Thus, such a subgraph H is an elementary spanning subgraph.

The sign of the permutation is given by $\text{sgn}(\sigma) = (-1)^{n-d}$, where d is the number of disjoint cycles in the decomposition of σ. Here, $d = c_1(H) + c(H)$, where $c_1(H)$ is the number of cycles of length 2 (K_2) and $c(H)$ is the number of cycles of length ≥ 3.

Regarding the magnitude of the contribution:

- For a K_2 component (transposition (ij)), the term is $a_{ij}a_{ji} = 1 \cdot 1 = 1$. There is only one way to traverse an edge.
- For a cycle component C_k ($k \geq 3$) with vertices $v_1, v_2, \ldots, v_k$, the permutation can correspond to the cycle $(v_1 v_2 \ldots v_k)$ or the reverse $(v_k v_{k-1} \ldots v_1)$. Both result in the product of adjacency entries being 1. Thus, every cycle component of length ≥ 3 contributes a factor of 2 to the sum (accounting for the two possible directions of traversal).

Therefore, combining the sign and the multiplicity of cycle traversals:

$$|A| = \sum_{H} (-1)^{n-(c_1(H)+c(H))} \cdot 2^{c(H)}.$$

■

Theorem 4.1.19 (Sachs Theorem). *Let $P(G, \lambda) = \lambda^n + a_1\lambda^{n-1} + \cdots + a_n$ be the characteristic polynomial of G. The coefficient a_i is given by:*

$$a_i = \sum_{H_i \in \mathcal{H}_i} (-1)^{p(H_i)} 2^{c(H_i)}$$

where: H_i is the set of elementary subgraphs of G with exactly i vertices, $p(H_i)$ is the number of components in H_i, and $c(H_i)$ is the number of cycle components in H_i.

Proof. Let G be a graph with adjacency matrix $A(G)$. From Matrix Theory, the coefficient a_i of the characteristic polynomial is given by:

$$a_i = (-1)^i \,[\text{Sum of all } i \times i \text{ principal minors of } A]$$

An $i \times i$ principal minor of A corresponds to the determinant of the adjacency matrix of an induced subgraph of G with i vertices. Let such a subgraph be denoted by G'. Applying the previous **Determinant Expansion Theorem** to this subgraph G' of order i:

$$\det(A(G')) = \sum_{H \subseteq G'} (-1)^{i-c_1(H)-c(H)} 2^{c(H)}$$

where H represents an elementary spanning subgraph of the induced subgraph G' (which is simply an elementary subgraph of G with i vertices).

Substituting this back into the expression for a_i:

$$\begin{aligned} a_i &= (-1)^i \sum_{H_i \in \mathcal{H}_i} (-1)^{i-c_1(H_i)-c(H_i)} 2^{c(H_i)} \\ &= \sum_{H_i \in \mathcal{H}_i} (-1)^{2i-c_1(H_i)-c(H_i)} 2^{c(H_i)} \end{aligned}$$

Since $2i$ is even, $(-1)^{2i} = 1$. Also, let $p(H_i)$ be the total number of components in H_i, so $p(H_i) = c_1(H_i) + c(H_i)$. The exponent of -1 becomes $-(c_1(H_i) + c(H_i)) = -p(H_i)$. Since $(-1)^{-x} = (-1)^x$, we have:

$$a_i = \sum_{H_i \in \mathcal{H}_i} (-1)^{p(H_i)} 2^{c(H_i)}.$$

■

4.2 Isomorphism and Co-spectral Graphs

Definition 4.2.1. Two graphs G_1 and G_2 are **co-spectral** if $\mathrm{Spec}(G_1) = \mathrm{Spec}(G_2)$.

While isomorphic graphs are always co-spectral, the converse is not true.

Example 4.2.2. The smallest non-isomorphic co-spectral pair consists of:

- $G_1 = K_{1,4}$ (The Star graph)
- $G_2 = C_4 \cup K_1$ (The union of a cycle and an isolated vertex) — (See Figure 4.2)

Moreover, the connected non-isomorphic graphs of smallest order that are cospectral are given in Figure 4.3 having the characteristic polynomial

$$\phi(G, \lambda) = \lambda^6 - 7\lambda^4 - 4\lambda^3 + 7\lambda^2 + 4\lambda - 1.$$

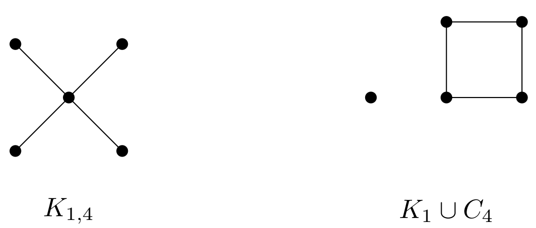


Figure 4.2: Non-isomorphic co-spectral graphs of smallest orders

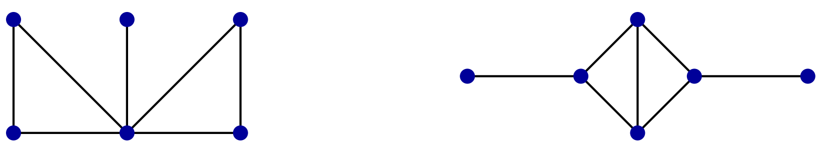


Figure 4.3: Smallest connected co-spectral non-isomorphic graphs

4.3 Regular Graphs

Definition 4.3.1. A graph G is said to be $k-$regular if each vertex of G has degree k.

Theorem 4.3.2. *Let G be a k-regular graph of order n. Then:*

1. *k is an eigenvalue of $A(G)$ with multiplicity at least 1, and $\mathbf{1} = (1, 1, \ldots, 1)^\top$ is a corresponding eigenvector.*
2. *For any eigenvalue λ of G, $|\lambda| \leq k$.*
3. *The multiplicity of k is 1 if and only if G is connected.*

Proof. Let $A(G)$ be the adjacency matrix of G. Since G is k-regular, every row sum and column sum of $A(G)$ is equal to k. Therefore, k is an eigenvalue of $A(G)$.

Observing the matrix-vector product:

$$A(G)\mathbf{1} = \begin{bmatrix} \cdots & \text{row } 1 & \cdots \\ \vdots & \vdots & \vdots \\ \cdots & \text{row } n & \cdots \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} k \\ k \\ \vdots \\ k \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Thus, $\mathbf{1} = (1, 1, \ldots, 1)^\top$ is an eigenvector of $A(G)$ corresponding to k.

Next, we consider the multiplicity. We show that if G is connected, the geometric multiplicity $GM(k) = 1$. Since $A(G)$ is real and symmetric, the algebraic multiplicity $AM(k)$ equals $GM(k)$.

Let $X = (x_1, x_2, \ldots, x_n)^\top$ be an eigenvector of $A(G)$ corresponding to k. By definition, $AX = kX$. Let x_j be an entry in X with the largest absolute value. Consider the j-th row of the eigenvalue equation:

$$(A(G)X)_j = kx_j \implies \sum_{v_i \sim v_j} x_i = kx_j,$$

where the sum runs over the k neighbors v_i adjacent to v_j. Since x_j has the maximum absolute value, $x_i \leq x_j$. For the average of the neighbors to equal the maximum value x_j, every neighbor must also equal x_j:

$$\sum_{v_i \sim v_j} x_i = \underbrace{x_j + x_j + \cdots + x_j}_{k \text{ times}}.$$

This implies $x_i = x_j$ for all neighbors v_i of v_j. Propagating this argument through the graph (assuming G is connected), we conclude that $x_i = x_j$ for all $i = 1, 2, \ldots, n$.

Therefore, any eigenvector for k is a scalar multiple of $\mathbf{1}$:

$$X = c \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

This implies the eigenspace has dimension 1, so $GM(k) = 1$, and consequently $AM(k) = 1$. ■

Short Proof for Bound. Since G is k-regular, $A\mathbf{1} = k\mathbf{1}$. To show $|\lambda| \leq k$, let X be an eigenvector for an eigenvalue λ. Let x_j be the entry of X with the largest absolute value ($|x_j| > 0$). The eigenvalue equation implies:

$$|\lambda x_j| = \left| \sum_{v_i \sim v_j} x_i \right| \leq \sum_{v_i \sim v_j} |x_i|.$$

Since $|x_i| \leq |x_j|$, we have:

$$|\lambda||x_j| \leq \sum_{v_i \sim v_j} |x_j| = k|x_j|.$$

Dividing by $|x_j|$, we obtain $|\lambda| \leq k$. ■

4.3.1 Spectrum of the Complement

Theorem 4.3.3. *If G is a k-regular graph with spectrum $\{k, \lambda_2, \ldots, \lambda_n\}$, then the spectrum of the complement graph $\overline{G}$ is:*

$$\operatorname{Spec}(\overline{G}) = \{n-k-1, -\lambda_2 - 1, \ldots, -\lambda_n - 1\}.$$

Proof. Since G is k-regular, k is an eigenvalue of G with the corresponding eigenvector $u = (1, 1, \ldots, 1)^\top$. Let A and $\overline{A}$ be the adjacency matrices of G and $\overline{G}$ respectively. We have the relation:

$$A + \overline{A} = J_n - I_n,$$

where J_n is the all-ones matrix and I_n is the identity matrix.

First, apply this to the eigenvector u:

$$\begin{aligned}(A + \overline{A})u &= (J_n - I_n)u \\ Au + \overline{A}u &= J_n u - I_n u.\end{aligned}$$

Using the facts that $Au = ku$, $I_n u = u$, and $J_n u = nu$:

$$\begin{aligned}ku + \overline{A}u &= nu - u \\ \overline{A}u &= (n-1)u - ku \\ \overline{A}u &= (n-k-1)u.\end{aligned}$$

Hence, $n - k - 1$ is an eigenvalue of $\overline{A}$ with eigenvector u.

Now, let X be any other eigenvector of A corresponding to an eigenvalue λ_i (for $i = 2, \ldots, n$). Since A is symmetric, eigenvectors corresponding to distinct eigenvalues are orthogonal. Thus, X is orthogonal to u, which implies the sum of entries of X is zero. Consequently, $J_n X = \mathbf{0}$. (Note that $J_n X = 0 \cdot X$ because any vector in $\mathbb{R}^n$, not a scalar multiple of $(1, 1, \ldots, 1)$, is an eigenvector of J_n corresponding to the eigenvalue 0).

Consider the action of $\overline{A}$ on X:

$$\begin{aligned}(A + \overline{A})X &= (J_n - I_n)X \\ AX + \overline{A}X &= J_n X - I_n X \\ \lambda_i X + \overline{A}X &= 0 - X \\ \overline{A}X &= -X - \lambda_i X \\ \overline{A}X &= (-\lambda_i - 1)X.\end{aligned}$$

Hence, for each $\lambda_i \in \operatorname{Spec}(G) \setminus \{k\}$, the value $(-\lambda_i - 1)$ is an eigenvalue of $\overline{A}$. ∎

We now turn to line graphs. Let G be a graph with $V(G) = \{1, \ldots, n\}$ and $E(G) = \{e_1, \ldots, e_m\}$. Recall that the line graph G_ℓ of G has vertex set $E(G)$. For $i \neq j$, e_i and e_j are said to be adjacent if they have a common vertex. If G is k-regular then G_ℓ is $(2k-2)$-regular. We first prove a preliminary result.

Lemma 4.3.4. *Let G be a graph with n vertices. Let A and B be the adjacency matrices of G and of G_ℓ, respectively. If M is the incidence matrix of G, then $M'M = B + 2I$. Furthermore, if G is k-regular then $MM' = A + kI$.*

Proof. Any diagonal entry of $M'M$ clearly equals 2. If e_i and e_j are edges of G then the (i, j)-element of $M'M$ is 1 if e_i and e_j have a common vertex, and 0 otherwise. Hence, $M'M = B + 2I$. To prove the second part, note that for a k-regular graph, $MM' = -L + 2kI$, where L is the Laplacian of G. Hence, $A = kI - L = MM' - kI$. Therefore, $MM' = A + kI$. ∎

Corollary 4.3.5. *Let G be a graph. If μ is an eigenvalue of G_ℓ then $\mu \geq -2$.*

Proof. Let B be the adjacency matrix of G_ℓ. By Lemma 4.3.4, $B + 2I = M'M$ is positive semidefinite. If μ is an eigenvalue of B then $\mu + 2$, being an eigenvalue of a positive semidefinite matrix, must be nonnegative. Hence, $\mu \geq -2$. ∎

Theorem 4.3.6. *Let G be a k-regular graph with n vertices. Then*

$$\phi(G_\ell, \lambda) = (\lambda + 2)^{\frac{n}{2}(k-2)} \phi(G, \lambda + 2 - k).$$

Proof. Let A and B be the adjacency matrices of G and of G_ℓ, respectively. Let M be the incidence matrix of G. If G has m edges then M is of order $n \times m$. Let $k = \lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of A. By Lemma 4.3.4 the eigenvalues of MM' are $2k, \lambda_2 + k, \ldots, \lambda_n + k$. Note that the eigenvalues of $M'M$ are given by the eigenvalues of MM', together with 0 with multiplicity $m - n$. Therefore, again by Lemma 4.3.4, the eigenvalues of B are $2k - 2, \lambda_2 + k - 2, \ldots, \lambda_n + k - 2$, and -2 with multiplicity $m - n$. Since

$$\phi(G, \lambda) = (\lambda - k)(\lambda - \lambda_2) \cdots (\lambda - \lambda_n),$$

then

$$\phi(G, \lambda + 2 - k) = (\lambda + 2 - 2k)(\lambda + 2 - k - \lambda_2) \cdots (\lambda + 2 - k - \lambda_n).$$

Also,

$$\phi(G_\ell, \lambda) = (\lambda + 2 - 2k)(\lambda + 2 - k - \lambda_2) \cdots (\lambda + 2 - k - \lambda_n)(\lambda + 2)^{m-n}.$$

$$\frac{\phi(G_\ell, \lambda)}{\phi(G, \lambda + 2 - k)} = (\lambda + 2)^{m-n}.$$

Since G is k-regular then $2m = nk$ and hence $m - n = \frac{n}{2}(k - 2)$. ■

4.4 The Laplacian Spectrum of a Graph

4.4.1 Definitions

Let $G = (V, E)$ be a simple undirected graph with n vertices and m edges.

- Let $A(G)$ be the **adjacency matrix** of G.
- Let $D(G)$ be the **degree matrix** of G, a diagonal matrix where $D_{ii} = \deg(v_i)$.

Definition 4.4.1 (Laplacian Matrix)**.** The Laplacian matrix $L(G)$ is defined as:

$$L(G) = D(G) - A(G)$$

Explicitly, the entries are given by:

$$L_{ij} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \sim v_j \\ 0 & \text{otherwise} \end{cases}$$

Definition 4.4.2 (Laplacian Spectrum)**.** Since $L(G)$ is a real symmetric matrix, its eigenvalues are real. The Laplacian spectrum is the set of eigenvalues of $L(G)$, usually denoted and ordered as:

$$\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n$$

Theorem 4.4.3 (Positive Semidefinite Property)**.** *The Laplacian matrix $L(G)$ is positive semidefinite. Consequently, $\mu_i \geq 0$ for all $i = 1, \ldots, n$.*

Proof. Let $\mathbf{x} = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^n$. Consider the quadratic form associated with L:

$$\begin{aligned}
\mathbf{x}^T L\mathbf{x} &= \mathbf{x}^T (D - A)\mathbf{x} \\
&= \mathbf{x}^T D\mathbf{x} - \mathbf{x}^T A\mathbf{x} \\
&= \sum_{i=1}^{n} \deg(v_i) x_i^2 - \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij} x_i x_j \\
&= \sum_{i=1}^{n} \left(\sum_{j\sim i} 1 \right) x_i^2 - \sum_{(i,j)\in E} 2x_i x_j \\
&= \sum_{(i,j)\in E} (x_i^2 + x_j^2 - 2x_i x_j) \\
&= \sum_{(i,j)\in E} (x_i - x_j)^2
\end{aligned}$$

Since $(x_i - x_j)^2 \geq 0$ for any edge, the sum is non-negative. Thus, $\mathbf{x}^T L\mathbf{x} \geq 0$ for all $\mathbf{x} \neq \mathbf{0}$. It follows that all eigenvalues μ_i are non-negative. ■

Theorem 4.4.4 (Smallest Eigenvalue)**.** *The smallest eigenvalue of $L(G)$ is always 0. That is, $\mu_n = 0$.*

Proof. Since the row sums of $L(G)$ are all zero (because the diagonal entry is the degree and the off-diagonal entries are -1 for each neighbor), the vector of all ones, $\mathbf{1} = (1, 1, \ldots, 1)^T$, is an eigenvector.

$$(L\mathbf{1})_i = \sum_{j=1}^{n} L_{ij} \cdot 1 = \deg(v_i) - \sum_{j\sim i} 1 = \deg(v_i) - \deg(v_i) = 0$$

Thus, $L\mathbf{1} = 0 \cdot \mathbf{1}$. Since L is positive semidefinite, 0 is the smallest possible eigenvalue. ■

Theorem 4.4.5 (Trace Property)**.** *The sum of the Laplacian eigenvalues is twice the number of edges:*

$$\sum_{i=1}^{n} \mu_i = 2m$$

Proof. The sum of the eigenvalues is equal to the trace of the matrix.

$$\sum_{i=1}^{n} \mu_i = \mathbf{tr}(L) = \sum_{i=1}^{n} L_{ii} = \sum_{i=1}^{n} \deg(v_i)$$

By the Handshaking Lemma, $\sum \deg(v_i) = 2m$. ■

4.5 Laplacian Energy

The concept of graph energy $E(G)$ (sum of absolute values of adjacency eigenvalues) was introduced by Gutman in 1978. In 2006, Gutman and Zhou introduced the **Laplacian Energy**.

The average degree of the graph is $\frac{2m}{n}$. This is also the spectral mean of the Laplacian matrix (average of eigenvalues).

Definition 4.5.1 (Laplacian Energy)**.** The Laplacian energy of a graph G, denoted by $LE(G)$, is defined as:

$$LE(G) = \sum_{i=1}^{n} \left| \mu_i - \frac{2m}{n} \right|$$

Theorem 4.5.2 (Relation to Graph Energy for Regular Graphs)**.** *If G is a k-regular graph, then the Laplacian energy equals the ordinary graph energy.*

$$LE(G) = E(G)$$

Proof. Let G be a k-regular graph. The degree matrix is a scalar matrix $D = kI$. The Laplacian Matrix is $L = kI - A$. If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of the adjacency matrix A, then the eigenvalues of L are $\mu_i = k - \lambda_i$. The number of edges is $m = \frac{nk}{2}$, so the average degree is $\frac{2m}{n} = k$.

Substituting these into the definition of Laplacian Energy:

$$\begin{aligned} LE(G) &= \sum_{i=1}^{n} \left| \mu_i - \frac{2m}{n} \right| \\ &= \sum_{i=1}^{n} |(k - \lambda_i) - k| \\ &= \sum_{i=1}^{n} |-\lambda_i| \\ &= \sum_{i=1}^{n} |\lambda_i| \\ &= E(G) \end{aligned}$$

■

Theorem 4.5.3 (Basic Properties of Laplacian Energy)**.** *For any graph G:*

(i) $LE(G) \geq 0$.

(i) $LE(G) = 0$ *if and only if* $m = 0$ *(the graph has no edges).*

Proof. (i). The sum of absolute values is always non-negative.

(ii). If $m = 0$, then L is the zero matrix, all $\mu_i = 0$, and $\frac{2m}{n} = 0$, so $LE(G) = 0$. Conversely, if $LE(G) = 0$, then $\mu_i = \frac{2m}{n}$ for all i. Since $\mu_n = 0$ (smallest eigenvalue), it implies $\frac{2m}{n} = 0$, which means $m = 0$. ■

Theorem 4.5.4. *The Laplacian spectrum of the complete graph K_n is $\{0^{(1)}, n^{(n-1)}\}$.*

Proof. The Laplacian matrix is given by $L(K_n) = nI - J$, where I is the identity matrix and J is the all-ones matrix. The spectrum of J is known to be $\{n^{(1)}, 0^{(n-1)}\}$, corresponding to the eigenvector $\mathbf{1}$ and the subspace orthogonal to $\mathbf{1}$, respectively. Since $L(K_n)$ is a linear combination of I and J, its eigenvalues are obtained via the transformation $\mu_i = n - \lambda_i(J)$.

- For $\lambda(J) = n$, we have $\mu = n - n = 0$ (multiplicity 1).
- For $\lambda(J) = 0$, we have $\mu = n - 0 = n$ (multiplicity $n - 1$).

■

Theorem 4.5.5. *The Laplacian spectrum of the complete bipartite graph $K_{r,s}$ is $\{0^{(1)}, (r+s)^{(1)}, r^{(s-1)}, s^{(r-1)}\}$.*

Proof. Let V_1 and V_2 be the partition sets with $|V_1| = r$ and $|V_2| = s$. The row sums of L are zero, so $\mu_1 = 0$ is an eigenvalue with eigenvector $\mathbf{1}$.

Consider vectors $\mathbf{x}$ that sum to zero on V_1 and are zero on V_2. There are $r - 1$ such linearly independent vectors. For these vectors, $L\mathbf{x} = s\mathbf{x}$, yielding eigenvalue s with multiplicity $r - 1$. Similarly, considering vectors supported only on V_2 that sum to zero, we obtain eigenvalue r with multiplicity $s - 1$.

We find the final eigenvalue using the trace of the matrix. The trace of L is the sum of degrees: $\text{tr}(L) = r(s) + s(r) = 2rs$. Let μ_n be the unknown eigenvalue. Then:

$$0 + s(r-1) + r(s-1) + \mu_n = 2rs$$

$$(rs - s) + (rs - r) + \mu_n = 2rs$$

$$2rs - (r + s) + \mu_n = 2rs \implies \mu_n = r + s$$

Thus, the spectrum is established. ■

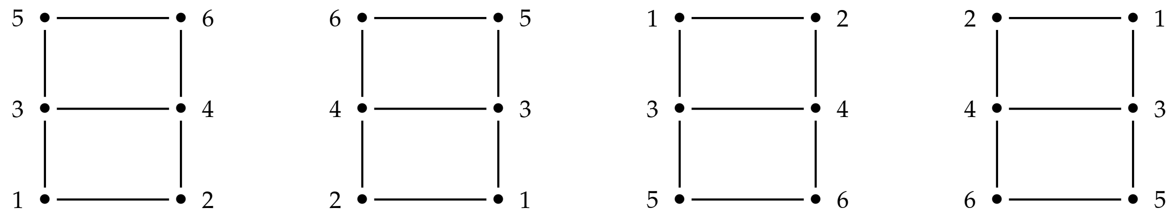


Figure 4.4: Automorphism group of $G_{2,3}$.

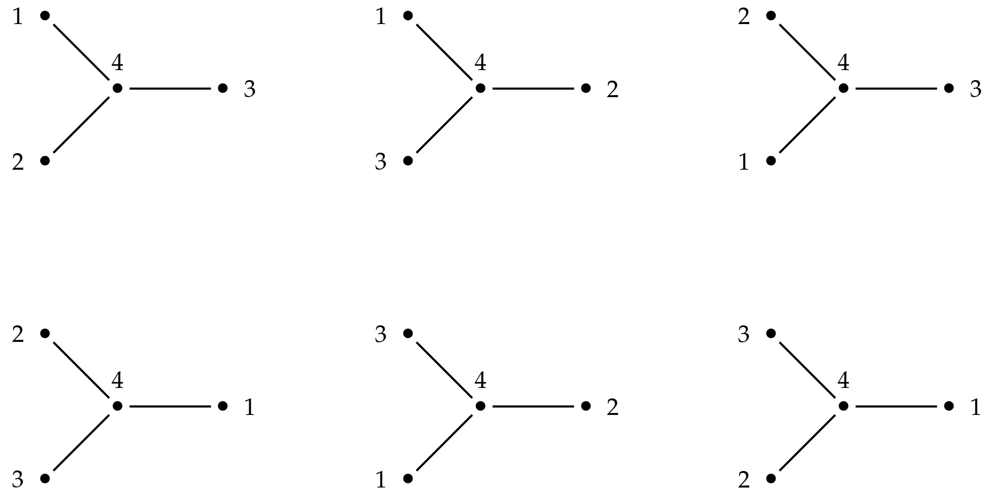


Figure 4.5: Automorphisms of star graph S_4.

4.6 Automorphism Groups of Graphs

Definition 4.6.1. A graph automorphism of G is a permutation φ on the set of vertices $V(G)$ that satisfies the property that $\{u_i, u_j\} \in E(G)$ if and only if $\{\varphi(u_i), \varphi(u_j)\} \in E(G)$.

Example 4.6.2. For example, the grid graph $G_{2,3}$ has four automorphisms: $(1,2,3,4,5,6)$, $(2,1,4,3,6,5)$, $(5,6,3,4,1,2)$, $(6,5,4,3,2,1)$. These correspond to the graph itself, the graph flipped from left to right, the graph flipped up down, and the graph flipped left to right and up down, respectively.

The star graph S_4 has six automorphisms: $(1,2,3,4)$, $(1,3,2,4)$, $(2,1,3,4)$, $(2,3,1,4)$, $(3,1,2,4)$, $(3,2,1,4)$.

Theorem 4.6.3. *The set* $\mathrm{Aut}(G)$ *of all graph automorphisms of a graph G forms a group under function composition.*

Definition 4.6.4 (Identity Graph)**.** A graph G whose automorphism group if $Aut(G) \cong S_1$ is called the identity graph.

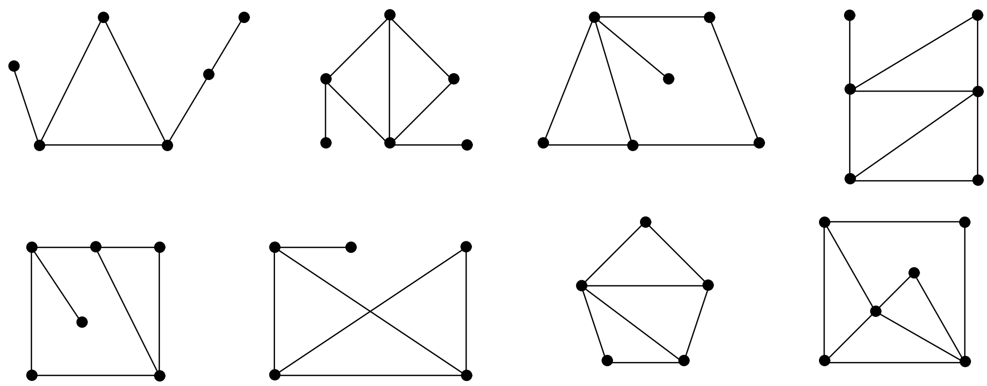


Figure 4.6: The Eight Identity Graphs of order Six.

The elements of $Aut(G)$ induce a group of permutations on the edge set E of G in the obvious way:

If $e = uv \in E$ and $\alpha \in Aut(G)$, then $\alpha_1(e) = \alpha_0(u)\alpha_0(v)$ where α_0 and α_1 denote the action of α on V and E respectively. This permutation group on E with degree $m = |E|$ and order $|Aut(G)|$ is called the induced edge group of G and is denoted by $Aut_I(G)$.

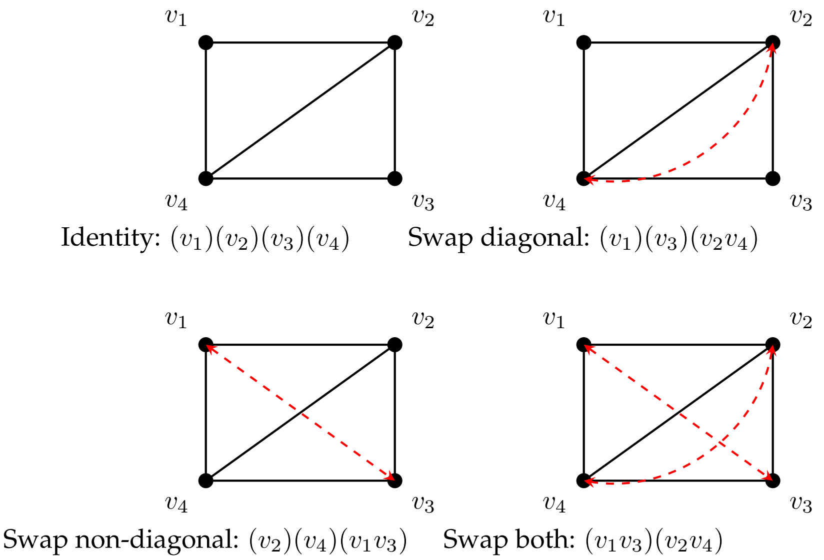


Figure 4.7: Automorphisms of $K_4 - e$

Example 4.6.5. Consider the graph $K_4 - e$ with vertices, Figure 4.7, labeled v_1, v_2, v_3, v_4 and edges e_1, e_2, e_3, e_4, e_5. The vertex group $Aut(K_4 - e)$ consists of the four permutations:

$$(v_1)(v_2)(v_3)(v_4), \quad (v_1)(v_3)(v_2v_4), \quad (v_2)(v_4)(v_1v_3), \quad (v_1v_3)(v_2v_4)$$

or

$$\begin{pmatrix} v_1 & v_2 & v_3 & v_4 \\ v_1 & v_2 & v_3 & v_4 \end{pmatrix}, \quad \begin{pmatrix} v_1 & v_3 & v_2 & v_4 \\ v_1 & v_3 & v_4 & v_2 \end{pmatrix}, \quad \begin{pmatrix} v_2 & v_4 & v_1 & v_3 \\ v_2 & v_4 & v_3 & v_1 \end{pmatrix}, \quad \begin{pmatrix} v_1 & v_3 & v_2 & v_4 \\ v_3 & v_1 & v_4 & v_2 \end{pmatrix}.$$

The identity permutation of the vertex-group induces the identity permutation on the edges, while $(v_1)(v_3)(v_2v_4)$ induces a permutation on the edges which fixes e_5, interchanges e_1 with e_4, and e_2 with e_3. In this way, the edge-group consists of the following permutations, induced respectively by the above members of the vertex-group.

$$(e_1)(e_2)(e_3)(e_4)(e_5), \quad (e_1e_4)(e_2e_3)(e_5), \quad (e_1e_2)(e_3e_4)(e_5), \quad (e_1e_3)(e_2e_4)(e_5)$$

or

$$\begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_1 & e_2 & e_3 & e_4 & e_5 \end{pmatrix}, \quad \begin{pmatrix} e_1 & e_4 & e_2 & e_3 & e_5 \\ e_4 & e_1 & e_3 & e_2 & e_5 \end{pmatrix}, \quad \begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_2 & e_1 & e_4 & e_3 & e_5 \end{pmatrix}, \quad \begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_3 & e_1 & e_4 & e_2 & e_5 \end{pmatrix}.$$

Of course, the edge-group and the vertex-group of $K_4 - e$ are isomorphic. But they are not identical permutation groups since $Aut_I(K_4 - e)$ has degree 5 and $Aut(K_4 - e)$ has degree 4. Note that edge e_5 is fixed by every member of the edge-group. Even the permutation group obtained from $Aut_I(K_4 - e)$ by restricting its object set to e_1, e_2, e_3, e_4 is not identical with $Aut(K_4 - e)$, since these two isomorphic permutation groups of the same degree have different cycle structure.

Theorem 4.6.6. *The vertex automorphism group Aut(G) and the induced edge automorphism group Aut$_I(G)$ of a graph G are isomorphic if and only if G contains at most one isolated vertex and no connected component isomorphic to K_2.*

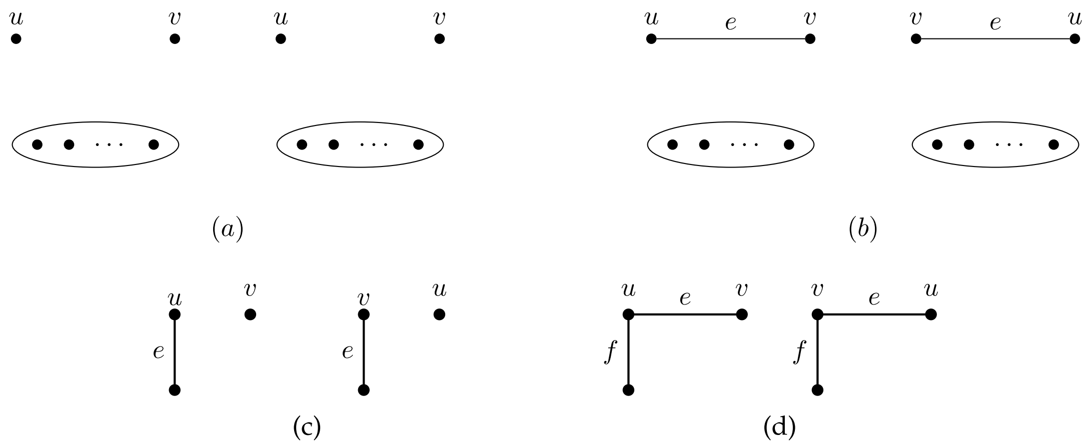


Figure 4.8: Theorem 3.4.5

Proof. Let $\phi : \mathrm{Aut}(G) \to \mathrm{Aut}_I(G)$ be the natural homomorphism defined by $\phi(\alpha_0) = \alpha_1$, where the edge automorphism α_1 is induced by the vertex automorphism α_0 according to the rule $\alpha_1(\{u, v\}) = \{\alpha_0(u), \alpha_0(v)\}$. Since every element of $\mathrm{Aut}_I(G)$ arises from some vertex automorphism, ϕ is surjective. Consequently, $\mathrm{Aut}(G) \cong \mathrm{Aut}_I(G)$ if and only if ϕ is injective, which requires the kernel of ϕ to be trivial. We examine the conditions under which a non-identity vertex automorphism induces the identity edge automorphism.

For the necessity of the stated conditions, assume first that G contains two distinct isolated vertices, u and v. The vertex automorphism α_0 that transposes u and v while fixing all other vertices leaves the edge set entirely unchanged, as no edges are incident to u or v. Thus $\phi(\alpha_0)$ is the identity edge map, implying a non-trivial kernel. Similarly, assume G contains a component isomorphic to K_2, consisting of vertices u and v and the single edge $e = \{u, v\}$. The automorphism that swaps u and v maps the edge e to $\{v, u\} = e$. Since this automorphism acts trivially on all other components, it fixes every edge in the graph, again resulting in a non-trivial kernel. Therefore, for the groups to be isomorphic, G must have at most one isolated vertex and no K_2 components.

For sufficiency, assume G satisfies the conditions. Let α_0 be a vertex automorphism in the kernel of ϕ, meaning α_0 maps every edge $\{x, y\}$ to the set $\{x, y\}$. Suppose, for the sake of contradiction, that α_0 is not the identity, so there exists a vertex u such that $\alpha_0(u) = v$ with $u \neq v$. Since automorphisms preserve degree, $\deg(u) = \deg(v)$. If $\deg(u) = 0$, then both u and v are isolated vertices, contradicting the assumption that G has at most one isolated vertex. Thus, $\deg(u) \geq 1$, and there exists an edge $e = \{u, w\}$. Because α_0 fixes all edges, it must map e to itself, so $\{\alpha_0(u), \alpha_0(w)\} = \{v, \alpha_0(w)\} = \{u, w\}$. Since $v \neq u$, this equality forces $v = w$, implying that u and v are adjacent via edge $e = \{u, v\}$. Furthermore, if there existed any other edge $f = \{u, z\}$ incident to u (where $z \neq v$), the condition that α_0 fixes f would similarly imply $\{v, \alpha_0(z)\} = \{u, z\}$. This requires $v \in \{u, z\}$, which is impossible since $v \neq u$ and $z \neq v$. Consequently, u has no neighbors other than v, and by symmetry, v has no neighbors other than u. This implies that u and v form an isolated K_2 component, contradicting the hypothesis. Therefore, no such vertex u exists, α_0 must be the identity, and ϕ is an isomorphism. ∎

Corollary 4.6.7. *If G is a connected graph with $n \geq 3$, then $Aut(G) \cong Aut_I(G)$.*

Theorem 4.6.8. *Given any graph G, $\mathrm{Aut}(G) = \mathrm{Aut}(\overline{G})$.*

Proof. Let V be the vertex set of G. Recall that for the complement graph $\overline{G}$, an edge $\{u, v\}$ exists in $E(\overline{G})$ if and only if $\{u, v\} \notin E(G)$.

Let σ be a permutation of V. We show that $\sigma \in \mathrm{Aut}(G)$ if and only if $\sigma \in \mathrm{Aut}(\overline{G})$. By the definition of an automorphism, $\sigma \in \mathrm{Aut}(G)$ if and only if for all distinct vertices $u, v \in V$:

$$\{u, v\} \in E(G) \iff \{\sigma(u), \sigma(v)\} \in E(G).$$

Taking the logical negation of both sides of the equivalence, we obtain:

$$\{u, v\} \notin E(G) \iff \{\sigma(u), \sigma(v)\} \notin E(G).$$

By the definition of the complement graph, $\{x, y\} \notin E(G)$ is equivalent to $\{x, y\} \in E(\overline{G})$. Substituting this into the expression above yields:

$$\{u, v\} \in E(\overline{G}) \iff \{\sigma(u), \sigma(v)\} \in E(\overline{G}).$$

This is precisely the condition for σ to be an automorphism of $\overline{G}$. Thus, $\sigma \in \mathrm{Aut}(G) \iff \sigma \in \mathrm{Aut}(\overline{G})$, proving that $\mathrm{Aut}(G) = \mathrm{Aut}(\overline{G})$. ■

Theorem 4.6.9. *1. $Aut(K_n) \cong S_n$*

2. *$Aut(K_n - e) \cong S_{n-2} \times S_2$*
3. *Automorphism group of K_n with two adjacent edges removed is $S_{n-3} \times S_2$*
4. *Automorphism group of K_n with two non -adjacent edges removed is $S_{n-4} \times D_4$*
5. *$Aut(C_n) \cong D_n$*
6. *For $m, n \geq 1$, $Aut(K_{m,n}) \cong S_m \times S_n$ $m \neq n$ and $Aut(K_{n,n}) \cong (S_n \times S_n) \rtimes \mathbb{Z}_2$.*
7. *$Aut(P_n) \cong S_2$.*

Definition 4.6.10. Let S_n be the symmetric group of degree n, and let $\mathbb{Z}_2 = \{0, 1\}$ be the cyclic group of order 2 (under addition modulo 2).

The **semidirect product** $S_n \rtimes_\varphi \mathbb{Z}_2$ is the set of ordered pairs $S_n \times \mathbb{Z}_2$ equipped with a group operation defined by a homomorphism:

$$\varphi : \mathbb{Z}_2 \to \mathrm{Aut}(S_n)$$

Let $\varphi_0 = \mathrm{id}_{S_n}$ and $\varphi_1 = \theta$, where θ is an automorphism of S_n such that $\theta^2 = \mathrm{id}$ (an involution).

The group operation for two elements (σ, a) and (τ, b) in $S_n \rtimes_\varphi \mathbb{Z}_2$ is defined as:

$$(\sigma, a) \cdot (\tau, b) = (\sigma \circ \varphi_a(\tau), a + b)$$

where $\circ$ denotes permutation multiplication and $+$ denotes addition modulo 2.

Note:

- If φ is the trivial homomorphism (maps everything to the identity automorphism), the group becomes the **Direct Product** $S_n \times \mathbb{Z}_2$.
- To obtain a non-trivial semidirect product, θ is typically chosen to be an inner automorphism (conjugation by an element of order 2) or, for $n = 6$, an outer automorphism.

Proof. 1. **Proof for K_n:**
The complete graph K_n contains every possible edge between n distinct vertices. By definition, an automorphism is a permutation of vertices that preserves adjacency. Since every pair of distinct vertices in K_n is adjacent, any bijection (permutation) from the vertex set V to itself preserves adjacency. There are $n!$ such permutations on n elements. The group of all permutations on n elements is the symmetric group S_n. Thus, $\mathrm{Aut}(K_n) \cong S_n$.

2. **Proof for $K_n - e$:**
Let $G = K_n - e$. Let the deleted edge be $e = \{u, v\}$. In G, the vertices u and v have degree $n - 2$ (connected to everyone except each other). All other $n - 2$ vertices (let's call this set W) have degree $n-1$. Any automorphism ϕ must map vertices to vertices of the same degree. Therefore, ϕ must map the set $\{u, v\}$ to itself and the set W to itself.

- On the set W, the induced subgraph is a complete graph K_{n-2}. Any permutation of these vertices is valid. This gives the group S_{n-2}.
- On the set $\{u, v\}$, we can either fix them ($u \to u, v \to v$) or swap them ($u \to v, v \to u$). This gives the group S_2.

Since these actions are independent, $\mathrm{Aut}(G) \cong S_{n-2} \times S_2$.

3. **Proof for K_n with two adjacent edges removed:**
Let the removed edges be $\{u, v\}$ and $\{v, w\}$. These edges share the common vertex v. We analyze the degrees of the vertices in G:

   - Vertex v is not connected to u or w. Its degree is $n - 1 - 2 = n - 3$.
   - Vertices u and w are not connected to v. Their degree is $n - 1 - 1 = n - 2$.
   - The remaining $n - 3$ vertices (set Z) are connected to everyone. Their degree is $n - 1$.

   Since vertex degrees must be preserved: 1. Vertex v is unique (degree $n - 3$) and must be fixed ($\phi(v) = v$). 2. The set $\{u, w\}$ (degree $n - 2$) must map to itself. We can swap them or fix them ($\cong S_2$). 3. The set Z (degree $n - 1$) forms a K_{n-3}. Any permutation is allowed ($\cong S_{n-3}$). Thus, $\mathrm{Aut}(G) \cong S_{n-3} \times S_2$.

4. **Proof for K_n with two non-adjacent edges removed:**
Let the removed edges be $e_1 = \{u, v\}$ and $e_2 = \{x, y\}$, where $\{u, v\} \cap \{x, y\} = \emptyset$. The vertices u, v, x, y have degree $n - 2$. The remaining $n - 4$ vertices (set Z) have degree $n - 1$. Any automorphism must map Z to Z and $\{u, v, x, y\}$ to $\{u, v, x, y\}$.

   - Action on Z: The subgraph induced by Z is K_{n-4}, allowing any permutation ($\cong S_{n-4}$).
   - Action on $\{u, v, x, y\}$: The induced subgraph on these four vertices is K_4 minus two disjoint edges (which is equivalent to a cycle C_4: $u - x - v - y - u$), which gives rise to a group of rotations and reflections of a square. This group of symmetries is isomorphic to the Dihedral group of order 8, D_4.

   Thus, $\mathrm{Aut}(G) \cong S_{n-4} \times D_4$.

5. **Proof for C_n:**
Let the vertices of the cycle be $1, 2, \ldots, n$ in clockwise order. The cycle graph C_n allows for two types of symmetries: 1. Rotations: We can shift every vertex by k positions (n possibilities). 2. Reflections: We can reflect the graph across an axis of symmetry (n possibilities). A group generated by rotations and reflections of a regular n-gon is the Dihedral group D_n, which has order $2n$. Thus, $\mathrm{Aut}(C_n) \cong D_n$.

6. **Proof for $K_{m,n}$:**
Let V_1 and V_2 be the partite sets with $|V_1| = m$ and $|V_2| = n$. In a complete bipartite graph, every vertex in V_1 is connected to every vertex in V_2. First assume $m \neq n$;

   - Since $m \neq n$, the degrees or the sizes of the partitions distinguish the sets V_1 and V_2. An automorphism cannot map a vertex from V_1 to V_2.
   - Therefore, any automorphism ϕ acts as a permutation σ on V_1 and a permutation τ on V_2.
   - Since all vertices in V_1 are connected to all in V_2, any choice of $\sigma \in S_m$ and $\tau \in S_n$ preserves adjacency.

   Consequently, $\mathrm{Aut}(K_{m,n}) \cong S_m \times S_n$.

   Next, assume that $m = n$; Let $G = K_{n,n}$ with partite sets V_1 and V_2. Since $\mathrm{Aut}(G) = \mathrm{Aut}(\overline{G})$, consider the complement $\overline{G}$. $\overline{G}$ consists of edges only within V_1 and within V_2, making it the disjoint union of two complete graphs $K_n \cup K_n$.

   Any automorphism of $\overline{G}$ must map connected components to connected components. Since the two components are isomorphic:

(a) We can permute vertices within V_1 arbitrarily (S_n) and within V_2 arbitrarily (S_n). This forms the subgroup $S_n \times S_n$.

(b) We can swap the sets V_1 and V_2 entirely. This action corresponds to $\mathbb{Z}_2$.

The group generated by these actions is the semi-direct product $(S_n \times S_n) \rtimes \mathbb{Z}_2$, also known as the wreath product $S_n \wr S_2$.

7. **Proof of P_n:** Let the vertices of P_n be labeled $1, 2, \ldots, n$ in sequence. The vertices 1 and n are the only vertices with degree 1. Since automorphisms preserve degrees, any $\phi \in \text{Aut}(P_n)$ must map the set $\{1, n\}$ to itself. This allows only two possibilities for $\phi(1)$:

(a) $\phi(1) = 1$: Since automorphisms preserve adjacency and the path is connected, fixing the first vertex fixes the entire path. Thus, $\phi(i) = i$ for all i (the identity map, id).

(b) $\phi(1) = n$: To preserve the path structure, the neighbor of 1 must map to the neighbor of n, and so on. This forces $\phi(i) = n - i + 1$ (the reflection map, σ).

Thus, $\text{Aut}(P_n) = \{id, \sigma\}$. Since $\sigma^2 = id$ and $\sigma \neq id$, this group is cyclic of order 2, which is isomorphic to the symmetric group S_2.

■

4.7 Edge Isomorphisms and Automorphisms

Definition. Two graphs $G(V, E)$ and $H(U, F)$ are said to be **edge-isomorphic** if there exists a bijection $\eta : E \to F$ that preserves edge-adjacency and edge-non-adjacency. Specifically, two edges $e, f \in E$ share a common vertex in G if and only if their images $\eta(e), \eta(f) \in F$ share a common vertex in H. Such a map η is called an *edge-isomorphism*.

An edge-isomorphism from a graph to itself is called an **edge-automorphism**. The set of all edge-automorphisms of a graph G forms a group under composition, known as the *edge-automorphism group* of G, denoted by $\text{Aut}_E(G)$.

It is important to observe that for a given graph G, the vertex automorphism group $\text{Aut}(G)$ and the edge-automorphism group $\text{Aut}_E(G)$ need not be isomorphic. This discrepancy arises because there may exist edge-automorphisms that are not induced by any vertex automorphism (i.e., they are not in the induced edge group $\text{Aut}_I(G)$).

Figure 1 illustrates three specific cases—$K_{1,3} + e$, $K_4 - e$, and K_4—where each graph admits an edge-automorphism η that cannot be induced by any permutation of the vertices.

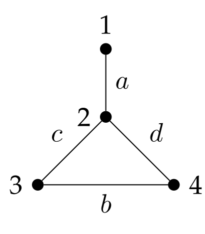


Figure 4.9: *
$K_{1,3} + e$
$\eta = (a\, b)(c\, d)$

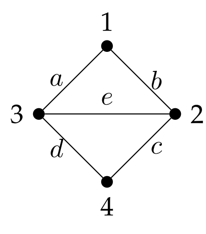


Figure 4.10: *
$K_4 - e$
$\eta = (a\, b\, c\, d)(e)$

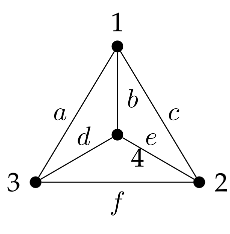


Figure 4.11: *
K_4
$\eta = (b\, f)(d)(e)(a)(c)$

Figure 4.12: Edge-automorphisms not induced by vertex automorphisms.

Theorem 4.7.1. *Let η be an edge-isomorphism from a connected graph G to a connected graph H. If G is not isomorphic to any of the graphs K_3, $K_{1,3}$, $K_{1,3} + e$, $K_4 - e$, or K_4, then η is induced by a vertex isomorphism from G to H.*

Theorem 4.7.2. *For a non-empty graph G, $\Gamma_I(G) \cong \Gamma_E(G)$ if and only if the following two conditions hold.*

(a) Both K_3 and $K_{1,3}$ are not components of G.

(b) None of the graphs $K_{1,3} + e$, $K_4 - e$ and K_4 is a component of G.

Proof. The necessity of these conditions follows from the existence of edge-automorphisms for the excluded graphs that are not induced by vertex automorphisms. The sufficiency is immediate for connected graphs by the preceding theorem.

We now establish sufficiency for a disconnected graph G. Let α be an arbitrary edge-automorphism of G. Our goal is to demonstrate that a vertex-automorphism of G induces α.

Consider any connected component H of G. The image of its edge set under α, denoted $\alpha(E(H))$, induces a subgraph $\langle \alpha(E(H)) \rangle$ which must also be a connected component of G.

If a component H is isomorphic to K_3 or $K_{1,3}$, condition (a) ensures that any such component maps to another isomorphic component in a way that preserves the structure. Specifically, we must have $H \cong \langle \alpha(E(H)) \rangle$. Although these specific graphs admit exceptional edge-automorphisms, within the context of the whole graph and the constraints on components, the restriction of α to H aligns with a vertex-automorphism of H.

If a component H is not isomorphic to K_3 or $K_{1,3}$, and by condition (b) is also not isomorphic to $K_{1,3} + e$, $K_4 - e$, or K_4, then H falls under the purview of Theorem 1. Consequently, the restriction of α to the edges of H is induced by a vertex-isomorphism from H to its image component.

Since α decomposes into independent mappings on the disjoint components of G, and each restriction is induced by a vertex-isomorphism, we can construct a global vertex-automorphism for G by combining these individual maps. Thus, α is induced by a vertex-automorphism of G. ■

Corollary 4.7.3. *If G is a connected graph of order $n \geq 3$, then $Aut(G), Aut_I(G)$ and $Aut_E(G)$ are all isomorphic if and only if G is none of the graphs $K_{1,3} + e, K_4 - e, K_4$.*

4.8 Cayley Graphs of Groups

Let G be a group and let $S \subseteq G$ be a generating set of G. The **Cayley graph** $\Gamma = \text{Cay}(G, S)$ is a graph constructed as follows:

1. The set of vertices $V(\Gamma)$ corresponds to the elements of the group G. That is, $V(\Gamma) = G$.
2. The set of edges $E(\Gamma)$ is defined such that there is a directed edge from g to h if and only if $h = gs$ for some $s \in S$.

Notation: An edge is typically denoted as (g, gs) and acts as a directed edge colored or labeled by the generator s.

- Usually, the generating set S is chosen such that the identity element $1 \notin S$.
- If S is symmetric (i.e., $s \in S \implies s^{-1} \in S$), the graph is viewed as undirected, because an edge (g, gs) implies the existence of an edge $(gs, gss^{-1}) = (gs, g)$.

Example 4.8.1 (The Cyclic Group $\mathbb{Z}_4$)**.** Let $G = \mathbb{Z}_4 = \{0, 1, 2, 3\}$ under addition modulo 4. Let the generating set be $S = \{1\}$.

- **Vertices:** $0, 1, 2, 3$.
- **Edges:** Since the operation is $+$, edges are $(g, g + 1)$.

$$0 \to 1, \quad 1 \to 2, \quad 2 \to 3, \quad 3 \to 0$$

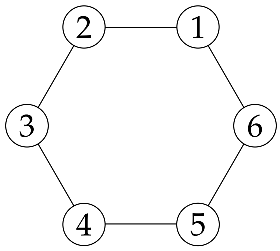
$$\begin{tikzcd} 0 \arrow[r, "+1"] & 1 \arrow[d, "+1"] \\ 3 \arrow[u, "+1"] & 2 \arrow[l, "+1"] \end{tikzcd}$$

This forms a directed cycle C_4.

Example 4.8.2 (Cyclic Group $\mathbb{Z}_6$). Let $G = \mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ (under addition modulo 6). Let the generating set be $S = \{1\}$.

- Vertices: $\{0, 1, 2, 3, 4, 5\}$
- Edges: $(i, i + 1 \pmod 6)$.

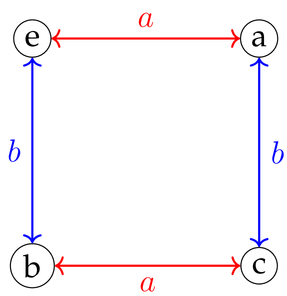
This forms a simple directed cycle C_6.



Example 4.8.3 (The Klein 4-Group (V_4)). Let $G = V_4 = \{e, a, b, c\}$ where $a^2 = b^2 = c^2 = e$ and $ab = c$. Let the generating set be $S = \{a, b\}$.

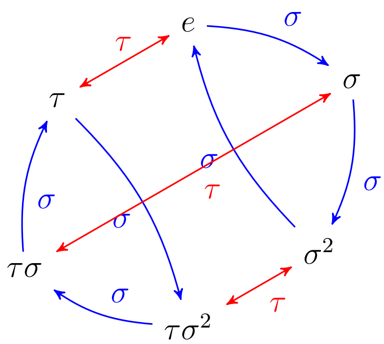
- Edges generated by a (Red): $(e, a), (a, e), (b, c), (c, b)$.
- Edges generated by b (Blue): $(e, b), (b, e), (a, c), (c, a)$.

Since $a = a^{-1}$ and $b = b^{-1}$, for every arrow $x \to y$, there is an arrow $y \to x$. We often draw these as undirected edges.



Example 4.8.4 (Symmetric Group S_3). Let $G = S_3$, the permutations of $\{1, 2, 3\}$. Let $S = \{\tau, \sigma\}$ where $\tau = (12)$ and $\sigma = (123)$.

- σ has order 3 (forms hexagons).
- τ has order 2 (forms bi-directional swaps).



Note: The red edges represent τ. Since $\tau^2 = e$, these are technically bidirectional arrows $x \leftrightarrow y$.

4.9 Undirected Cayley Graphs

While the definition implies a digraph (directed graph), we often treat them as undirected graphs under specific conditions.

Definition 4.9.1. Cay(G, S) is considered an undirected graph if S is inverse-closed. That is, for every $s \in S$, we have $s^{-1} \in S$.

In this case, if there is an edge $g \to gs$, there is also an edge $gs \to (gs)s^{-1} = g$. These pairs of directed edges are treated as a single undirected edge.

Theorem 4.9.2. *If S generates the group G, then the Cayley graph Cay(G, S) is connected.*

Proof. Let $u, v \in G$ be any two vertices in the graph. We wish to show there is a path from u to v.
Consider the element $g = u^{-1}v$. Since S generates G, the element g can be written as a product of generators and their inverses (if we consider the undirected case or if S is closed under inverses).

$$g = s_1 s_2 \dots s_k \quad \text{where } s_i \in S \cup S^{-1}$$

Therefore, $v = u(s_1 s_2 \dots s_k)$. We can construct a walk starting at u:

$$\begin{aligned} w_0 &= u \\ w_1 &= us_1 \\ w_2 &= us_1 s_2 \\ &\vdots \\ w_k &= us_1 s_2 \dots s_k = v \end{aligned}$$

Since there is a sequence of edges connecting u to v, the graph is connected. ∎

Theorem 4.9.3. *The Cayley graph $\Gamma =$ Cay(G, S) is $|S|$-regular (in terms of out-degree).*

Proof. Let $g \in G$ be any vertex. The edges originating from g are determined by the definition:

$$E_g = \{(g, gs) \mid s \in S\}$$

Since group cancellation holds ($gs_1 = gs_2 \implies s_1 = s_2$), the number of distinct edges starting at g is exactly equal to the number of distinct elements in S. Thus, the out-degree of every vertex is $|S|$. ∎

This is the most fundamental property of Cayley graphs. It implies that the graph "looks the same" from every vertex.

Theorem 4.9.4. *Every Cayley graph Cay(G, S) is vertex-transitive.*

Proof. A graph Γ is vertex-transitive if, for any two vertices $u, v \in V(\Gamma)$, there exists an automorphism $\phi \in \text{Aut}(\Gamma)$ such that $\phi(u) = v$.
Let $u, v \in G$. Let $h = vu^{-1}$. We define the map $\lambda_h : G \to G$ by left multiplication:

$$\lambda_h(x) = hx$$

1. λ_h is a bijection: This follows from group theory axioms. The inverse is $\lambda_{h^{-1}}$.
2. λ_h preserves adjacency (is a graph homomorphism): Suppose there is an edge from x to y. By definition, this means $y = xs$ for some $s \in S$. We check the relationship between $\lambda_h(x)$ and $\lambda_h(y)$:

$$\lambda_h(y) = hy = h(xs) = (hx)s = \lambda_h(x)s$$

Since $\lambda_h(y) = \lambda_h(x)s$, there exists an edge from $\lambda_h(x)$ to $\lambda_h(y)$ with the same label s. Thus, λ_h is a graph automorphism.
3. Mapping u to v:

$$\lambda_h(u) = hu = (vu^{-1})u = v$$

Since we found an automorphism mapping u to v for arbitrary u, v, the graph is vertex-transitive. ∎

Definition 4.9.5. In the Cayley digraph $D(\Gamma, \Lambda)$, if (g, k) is an arc, $k = gh$ for some $h \in \Lambda$, that is $g^{-1}k \in \Lambda$ and $g^{-1}k$ is called the color of (g, k).
An automorphism of D is said to be color preserving if it preserves the colors of the arcs.

Theorem 4.9.6 (Automorphism Theorem for Colored Cayley Graphs)**.** *The group of color-preserving automorphisms of the Cayley digraph $D(\Gamma, \Lambda)$ is isomorphic to Γ.*

Proof. Let $C(D)$ denote the group of color-preserving automorphisms of the digraph $D(\Gamma, \Lambda)$. We first establish that the group of left multiplications is a subgroup of $C(D)$. For each element $g \in \Gamma$, define the mapping $\alpha_g : \Gamma \to \Gamma$ by $\alpha_g(x) = gx$. Since Γ is a group, α_g is clearly a bijection. To see that α_g preserves colors, consider an arbitrary arc (u, v) in D with color $h = u^{-1}v \in \Lambda$. The image of this arc under α_g is $(\alpha_g(u), \alpha_g(v)) = (gu, gv)$. The color of the image arc is determined by

$$(gu)^{-1}(gv) = u^{-1}g^{-1}gv = u^{-1}v = h.$$

Since the image arc has the same color h as the original arc, α_g is a color-preserving automorphism. Thus, the set of left multiplications $L(\Gamma) = \{\alpha_g \mid g \in \Gamma\}$ is contained in $C(D)$.

Conversely, we show that every colour-preserving automorphism belongs to $L(\Gamma)$. Let $\phi \in C(D)$ be an arbitrary automorphism. Let e denote the identity of Γ, and let $\phi(e) = g$. We claim that $\phi(x) = gx$ for all $x \in \Gamma$. We proceed by induction on the length of the shortest path from e to x in terms of the generating set Λ. The base case $x = e$ holds by definition, as $\phi(e) = g = ge$.

For the inductive step, assume that $\phi(u) = gu$ for some vertex u, and let v be a vertex adjacent to u such that (u, v) is an arc with color $h \in \Lambda$. By the definition of color, $u^{-1}v = h$, or $v = uh$. Since ϕ is color-preserving, the arc $(\phi(u), \phi(v))$ must also have color h. Therefore,

$$\phi(u)^{-1}\phi(v) = h.$$

Substituting the inductive hypothesis $\phi(u) = gu$, we have

$$(gu)^{-1}\phi(v) = h \implies u^{-1}g^{-1}\phi(v) = h \implies \phi(v) = guh.$$

Since $v = uh$, it follows that $\phi(v) = gv$. Because the Cayley graph is connected (assuming Λ generates Γ), this property extends to all vertices $x \in \Gamma$. Thus, $\phi = \alpha_g$, implying $C(D) \subseteq L(\Gamma)$.

We have established that $C(D) = L(\Gamma)$. Finally, consider the mapping $\Psi : \Gamma \to C(D)$ defined by $\Psi(g) = \alpha_g$. This map is surjective by the argument above and injective because $\alpha_g(e) = g$ implies distinct group elements define distinct automorphisms. Furthermore, Ψ is a homomorphism because the composition of left multiplications corresponds to multiplication in the group:

$$(\alpha_g \circ \alpha_k)(x) = \alpha_g(kx) = g(kx) = (gk)x = \alpha_{gk}(x).$$

Therefore, Ψ is an isomorphism, and $\Gamma \cong C(D)$. ■

Theorem 4.9.7 (Frucht, 1939)**.** *Let Γ be a finite group. There exists a finite, undirected graph G such that the automorphism group of G is isomorphic to Γ:*

$$Aut(G) \cong \Gamma$$

Proof. Let $\Lambda = \{h_1, h_2, \ldots, h_n\}$ be a set of generators for the group Γ, and let $D = D(\Gamma, \Lambda)$ be the Cayley digraph of Γ. We have previously established that the group of colour-preserving automorphisms of this digraph, denoted $C(D)$, is isomorphic to Γ. To prove the theorem, we construct an undirected, uncoloured graph G whose full automorphism group coincides with $C(D)$.

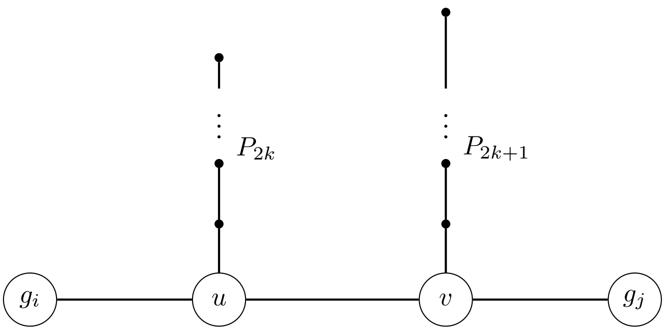
The graph G is obtained from D by a specific topological substitution that encodes the direction and colour of the original arcs. For every directed arc (g_i, g_j) in D with color h_k (where $g_i^{-1}g_j = h_k$), we replace the arc with a subgraph connecting g_i and g_j. This subgraph consists of a path of length three, $g_i - u - v - g_j$, along with two identifying paths rooted at the internal vertices: a path P_{2k} of length $2k$ attached to u, and a path P_{2k+1} of length $2k+1$ attached to v. This construction is illustrated in Figure 3.1. Since this procedure is performed for every arc, the resulting graph G is undirected.

We now define the isomorphism between the automorphism groups. Any colour-preserving automorphism $\alpha \in C(D)$ naturally induces a unique automorphism on G. Since α maps an arc (g_i, g_j) of color h_k to another arc (g_x, g_y) of the same color h_k, it maps the substituting subgraph of the graph G to the identical subgraph of the isomorphic copy of G under α, preserving adjacency throughout. Thus, $C(D)$ is isomorphic to a subgroup of $\mathrm{Aut}(G)$.

Conversely, let $\beta \in \mathrm{Aut}(G)$ be an arbitrary automorphism of the constructed graph. We observe that the vertices of G can be partitioned based on their degrees and structural properties. The original vertices from Γ have degree at least $2|\Lambda|$, whereas the internal vertices u and v have degree 3, and the vertices on the attached paths have degree 1 or 2. Consequently, β must map the set of original vertices Γ to itself. Furthermore, β must map the internal structures connecting these vertices to isomorphic structures.

Consider the subgraph connecting two adjacent original vertices. Because the vertex u has an attached path of even length $2k$, and v has an attached path of odd length $2k + 1$, β cannot map u to v. This asymmetry forces β to preserve the orientation of the original arc (mapping u to a u-type vertex and v to a v-type vertex). Similarly, since the lengths of the attached paths depend strictly on the index k, β can only map a subgraph corresponding to generator h_k to another subgraph corresponding to h_k. This implies that β effectively acts on the original vertices Γ in a way that preserves both the adjacency direction and the edge colors. Thus, β corresponds to a unique element in $C(D)$.

We conclude that $\mathrm{Aut}(G) \cong C(D)$. Since $C(D) \cong \Gamma$, it follows that $\mathrm{Aut}(G) \cong \Gamma$. ■



Replaces arc $g_i \xrightarrow{h_k} g_j$

Figure 4.13: Frucht Graph: The subgraph replacing the directed edge from g_i to g_j with color h_k. The asymmetry of the paths P_{2k} and P_{2k+1} enforces the direction and color preservation.

Corollary 4.9.8. *Let Γ be a finite group of order n, and let Λ be a generating set of size $s = |\Lambda|$. The Frucht graph $G(\Gamma)$ (Fig 4.13) has exactly $n(s + 1)(2s + 1)$ vertices.*

Proof. To determine the order of the Frucht graph $G(\Gamma)$, we consider the vertices contributed by the original group elements and those introduced by the topological construction. The underlying Cayley digraph $D(\Gamma, \Lambda)$ contains n vertices, and each vertex possesses exactly s outgoing arcs corresponding to the k generators indexed $k = 1$ to s. In the transition to $G(\Gamma)$, each arc associated with the k-th generator is replaced by a subgraph that introduces $4k + 1$ new internal vertices. Consequently, for every element in the group, the set of outgoing arcs contributes a total of $\sum_{k=1}^{s} (4k + 1)$ additional vertices. Evaluating this sum yields $4\frac{s(s+1)}{2} + s = 2s^2 + 3s$ new vertices per group element. Across the entire group of n elements, the total number of added vertices is $n(2s^2 + 3s)$. Adding the n original vertices to this count, we find that the total order of the graph is $n(2s^2 + 3s + 1)$. Factoring the quadratic expression, we conclude that $|V(G(\Gamma))| = n(s + 1)(2s + 1)$. ■

Example 4.9.9. Frucht Graph Order for $n = 3,$ and $s = 2$.

Solution. First, calculating directly using the formula:

$$|V(G)| = 3(2 + 1)(2(2) + 1) = 3(3)(5) = 45.$$

We now construct the count step-by-step based on the topological definition of the graph.

Every vertex in Γ has $s = 2$ outgoing arcs, one for each generator. The construction replaces these arcs with "gadgets" of different sizes depending on the index k. The number of internal vertices added for the k-th generator is $4k + 1$.

- For generator h_1 ($k = 1$): The gadget adds $4(1) + 1 = 5$ vertices.
- For generator h_2 ($k = 2$): The gadget adds $4(2) + 1 = 9$ vertices.

Since every group element (e, x, y) has exactly one outgoing arc for h_1 and one outgoing arc for h_2, the total number of vertices added **per group element** is:

$$\text{Added per vertex} = 5 + 9 = 14.$$

Therefore,

$$\begin{aligned}
\text{Total Vertices} &= n + n\sum_{k=1}^{2}(4k+1) \\
&= 3 + 3[(4(1)+1) + (4(2)+1)] \\
&= 3 + 3[5+9] \\
&= 3 + 42 \\
&= 45.
\end{aligned}$$

■